# Reflection Li–Weil Matrices: Determinant Geometry, Local Pivot Dynamics, and Jacobi Spectral Encoding

Víctor Casaña-Giner[a,*]

[a] *Universidad Nacional de Educación a Distancia, Madrid, Spain*

**Abstract**

We study two even/odd reflection compressions of the bilateral Li–Weil kernel, separating results independent of the Riemann Hypothesis (RH) from those that assume it. Unconditionally, we establish the Toeplitz–Hankel reflection form, exact reconstruction of the Li sequence, reflected Weil-pairing identities, discrete wave equations, rank-two displacement, closed bordering recursions, and a local $LDL^T$ pivot law; the even family alone also yields a matrix criterion equivalent to RH. Under RH, the matrices become positive sine Gram systems on the Cayley images of the zeta zeros. This gives positive Chebyshev moment measures, exact Cauchy–Binet–Vandermonde expansions, a fixed-size projection determinantal process, and infinite Jacobi operators whose spectral supports and scaled cyclic measures recover distinct ordinates and multiplicities. We also obtain the critical Schatten exponent 1/2 and exact Christoffel/Darboux and Fredholm determinant relations. Independently of RH, an endpoint theorem proves $\log D_N = -\beta N^2 \log N + o(N^2 \log N)$ under explicit node-scale, separation, and weight hypotheses; $\beta$ alone is insufficient. For the Li–Cayley measures these hypotheses give coefficient $-2$ under RH. A second-order $N^2 \log\log N$ law requires additional gap and weight assumptions and is not deduced from RH. Finally, the independent unconditional zeta-value Hankel family satisfies $\log\det(\zeta(a(i+j)+b))_{1\le i,j\le n} = -an^2 \log n + O(n^2)$.

*Keywords:* Li coefficients, Riemann Hypothesis, Weil criterion, Toeplitz–Hankel matrices, Hankel determinants, Jacobi operators

*Corresponding author

*Email address:* vcasana1@alumno.uned.es (Víctor Casaña-Giner)

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## 1. Introduction

The aim of this paper is to isolate, with complete proofs, the exact mathematical consequences of a pair of reflection compressions of the bilateral Li–Weil kernel. The scalar Li criterion and the bilateral Weil kernel are classical. The objects developed here are the even and odd reflection arrays and the finite- and infinite-dimensional structures generated by them.

The logical status of every result is separated explicitly. Reflection definitions, reconstruction formulas, wave and displacement identities, bordering recursions, and local elimination identities are unconditional. Positive zero-side Gram, moment, determinantal-process, and Jacobi descriptions assume RH. The abstract endpoint theorem is independent of RH, but it requires explicit geometric hypotheses on node scale, separation, and weights. Its Li–Cayley specialization uses RH to obtain a positive moment representation, whereas the final zeta-value Hankel application is unconditional.

The matrix-theoretic viewpoint also places the present work within several themes developed in *Linear Algebra and its Applications*. The direct link between semi-infinite Hankel and Jacobi matrices through commutation and orthogonal-polynomial structures was investigated by Štampach and Šťovíček [37]; our construction uses a different mechanism, namely the moment matrix of a positive measure and its three-term Jacobi recurrence. Structured rank deficiency and low-rank approximation for Hankel matrices were studied by Fazzi, Guglielmi and Markovsky [15], while fixed-rank matrix perturbations and their determinant constraints were treated by Baragaña and Roca [2]. At the level of determinantal algebra, compound-matrix and Binet–Cauchy-type identities also form an active line of research in the journal, including the tropical framework of Akian, Gaubert and Niv [1]. These works address different problems, but they provide a natural linear-algebraic context for the Hankel, Jacobi, low-rank displacement, compound-minor and determinant structures developed below.

### *1.1. Main contributions*

The main contributions are the following.

(1) We introduce the even and odd reflection arrays

$$e_{mn} = \lambda_{m+n} - \lambda_{|m-n|}, \qquad o_{mn} = \lambda_{m+n-1} - \lambda_{|m-n|},$$

and determine their unconditional information content. The odd semi-infinite first row reconstructs the complete Li sequence; the odd reflection map is injective; and the kernel of the even reflection map is identified exactly.

(2) We prove unconditional reflected Weil-pairing formulas, discrete wave equations, rank-two displacement identities, inverse-displacement identities whenever the relevant blocks are invertible, closed bordering recursions, and a local $LDL^T$ pivot dynamics controlled by the first two elimination subdiagonals.

(3) We obtain matrix criteria equivalent to RH. Positivity of all even and odd leading reflection blocks is equivalent to the Li criterion, and the even family alone is shown to be sufficient through a transformed Bombieri–Lagarias multiset.

(4) Assuming RH, we construct the positive sine Gram and Chebyshev moment realizations. They lead to exact Cauchy–Binet–Vandermonde formulas, a fixed-order projection determinantal ensemble, zero deletion and transport identities, and high-zero leverage estimates whose scope is explicitly restricted to fixed matrix order.

(5) We prove an abstract endpoint-exponent theorem for positive discrete Gram determinants. Its conclusion is not a consequence of $\beta$ alone: the theorem identifies the precise node-scale, separation, and weight assumptions under which the leading coefficient is $-\beta$. Applying it to the Li–Cayley support gives coefficient $-2$ under RH.

(6) Under RH, we construct infinite Jacobi operators with exact spectral support, distinguish support recovery from multiplicity recovery, determine the critical Schatten exponent, and derive the Christoffel/Darboux and Fredholm determinant relations between the even and odd systems.

(7) We isolate a second-order law under hypotheses strictly stronger than RH, and we give an independent unconditional application to Hankel determinants of zeta values with leading term $-an^2 \log n$.

**Novelty boundary.** No novelty is claimed for the scalar Li criterion, the Bombieri–Lagarias multiset criterion, the bilateral Weil kernel, standard

moment-matrix or Jacobi theory, or the generic occurrence of $N^2 \log N$ scales in discrete determinants. The contribution developed here is the specific even/odd reflection package, its exact Toeplitz–Hankel, wave, displacement, elimination, determinantal, and spectral structures, and the endpoint analysis of the Li–Cayley measures. All priority claims are limited to the precise statements proved in this paper.

For orientation, other zeta-associated matrix programs include Roesler's eigenvalue formulations [32, 33], spectral analysis of Redheffer matrices [43], and multiplicative Toeplitz matrices tied to the zeta function [18, 19]. Those constructions are structurally distinct from the reflection matrices studied here; they are mentioned to position the present matrix framework within the broader linear-algebraic literature surrounding the zeta function.

*1.2. Guide to hypotheses and proofs*

The proof architecture can be audited in the following order.

(a) The algebraic reflection results use only the Li sequence and explicitly stated nonsingularity assumptions when an inverse or pivot is formed.

(b) The even-only equivalence uses the Bombieri–Lagarias criterion, but does not assume RH.

(c) The sine Gram, positive moment, determinantal-process, and Jacobi layers begin only after RH is assumed.

(d) The abstract endpoint theorem uses its four displayed hypotheses. In the Li–Cayley application, standard zero counting verifies the first-order node and separation requirements; no unproved gap hypothesis is used for the coefficient $-2$.

(e) The second-order refinement adds explicit scale, aggregate-gap, and weight hypotheses and is not deduced from RH.

(f) The zeta-value Hankel application uses a separate positive Dirichlet-series measure and does not use RH or the zeta-zero measures.

Subsection 2.1 gives the complete hypothesis table, and Figure 1 displays the same dependency structure graphically.

For later reference, Lagarias's bilateral Li test functions are

$$G_k(s) := 1 - \left(1 - \frac{1}{s}\right)^k, \qquad k \in \mathbb{Z}, \tag{1.1}$$

and their Weil Gram kernel is

$$\langle G_r, G_s\rangle_W = \lambda_r + \lambda_s - \lambda_{|r-s|}. \tag{1.2}$$

The reflected test vectors used throughout are

$$F_n^E = \frac{G_n - G_{-n}}{\sqrt{2}}, \qquad F_n^O = \frac{G_n - G_{1-n}}{\sqrt{2}}. \tag{1.3}$$

## 2. Definitions and conventions

The following conventions are used throughout.

**Zeros and multiplicity**

A nontrivial zero of $\zeta$ is denoted by $\rho$. Under RH we write $\rho = \frac{1}{2} + i\gamma$ with $\gamma > 0$. The symbol $m_\gamma$ denotes the multiplicity of the ordinate $\gamma$. Sums over $\gamma > 0$ count multiplicity unless the set $\Gamma_+^*$ of distinct positive ordinates is displayed explicitly.

**Li coefficients**

The coefficients $\lambda_n$ are defined by (3.3) below, with $\lambda_0 = 0$. For the bilateral Li–Weil kernel we use the even extension $\lambda_{-n} = \lambda_n$.

**Reflection matrices**

For $m, n \geq 1$,

$$e_{mn} := \lambda_{m+n} - \lambda_{|m-n|}, \qquad o_{mn} := \lambda_{m+n-1} - \lambda_{|m-n|},$$

and $\mathsf{E}_N = (e_{mn})_{m,n\leq N}$, $\mathsf{O}_N = (o_{mn})_{m,n\leq N}$. Their leading determinants are

$$\Delta_N^E := \det \mathsf{E}_N, \qquad \Delta_N^O := \det \mathsf{O}_N, \qquad \Delta_0^E = \Delta_0^O := 1.$$

Whenever the preceding leading block is nonsingular, the corresponding Schur or $LDL^T$ pivot is

$$p_N^E := \frac{\Delta_N^E}{\Delta_{N-1}^E}, \qquad p_N^O := \frac{\Delta_N^O}{\Delta_{N-1}^O}.$$

**Li–Cayley coordinates**

For a zero $\rho$ put $q_\rho := 1 - \rho^{-1}$. Under RH, $q_\rho = e^{i\theta_\gamma}$ and

$$x_\gamma := \cos\theta_\gamma = \frac{4\gamma^2 - 1}{4\gamma^2 + 1}, \qquad 1 - x_\gamma = \frac{2}{4\gamma^2 + 1}.$$

Thus $x_\gamma \uparrow 1$ as $\gamma \to \infty$.

**Moment determinants**

For a finite positive discrete measure $\mu = \sum_j w_j \delta_{x_j}$, its monomial Gram or Hankel determinant is

$$D_N[\mu] := \det\left(\int x^{r+s}\, \mathrm{d}\mu(x)\right)_{r,s=0}^{N-1}.$$

The endpoint exponent $\beta$ refers only to the local support law $1 - x(t) \asymp t^{-\beta}$, or its regularly varying form. By itself this exponent does *not* determine the determinant asymptotic; the endpoint theorem also imposes separate hypotheses on the node scale, a separated subconfiguration, and the selected weights.

**Jacobi operators**

For either normalized reflection measure, $\mathcal{J}$ denotes the Jacobi operator unitarily equivalent to multiplication by $x$. Its defect is $\mathcal{D} := I - \mathcal{J}$. The Schatten ideal of order $p > 0$ is denoted by $\mathcal{S}_p$.

**Zero-index kernels**

At fixed matrix order $N$, the zero-side Gram factorization is $M_N = AA^T$ and

$$K_N := A^T M_N^{-1} A.$$

A repeated zero is represented by repeated identical columns, one for each multiplicity index.

**Asymptotic notation**

All $O(\cdot)$, $o(\cdot)$, $\ll$, and $\asymp$ statements are taken in the indicated asymptotic variable; constants may depend on fixed matrix order or fixed parameters when this is explicitly stated.

### *2.1. Logical status of the main results*

The phrases "unconditional", "under RH", and "under endpoint hypotheses" are used in a literal logical sense throughout the paper. In particular, the abstract endpoint theorem is not a consequence of RH, and the positive zero-side moment/Jacobi constructions are not asserted without RH. The main blocks are as follows.

| **Block** | **Logical status** | **Input beyond definitions** |
|---|---|---|
| Reflection reconstruction, wave equation, displacement and local $LDL^T$ identities | Unconditional | Algebra of the Li sequence; nonsingularity only where an inverse or pivot is used. |
| Exact reflected Weil pairing | Unconditional identity | Classical bilateral Li–Weil kernel. |
| Even-family matrix criterion | Equivalence to RH | Bombieri–Lagarias multiset criterion; no prior assumption of RH. |
| Sine Gram, positive moment measures, Cauchy–Binet zero ensemble | Under RH | Zeros written as $1/2 + i\gamma$, hence positive Cayley weights. |
| Jacobi zero spectrum, cyclic masses, Schatten and Christoffel/Darboux theory | Under RH | Positive reflection measures from the preceding row. |
| Endpoint-exponent theorem | Unconditional theorem under hypotheses | Endpoint law plus node-density, separated-subconfiguration and weight hypotheses. |
| Li–Weil coefficient $-2$ | Under RH | Positive Li–Cayley measures; the endpoint hypotheses are then verified from zero counting. |
| Second-order $N^2 \log\log N$ law | RH plus extra hypotheses | Strengthened node scale, gap regularity and weight asymptotics. |
| Zeta-value Hankel law $-an^2 \log n + O(n^2)$ | Unconditional | Positive Dirichlet-series moment measure with node scale $t_j = j$. |

Figure 1 summarizes these dependencies. Blue arrows in the figure are unconditional, green arrows are implications made after assuming RH, orange dashed arrows use the abstract endpoint hypotheses, and the red branch is the external unconditional zeta-value Hankel application.

## 3. The xi-function and Li coefficients

Define the completed zeta-function

$$\xi(s) = \frac{1}{2} s(s-1) \pi^{-s/2} \Gamma\left(\frac{s}{2}\right) \zeta(s). \tag{3.1}$$

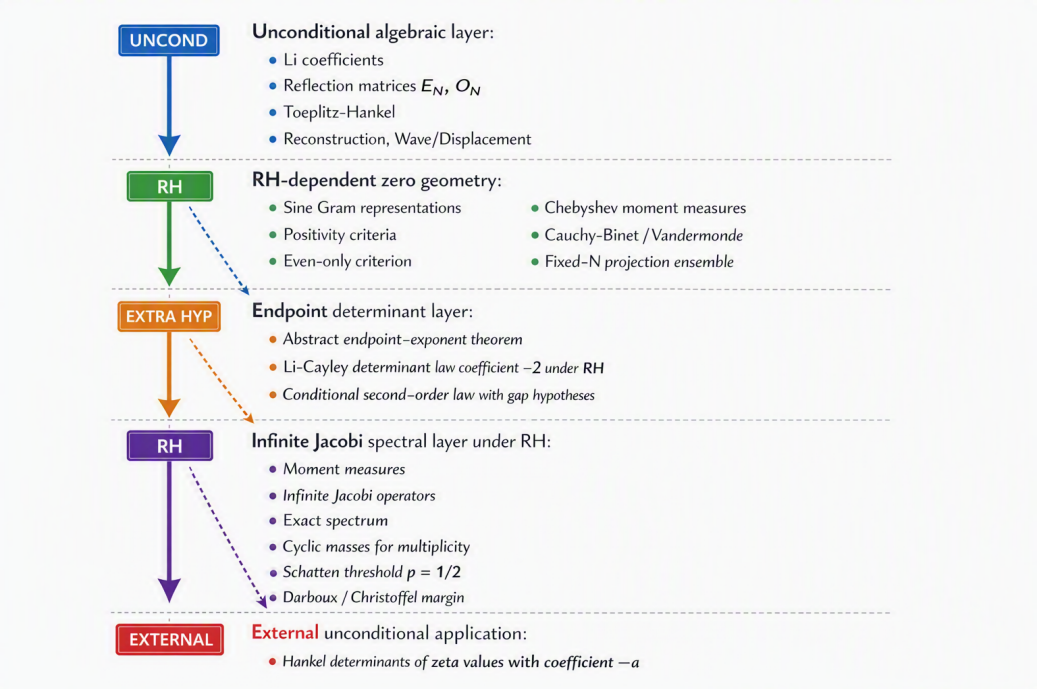



Figure 1: Dependency map with logical status made explicit. Blue arrows are unconditional; green arrows require RH; orange dashed arrows invoke the abstract endpoint hypotheses; the red branch is the external unconditional zeta-value Hankel application. The $-2$ Li–Weil law therefore belongs to the RH branch because the positive Li–Cayley moment measures do, whereas the endpoint theorem itself does not assume RH.

It is entire of order one and satisfies

$$\xi(s) = \xi(1-s) = \overline{\xi(\overline{s})}. \tag{3.2}$$

Riemann's original memoir introduced the analytic framework behind this completion [31]; a modern overview of the hypothesis and its context is given by Conrey [12]. The standard analytic theory may be found in [14, 41, 5].

**Definition 3.1** (Li coefficients)**.** For $n \geq 1$, define

$$\lambda_n = \frac{1}{(n-1)!}\,\frac{\mathrm{d}^n}{\mathrm{d}s^n}\Big[s^{n-1}\log\xi(s)\Big]\Big|_{s=1}, \qquad \lambda_0 := 0. \tag{3.3}$$

Equivalently, with the symmetric limiting convention over nontrivial zeros,

$$\lambda_n = \sum_{\rho}\left[1-\left(1-\frac{1}{\rho}\right)^n\right]. \tag{3.4}$$

These numbers are also called the Keiper–Li coefficients: the power-series expansion of $\log \xi$ underlying (3.3) and its high-precision evaluation were studied by Keiper [22] before Li's positivity criterion appeared. Subsequent formulations and computational perspectives on these coefficients include [9, 10, 11, 25, 26].

For the zeta xi-function the zero symmetry implies an integer-index extension

$$\lambda_{-n} = \lambda_n \qquad (n \geq 1), \tag{3.5}$$

as recorded, for example, in [23, Eq. (1.2)]. This identity is unconditional and will be used whenever differences of integer indices occur.

**Theorem 3.2** (Li)**.** *The Riemann hypothesis is equivalent to*

$$\lambda_n \geq 0 \qquad \textit{for every } n \geq 1. \tag{3.6}$$

*Proof.* This is Li's positivity criterion [24]; see also [6, 23]. □

Li-type criteria have also been developed for zero-free half-planes, extensions of the Selberg class, Hecke and related $L$-functions, and generalized Bombieri–Lagarias settings; see [16, 13, 29, 28, 35, 7, 30]. None of those variants is used below.

Define

$$R_\xi := \min\left\{1, \inf_\rho \left|1 - \frac{1}{\rho}\right|\right\}. \tag{3.7}$$

The Li generating identity is the formal power-series identity

$$\Lambda(z) := \sum_{n\geq 1} \lambda_n z^n = \frac{z}{(1-z)^2} \frac{\xi'}{\xi}\Big(\frac{1}{1-z}\Big). \tag{3.8}$$

Analytically, (3.8) holds in the disk $|z| < R_\xi$. Under RH, every nontrivial zero satisfies $|1-1/\rho| = 1$, and therefore $R_\xi = 1$. If RH fails, a zero with real part $> 1/2$ produces a singularity with $|1 - 1/\rho| < 1$, so the unconditional radius must not be stated as 1.

Under RH we may parameterize each zero in the upper half-plane by

$$\rho = \frac{1}{2} + i\gamma, \qquad 1 - \frac{1}{\rho} = \mathrm{e}^{i\theta_\gamma}, \qquad 0 < \theta_\gamma < \pi. \tag{3.9}$$

A direct calculation gives

$$\cos\theta_\gamma = \frac{4\gamma^2 - 1}{4\gamma^2 + 1}, \tag{3.10}$$

$$\sin\theta_\gamma = \frac{4\gamma}{4\gamma^2 + 1}. \tag{3.11}$$

Pairing conjugate zeros in (3.4) yields the absolutely convergent representation

$$\lambda_n = 2\sum_{\gamma>0} \left(1 - \cos(n\theta_\gamma)\right). \tag{3.12}$$

For fixed $n$, the summands are $O_n(\gamma^{-2})$.

## 4. Definition of the reflection matrices

**Definition 4.1** (Even and odd reflection matrices)**.** For $N \geq 1$, define

$$\mathsf{E}_N := (e_{mn})_{m,n=1}^N, \qquad e_{mn} := \lambda_{m+n} - \lambda_{|m-n|}, \tag{4.1}$$

$$\mathsf{O}_N := (o_{mn})_{m,n=1}^N, \qquad o_{mn} := \lambda_{m+n-1} - \lambda_{|m-n|}. \tag{4.2}$$

Both are real symmetric. Their first $4 \times 4$ truncations are

$$\mathsf{E}_4 = \begin{pmatrix} \lambda_2 & \lambda_3 - \lambda_1 & \lambda_4 - \lambda_2 & \lambda_5 - \lambda_3 \\ \lambda_3 - \lambda_1 & \lambda_4 & \lambda_5 - \lambda_1 & \lambda_6 - \lambda_2 \\ \lambda_4 - \lambda_2 & \lambda_5 - \lambda_1 & \lambda_6 & \lambda_7 - \lambda_1 \\ \lambda_5 - \lambda_3 & \lambda_6 - \lambda_2 & \lambda_7 - \lambda_1 & \lambda_8 \end{pmatrix}, \tag{4.3}$$

and

$$\mathsf{O}_4 = \begin{pmatrix} \lambda_1 & \lambda_2 - \lambda_1 & \lambda_3 - \lambda_2 & \lambda_4 - \lambda_3 \\ \lambda_2 - \lambda_1 & \lambda_3 & \lambda_4 - \lambda_1 & \lambda_5 - \lambda_2 \\ \lambda_3 - \lambda_2 & \lambda_4 - \lambda_1 & \lambda_5 & \lambda_6 - \lambda_1 \\ \lambda_4 - \lambda_3 & \lambda_5 - \lambda_2 & \lambda_6 - \lambda_1 & \lambda_7 \end{pmatrix}. \tag{4.4}$$

**Proposition 4.2** (Toeplitz–Hankel decomposition)**.** *Let*

$$\mathsf{T}_N := \left(\lambda_{|m-n|}\right)_{m,n=1}^N, \tag{4.5}$$

$$\mathsf{H}_N^{(0)} := \left(\lambda_{m+n}\right)_{m,n=1}^N, \qquad \mathsf{H}_N^{(-1)} := \left(\lambda_{m+n-1}\right)_{m,n=1}^N. \tag{4.6}$$

*Then*

$$\mathsf{E}_N = \mathsf{H}_N^{(0)} - \mathsf{T}_N, \qquad \mathsf{O}_N = \mathsf{H}_N^{(-1)} - \mathsf{T}_N. \tag{4.7}$$

*Moreover,*

$$\mathsf{E}_N - \mathsf{O}_N = \left(\lambda_{m+n} - \lambda_{m+n-1}\right)_{m,n=1}^{N}, \tag{4.8}$$

*which is a pure Hankel matrix.*

*Proof.* All statements follow directly from (4.1) and (4.2). □

## 5. Algebraic reconstruction and information content

The two arrays contain substantial algebraic redundancy. This can be described exactly without assuming RH.

**Proposition 5.1** (Reconstruction from the infinite first rows)**.** *Since*

$$o_{1j} = \lambda_j - \lambda_{j-1}, \qquad j \geq 1, \tag{5.1}$$

*one has*

$$\lambda_n = \sum_{j=1}^{n} o_{1j} \qquad (n \geq 1), \tag{5.2}$$

*and, for all* $m, n \geq 1$*,*

$$o_{mn} = \sum_{j=|m-n|+1}^{m+n-1} o_{1j}. \tag{5.3}$$

*If* $r := \min(m, n)$*, then*

$$e_{mn} = \sum_{j=0}^{r-1} e_{1,\,|m-n|+1+2j}. \tag{5.4}$$

*Finally,*

$$e_{mn} = \sum_{j=|m-n|+1}^{m+n} o_{1j}. \tag{5.5}$$

*Thus the semi-infinite odd array determines the complete Li sequence and the semi-infinite even array.*

*Proof.* Equation (5.2) is telescoping. Substituting it into the definition of $o_{mn}$ gives (5.3). Since

$$e_{1k} = \lambda_{k+1} - \lambda_{k-1}, \tag{5.6}$$

the right side of (5.4) telescopes from $\lambda_{|m-n|}$ to $\lambda_{m+n}$. Equation (5.5) is the same telescoping identity written in the unit-step differences $o_{1j}$. □

*Remark* 5.2 (Finite truncations versus infinite first rows). Proposition 5.1 concerns the first row of the *semi-infinite* arrays. The leading block $\mathsf{O}_N$ is not determined by its own $N$ displayed first-row entries: reconstructing all of $\mathsf{O}_N$ from (5.3) requires the prefix $o_{1j}$ through $j = 2N-1$. Likewise, reconstructing $\mathsf{E}_N$ from its infinite first row requires $e_{1j}$ through $j = 2N-1$. Reconstructing $\mathsf{E}_N$ from the odd first row by (5.5) requires $o_{1j}$ through $j = 2N$.

**Proposition 5.3** (Injectivity of the odd array and one-dimensional kernel of the even array)**.** *On real sequences $(\lambda_n)_{n\geq 0}$ with $\lambda_0 = 0$, the map*

$$(\lambda_n)_{n\geq 0} \longmapsto (o_{mn})_{m,n\geq 1} \tag{5.7}$$

*is injective. The kernel of*

$$(\lambda_n)_{n\geq 0} \longmapsto (e_{mn})_{m,n\geq 1} \tag{5.8}$$

*consists exactly of the one-parameter family*

$$\lambda_{2k} = 0, \qquad \lambda_{2k+1} = c \qquad (k \geq 0), \tag{5.9}$$

*with $c \in \mathbb{R}$. Consequently $(\mathsf{E}, \lambda_1)$ determines the whole Li sequence and hence $\mathsf{O}$.*

*Proof.* If the odd array vanishes, then its first row gives $\lambda_j - \lambda_{j-1} = 0$ for every $j$, and $\lambda_0 = 0$ gives $\lambda_j = 0$. Thus the odd map is injective.

If the even array vanishes, its diagonal gives $\lambda_{2n} = 0$ for all $n \geq 1$. Its entries $e_{1,2k} = 0$ give $\lambda_{2k+1} = \lambda_{2k-1}$, so every odd term equals $\lambda_1$. Conversely, a sequence of the form (5.9) gives $e_{mn} = 0$ because $m+n$ and $m-n$ have the same parity. This proves the kernel statement. □

**Proposition 5.4** (Reflection boundary conditions)**.** *Using the even extension $\lambda_{-n} = \lambda_n$, the bilateral kernels satisfy*

$$e(-m, n) = -e(m, n), \tag{5.10}$$

$$o(1-m, n) = -o(m, n). \tag{5.11}$$

*Hence*

$$e(0, n) = 0, \qquad o(0, n) = -o(1, n). \tag{5.12}$$

*Together with symmetry in $m, n$ and the discrete wave equation of Section 8, these are the natural reflection boundary conditions at 0 and 1/2.*

*Proof.* The identities follow by direct substitution in $e(m, n) = \lambda_{m+n} - \lambda_{m-n}$ and $o(m, n) = \lambda_{m+n-1} - \lambda_{m-n}$, using $\lambda_{-k} = \lambda_k$. □

## 6. Exact Weil realization and critical-line Gram representations

**Logical status.** The reflected Weil identities in this section are unconditional. The sine Gram representations and the resulting positive-definiteness statements begin only after RH is assumed explicitly.

Lagarias defines on the Li test-function class a Hermitian Weil scalar product for which the functions (1.1) satisfy (1.2); see [23, Section 3, Theorem 3.1]. This construction belongs to the explicit-formula tradition initiated by Weil [44]; Bombieri's analysis of Weil's quadratic functional provides a further reference for the positivity framework [4]. In particular, for the Riemann zeta-function,

$$\langle G_r, G_s\rangle_W = \lambda_r + \lambda_s - \lambda_{|r-s|}, \qquad r, s \in \mathbb{Z}, \tag{6.1}$$

where (3.5) has been used.

**Theorem 6.1** (Exact reflected Weil Gram matrices)**.** *For every $m, n \geq 1$, without assuming* RH*,*

$$e_{mn} = \langle F_m^E, F_n^E\rangle_W, \tag{6.2}$$
$$o_{mn} = \langle F_m^O, F_n^O\rangle_W, \tag{6.3}$$

*where $F_n^E, F_n^O$ are given by* (1.3)*.*

*Proof.* Using (6.1) and bilinearity/polarization,

$$\langle F_m^E, F_n^E\rangle_W = \frac{1}{2}\Big[K(m,n) - K(m,-n) - K(-m,n) + K(-m,-n)\Big] \tag{6.4}$$
$$= \lambda_{m+n} - \lambda_{|m-n|} = e_{mn}, \tag{6.5}$$

where $K(r,s) := \lambda_r + \lambda_s - \lambda_{|r-s|}$. Similarly,

$$\langle F_m^O, F_n^O\rangle_W = \frac{1}{2}\Big[K(m,n) - K(m,1-n) - K(1-m,n) + K(1-m,1-n)\Big] \tag{6.6}$$
$$= \lambda_{m+n-1} - \lambda_{|m-n|} = o_{mn}. \tag{6.7}$$

□

The same reflection is transparent in the Cayley coordinate

$$q(s) := 1 - \frac{1}{s}. \tag{6.8}$$

Indeed,

$$F_n^E(s) = \frac{q(s)^{-n} - q(s)^n}{\sqrt{2}}, \tag{6.9}$$

$$F_n^O(s) = \frac{q(s)^{1-n} - q(s)^n}{\sqrt{2}}. \tag{6.10}$$

**Theorem 6.2** (Integer and half-integer sine Gram representations)**.** *Assume* RH. *Enumerate the positive ordinates of the nontrivial zeros with multiplicity. Then for every $m, n \geq 1$,*

$$e_{mn} = 4 \sum_{\gamma>0} \sin(m\theta_\gamma) \sin(n\theta_\gamma), \tag{6.11}$$

$$o_{mn} = 4 \sum_{\gamma>0} \sin\Big((m - \tfrac{1}{2})\theta_\gamma\Big) \sin\Big((n - \tfrac{1}{2})\theta_\gamma\Big). \tag{6.12}$$

*The two series converge absolutely for fixed $m, n$.*

*Proof.* Under RH, $q(\rho) = \mathrm{e}^{i\theta_\gamma}$ for $\rho = 1/2 + i\gamma$. Equations (6.9)–(6.10) become

$$F_n^E(\rho) = -\sqrt{2}\, i \sin(n\theta_\gamma), \tag{6.13}$$

$$F_n^O(\rho) = -\sqrt{2}\, i \mathrm{e}^{i\theta_\gamma/2} \sin\Big((n - \tfrac{1}{2})\theta_\gamma\Big). \tag{6.14}$$

For the numerical normalization, it is safest to use the already established Li cosine formula (3.12). Subtracting the terms with indices $m+n$ and $|m-n|$ and applying

$$\cos((m-n)\theta) - \cos((m+n)\theta) = 2\sin(m\theta)\sin(n\theta) \tag{6.15}$$

gives (6.11). Likewise,

$$\cos((m-n)\theta) - \cos((m+n-1)\theta) = 2\sin\Big((m - \tfrac{1}{2})\theta\Big)\sin\Big((n - \tfrac{1}{2})\theta\Big) \tag{6.16}$$

gives (6.12). Since $\theta_\gamma = O(\gamma^{-1})$, every summand is $O_{m,n}(\gamma^{-2})$, and the zero-counting estimate gives absolute convergence. □

*Remark* 6.3 (Negative definite sequences and screw functions). Representation (3.12) says that, under RH, the extended sequence $\lambda$ is a *negative definite* function on $\mathbb{Z}$ in the sense of harmonic analysis on groups: it is real, even, vanishes at 0, and has the Lévy–Khinchine form $\int(1-\cos n\theta)\,\mathrm{d}\sigma(\theta)$ with $\sigma$ the atomic measure carried by the angles $\theta_\gamma$. For any such function the kernel $\lambda_r+\lambda_s-\lambda_{r-s}$ is positive definite, and the passage from that kernel to the sine Gram matrices of Theorem 6.2 is the standard consequence of the Lévy–Khinchine representation [34, 3]. Conversely, a negative definite $\lambda$ with $\lambda_0=0$ satisfies $\lambda_n\geq 0$ for all $n$, so Lagarias's kernel identity (6.1) amounts to the statement that RH is equivalent to negative definiteness of the Li sequence. The continuous counterpart of this circle of ideas is Krein's theory of screw functions, developed for the Riemann zeta-function by Suzuki [38, 39, 40]. The positivity mechanism exploited below is therefore classical; what is specific here is the pair of reflection compressions of Definition 4.1 and the structure theory derived from it.

**Theorem 6.4** (Li–Weil reflection criterion)**.** *The following are equivalent:*

*(i)* RH*;*

*(ii)* $\mathsf{E}_N\succeq 0$ *and* $\mathsf{O}_N\succeq 0$ *for every* $N\geq 1$*;*

*(iii)* $\mathsf{E}_N\succ 0$ *and* $\mathsf{O}_N\succ 0$ *for every* $N\geq 1$*;*

*(iv)* $\det\mathsf{E}_N>0$ *and* $\det\mathsf{O}_N>0$ *for every* $N\geq 1$.

*Proof.* Assume RH. Theorem 6.2 gives positive semidefiniteness. If $c^{T}\mathsf{E}_N c=0$, then

$$\sum_{m=1}^{N} c_m\sin(m\theta_\gamma)=0 \qquad \text{for every distinct positive zero ordinate } \gamma. \tag{6.17}$$

There are infinitely many distinct ordinates and $\theta_\gamma\to 0$. The finite trigonometric polynomial in (6.17) is analytic in $\theta$ and therefore must vanish identically; hence $c=0$. The half-integer system is handled in the same way. Thus (i) implies (iii), and (iii) implies (ii).

Assume (ii). For $N\geq n$ the diagonal entries satisfy

$$\lambda_{2n}=(\mathsf{E}_N)_{nn}\geq 0, \qquad \lambda_{2n-1}=(\mathsf{O}_N)_{nn}\geq 0. \tag{6.18}$$

Hence every Li coefficient is nonnegative and Theorem 3.2 gives RH. Finally, because the families are nested leading principal truncations, Sylvester's criterion gives the equivalence of (iii) and (iv). □

*Remark* 6.5 (High-precision warning for numerical positivity tests). The criterion in Theorem 6.4 is extremely ill-conditioned numerically. In the truncation to the first 1200 positive ordinates used in Remark 13.22, high-precision computation at $N = 12$ gives

$$\det \mathsf{E}_{12} \approx 4.3 \times 10^{-436}, \qquad \lambda_{\min}(\mathsf{E}_{12}) \approx 3.6 \times 10^{-87}.$$

Consequently, double precision—and even a computation formed from Li coefficients known to only 30 decimal digits—can produce spurious negative eigenvalues and nonpositive leading minors. Resolving the smallest eigenvalue against entries of order one requires at least about 90 reliable decimal digits, plus guard digits, while a direct determinant or principal-minor calculation must accommodate the roughly 436-decimal scale of $|\log_{10} \det \mathsf{E}_{12}|$. This rapid precision loss reflects the $N^2 \log N$ determinant scale proved below; low-precision indefiniteness is therefore not evidence against the exact positivity criterion.

**Lemma 6.6** (Bombieri–Lagarias admissibility of the quadratic transform)**.** *Let $\mathcal{R}$ be the multiset of nontrivial zeros of $\zeta$, counted with multiplicity, and define*

$$q_\rho := 1 - \frac{1}{\rho}, \qquad \sigma_\rho := \frac{1}{1 - q_\rho^2} = \frac{\rho^2}{2\rho - 1}. \tag{6.19}$$

*The transformed multiset $\mathcal{S} := \{\sigma_\rho : \rho \in \mathcal{R}\}$, with pushed-forward multiplicities, contains neither 0 nor 1, is invariant under $\sigma \mapsto \overline{\sigma}$ and $\sigma \mapsto 1 - \sigma$, and satisfies the summability and symmetric convergence hypotheses of the Bombieri–Lagarias multiset criterion. In particular,*

$$\sum_{\sigma \in \mathcal{S}} \frac{1 + |\Re\sigma|}{(1 + |\sigma|)^2} < \infty, \tag{6.20}$$

*and the symmetric sum $\sum^*_{\sigma \in \mathcal{S}} \sigma^{-1}$ exists.*

*Proof.* The denominator $2\rho - 1$ cannot vanish on the nontrivial zero set, since $\rho = \frac{1}{2}$ is not a nontrivial zero of $\zeta$. Moreover $\sigma_\rho = 0$ would force $\rho = 0$, and $\sigma_\rho = 1$ would force $(\rho - 1)^2 = 0$; neither value occurs in $\mathcal{R}$.

The exact identities

$$\frac{1}{\sigma_\rho} = \frac{2}{\rho} - \frac{1}{\rho^2}, \qquad \sigma_{\bar{\rho}} = \overline{\sigma_\rho}, \qquad \sigma_{1-\rho} = 1 - \sigma_\rho \tag{6.21}$$

give both symmetries. Also

$$\sigma_\rho = \frac{\rho}{2} + \frac{1}{4} + \frac{1}{4(2\rho-1)}. \tag{6.22}$$

Since every nontrivial zero satisfies $0 < \Re\rho < 1$, the last formula gives $|\sigma_\rho| \asymp |\rho|$ and $\Re\sigma_\rho = O(1)$ as $|\rho| \to \infty$. Therefore

$$\frac{1+|\Re\sigma_\rho|}{(1+|\sigma_\rho|)^2} \ll \frac{1}{1+|\rho|^2}.$$

The right-hand side is summable over the zeta zeros with multiplicity: by $N(T) = O(T\log T)$, partial summation gives $\sum_\rho (1+|\rho|^2)^{-1} < \infty$. This proves (6.20).

Finally, the first identity in (6.21) gives

$$\sum_\rho^* \frac{1}{\sigma_\rho} = 2\sum_\rho^* \frac{1}{\rho} - \sum_\rho \frac{1}{\rho^2}.$$

The second series is absolutely convergent, while the first symmetric sum is the standard one attached to the zero multiset of the completed zeta-function. Hence the transformed symmetric sum exists. These are precisely the convergence properties needed in Bombieri–Lagarias [6, Theorem 1 and its symmetric corollary]. □

**Theorem 6.7** (The even reflection family alone is an RH criterion)**.** *The following are equivalent:*

*(i)* RH*;*

*(ii)* $\mathsf{E}_N \succeq 0$ *for every* $N \geq 1$*;*

*(iii)* $\mathsf{E}_N \succ 0$ *for every* $N \geq 1$*.*

*In particular, the odd family is not required for the equivalence.*

*Proof.* The implication $(i) \Rightarrow (iii) \Rightarrow (ii)$ is contained in Theorem 6.4. Assume $\mathsf{E}_N \succeq 0$ for every $N$. Its diagonal gives

$$\lambda_{2N} = (\mathsf{E}_N)_{NN} \geq 0 \qquad (N \geq 1).$$

For the transform (6.19),

$$1 - \frac{1}{\sigma_\rho} = q_\rho^2 = \left(1 - \frac{1}{\rho}\right)^2, \tag{6.23}$$

and hence, with the same symmetric limiting convention,

$$\lambda_{2n} = \sum_\rho \left[1 - \left(1 - \frac{1}{\sigma_\rho}\right)^n\right]. \tag{6.24}$$

Lemma 6.6 verifies the multiset hypotheses. The symmetric Bombieri–Lagarias criterion therefore gives

$$\Re\sigma_\rho = \frac{1}{2} \qquad \text{for every nontrivial zero } \rho.$$

But

$$\begin{aligned} \Re\sigma_\rho = \frac{1}{2} &\iff \left|1 - \frac{1}{\sigma_\rho}\right| = 1 \iff \left|1 - \frac{1}{\rho}\right| = 1, \\ &\iff |\rho - 1| = |\rho| \iff \Re\rho = \frac{1}{2}. \end{aligned}$$

Thus every nontrivial zero lies on the critical line, proving RH. □

*Remark* 6.8 (Source of the even-only converse). The converse in Theorem 6.7 is an application of the general Bombieri–Lagarias multiset criterion after the elementary quadratic Cayley transform (6.19). The paper therefore claims no priority for the underlying positivity criterion. What is used later is only that the matrix diagonal exposes precisely these even Bombieri–Lagarias sums.

*Remark* 6.9 (Role of the odd family). The odd matrices remain useful because they supply the Christoffel/Darboux bridge, an independent pivot sequence, and a second moment system. None of those structures is logically necessary for the implication to RH: the even family is already a complete reflection criterion through $\mathsf{E}_N \succ 0$ for every $N$.

**Corollary 6.10** (Full rank)**.** *Under* RH*,*

$$\operatorname{rank} \mathsf{E}_N = \operatorname{rank} \mathsf{O}_N = N \qquad (N \geq 1). \tag{6.25}$$

*Proof.* Under RH, Theorem 6.4 gives $\mathsf{E}_N \succ 0$ and $\mathsf{O}_N \succ 0$. A positive definite $N \times N$ matrix is nonsingular, hence has rank $N$. □

**Corollary 6.11** (All principal minors)**.** *Under* RH*, every principal minor of* $\mathsf{E}_N$ *and* $\mathsf{O}_N$ *is strictly positive.*

*Proof.* A principal submatrix is the Gram matrix of a finite set of distinct integer, respectively half-integer, sine modes. The same analytic-zero argument as in Theorem 6.4 gives linear independence. □

## 7. Reflection geometry inside the Li test class

The exact matrix identities above can be phrased as reflections of Lagarias's bilateral Li-class Gram kernel. Define

$$\mathsf{K}(r,s) := \langle G_r, G_s \rangle_W = \lambda_r + \lambda_s - \lambda_{|r-s|}, \qquad r,s \in \mathbb{Z}. \tag{7.1}$$

Let

$$R_0(k) = -k, \qquad R_{1/2}(k) = 1 - k. \tag{7.2}$$

Then

$$R_{1/2}R_0(k) = k + 1, \tag{7.3}$$

so the two reflections generate the infinite dihedral action on the integer label set.

If $u_m = e_m - e_{-m}$ and $v_m = e_m - e_{1-m}$ in the finitely supported bilateral coordinate space, then

$$\frac{1}{2}\, u_m^* \mathsf{K} u_n = e_{mn}, \tag{7.4}$$

$$\frac{1}{2}\, v_m^* \mathsf{K} v_n = o_{mn}. \tag{7.5}$$

Thus $\mathsf{E}_N$ and $\mathsf{O}_N$ are exact congruence compressions of the known Lagarias Gram kernel. The point of the present coordinates is the resulting even–odd diagonal split, the Toeplitz–Hankel form, and the matrix structures developed below.

## 8. Discrete wave equations and displacement rank

The Toeplitz–Hankel structure has a useful equivalent formulation as a low-displacement-rank identity.

**Definition 8.1** (Universal tridiagonal matrices)**.** Let $\mathsf{J}_{E,\mathrm{ref}}^{(N)}$ be the $N \times N$ symmetric tridiagonal reflection operator with zero diagonal and $1/2$ on both adjacent diagonals:

$$\mathsf{J}_{E,\mathrm{ref}}^{(N)} = \frac{1}{2}\begin{pmatrix} 0 & 1 & & & \\ 1 & 0 & 1 & & \\ & 1 & 0 & \ddots & \\ & & \ddots & \ddots & 1 \\ & & & 1 & 0 \end{pmatrix}. \tag{8.1}$$

Let $\mathsf{J}_{O,\mathrm{ref}}^{(N)}$ be the corresponding odd reflection operator, with the same adjacent diagonals and upper-left entry $-1/2$:

$$\mathsf{J}_{O,\mathrm{ref}}^{(N)} = \begin{pmatrix} -\frac{1}{2} & \frac{1}{2} & & & \\ \frac{1}{2} & 0 & \frac{1}{2} & & \\ & \frac{1}{2} & 0 & \ddots & \\ & & \ddots & \ddots & \frac{1}{2} \\ & & & \frac{1}{2} & 0 \end{pmatrix}. \tag{8.2}$$

**Proposition 8.2** (Discrete wave relation)**.** *For $m, n \geq 2$, both arrays satisfy*

$$a_{m+1,n} + a_{m-1,n} = a_{m,n+1} + a_{m,n-1}, \tag{8.3}$$

*where $a = e$ or $a = o$. The boundary relations are exactly those encoded by $\mathsf{J}_{E,\mathrm{ref}}$ and $\mathsf{J}_{O,\mathrm{ref}}$.*

*Proof.* For $e_{mn} = \lambda_{m+n} - \lambda_{|m-n|}$, the four terms in (8.3) agree after matching the sum and difference indices. The same calculation applies to $o_{mn}$ because the sum index is shifted by one. The boundary $m = 1$ is described by the identities $xU_0 = U_1/2$ and $xP_0 = (P_1 - P_0)/2$ developed in Section 9. □

**Theorem 8.3** (Displacement rank at most two)**.** *Let*

$$r_E^{(N)} := \left(e_{1,N+1}, \ldots, e_{N-1,N+1}, 0\right)^T, \tag{8.4}$$

$$r_O^{(N)} := \left(o_{1,N+1}, \ldots, o_{N-1,N+1}, 0\right)^T, \tag{8.5}$$

*and let $e_N$ denote the last standard basis vector of $\mathbb{R}^N$. Then*

$$\mathsf{J}_{E,\mathrm{ref}}^{(N)}\mathsf{E}_N - \mathsf{E}_N\mathsf{J}_{E,\mathrm{ref}}^{(N)} = \frac{1}{2}\left(r_E^{(N)}e_N^T - e_N(r_E^{(N)})^T\right), \tag{8.6}$$

$$\mathsf{J}_{O,\mathrm{ref}}^{(N)}\mathsf{O}_N - \mathsf{O}_N\mathsf{J}_{O,\mathrm{ref}}^{(N)} = \frac{1}{2}\left(r_O^{(N)}e_N^T - e_N(r_O^{(N)})^T\right). \tag{8.7}$$

*In particular, each displacement has rank at most two.*

*Proof.* The semi-infinite arrays satisfy the commutator relation entrywise as a consequence of (8.3) and the boundary relation; no bounded-operator assertion is needed here. Finite truncation removes only the coupling from index $N$ to $N+1$. The omitted term is precisely the outer-product defect in (8.6) and (8.7). The final component of each boundary vector has been set to zero because it cancels identically in the skew outer product. □

**Corollary 8.4** (Displacement of the inverses)**.** *Whenever* $\mathsf{E}_N$ *and* $\mathsf{O}_N$ *are invertible,*

$$\mathsf{J}_{E,\mathrm{ref}}^{(N)}\mathsf{E}_N^{-1} - \mathsf{E}_N^{-1}\mathsf{J}_{E,\mathrm{ref}}^{(N)} = -\mathsf{E}_N^{-1}\left(\mathsf{J}_{E,\mathrm{ref}}^{(N)}\mathsf{E}_N - \mathsf{E}_N\mathsf{J}_{E,\mathrm{ref}}^{(N)}\right)\mathsf{E}_N^{-1}, \tag{8.8}$$

$$\mathsf{J}_{O,\mathrm{ref}}^{(N)}\mathsf{O}_N^{-1} - \mathsf{O}_N^{-1}\mathsf{J}_{O,\mathrm{ref}}^{(N)} = -\mathsf{O}_N^{-1}\left(\mathsf{J}_{O,\mathrm{ref}}^{(N)}\mathsf{O}_N - \mathsf{O}_N\mathsf{J}_{O,\mathrm{ref}}^{(N)}\right)\mathsf{O}_N^{-1}. \tag{8.9}$$

*Hence the inverse matrices also have displacement rank at most two with respect to the same tridiagonal operators. More precisely, if* $A$ *denotes either* $\mathsf{E}_N$ *or* $\mathsf{O}_N$ *and* $A$ *is invertible, then*

$$\operatorname{rank}\left(\mathsf{J}^{(N)}A^{-1} - A^{-1}\mathsf{J}^{(N)}\right) = \operatorname{rank}\left(\mathsf{J}^{(N)}A - A\mathsf{J}^{(N)}\right), \tag{8.10}$$

*with the appropriate choice of* $\mathsf{J}_{E,\mathrm{ref}}^{(N)}$ *or* $\mathsf{J}_{O,\mathrm{ref}}^{(N)}$*. Thus inversion preserves the exact displacement rank, whether it is* 0 *or* 2*.*

*Proof.* For an invertible matrix $A$,

$$\mathsf{J}A^{-1} - A^{-1}\mathsf{J} = -A^{-1}(\mathsf{J}A - A\mathsf{J})A^{-1},$$

by direct expansion. Since left and right multiplication by invertible matrices preserve rank, the rank equality follows as well. □

## 9. Chebyshev moment representations

Let $U_n$ denote the Chebyshev polynomial of the second kind, normalized by

$$U_n(\cos\theta) = \frac{\sin((n+1)\theta)}{\sin\theta}. \tag{9.1}$$

Define

$$P_n(x) := U_n(x) + U_{n-1}(x), \qquad U_{-1} := 0. \tag{9.2}$$

Then

$$P_{m-1}(\cos\theta) = \frac{\sin((m-\frac{1}{2})\theta)}{\sin(\theta/2)}. \tag{9.3}$$

Under RH, let $m_\gamma$ denote the multiplicity of the zero $1/2 + i\gamma$ and set

$$x_\gamma := \cos\theta_\gamma = \frac{4\gamma^2 - 1}{4\gamma^2 + 1}. \tag{9.4}$$

Define finite positive Borel measures on $[-1, 1]$ by

$$\mu_E := 4 \sum_{\gamma>0}^{\text{distinct}} m_\gamma \sin^2\theta_\gamma \, \delta_{x_\gamma}, \tag{9.5}$$

$$\mu_O := 4 \sum_{\gamma>0}^{\text{distinct}} m_\gamma \sin^2(\theta_\gamma/2) \, \delta_{x_\gamma}. \tag{9.6}$$

These are finite measures because their total masses are $\lambda_2$ and $\lambda_1$, respectively; equivalently the weights are $O(m_\gamma \gamma^{-2})$ and the standard zero-counting estimate gives convergence.

**Theorem 9.1** (Moment representation)**.** *Under* RH*,*

$$e_{mn} = \int_{-1}^{1} U_{m-1}(x) U_{n-1}(x) \, \mathrm{d}\mu_E(x), \tag{9.7}$$

$$o_{mn} = \int_{-1}^{1} P_{m-1}(x) P_{n-1}(x) \, \mathrm{d}\mu_O(x). \tag{9.8}$$

*Moreover,*

$$\mathrm{d}\mu_E(x) = 2(1+x) \, \mathrm{d}\mu_O(x). \tag{9.9}$$

*Proof.* We first verify that the integrals are well defined. For fixed $m, n$, the polynomials $U_{m-1}U_{n-1}$ and $P_{m-1}P_{n-1}$ are bounded on $[-1, 1]$, while $\mu_E$ and $\mu_O$ are finite positive measures. Thus both Gram integrals converge absolutely.

For the even family, using the atomic definition of $\mu_E$ and $U_{k-1}(\cos\theta) = \sin(k\theta)/\sin\theta$,

$$\int U_{m-1}(x) U_{n-1}(x) \, \mathrm{d}\mu_E(x) = 4 \sum_{\gamma>0}^{\text{distinct}} m_\gamma \sin^2\theta_\gamma U_{m-1}(x_\gamma) U_{n-1}(x_\gamma) \tag{9.10}$$

$$= 4 \sum_{\gamma>0}^{\text{distinct}} m_\gamma \sin(m\theta_\gamma) \sin(n\theta_\gamma). \tag{9.11}$$

Counting each ordinate according to multiplicity turns the last sum into the right side of (6.11), hence it equals $e_{mn}$. For the odd family, (9.3) gives

$$\int P_{m-1}(x)P_{n-1}(x)\,\mathrm{d}\mu_O(x) = 4\sum_{\gamma>0}^{\text{distinct}} m_\gamma \sin^2(\theta_\gamma/2)P_{m-1}(x_\gamma)P_{n-1}(x_\gamma) \tag{9.12}$$

$$= 4\sum_{\gamma>0}^{\text{distinct}} m_\gamma \sin((m-\tfrac{1}{2})\theta_\gamma)\sin((n-\tfrac{1}{2})\theta_\gamma), \tag{9.13}$$

which is (6.12).

It remains to compare the atom masses. At every support point $x_\gamma = \cos\theta_\gamma$,

$$\frac{4m_\gamma \sin^2\theta_\gamma}{4m_\gamma \sin^2(\theta_\gamma/2)} = \frac{\sin^2\theta_\gamma}{\sin^2(\theta_\gamma/2)} = 4\cos^2(\theta_\gamma/2) = 2(1+x_\gamma). \tag{9.14}$$

Thus the two atomic measures have the same support and their masses satisfy $\mu_E(\{x_\gamma\}) = 2(1+x_\gamma)\mu_O(\{x_\gamma\})$ at every atom, which is exactly (9.9). □

*Remark* 9.2 (Christoffel transform). Equation (9.9) says that the even measure is, up to the scalar factor 2, the Christoffel transform of the odd measure at the point $-1$. Here $-1$ is the left endpoint of the ambient interval $[-1,1]$ but lies outside the actual zeta support $\{x_\gamma\}\cup\{1\}$. Thus the two matrix families are linked at the level of measures, orthogonal polynomials, Jacobi matrices, determinants, and Cholesky pivots.

The weights can be written directly in terms of zero ordinates:

$$4\sin^2\theta_\gamma = \frac{64\gamma^2}{(4\gamma^2+1)^2}, \tag{9.15}$$

$$4\sin^2(\theta_\gamma/2) = \frac{4}{4\gamma^2+1}. \tag{9.16}$$

*Remark* 9.3 (Normalization ledger). The numerical factors used later are fixed by the following exact chain. The sine Gram representations (6.11)–(6.12) carry the common factor 4. That factor is absorbed into the masses of $\mu_E$ and $\mu_O$ in (9.5)–(9.6); no further scalar is introduced in the moment identities (9.7)–(9.8). The ratio of the two atomic masses is exactly $2(1+x)$,

as shown in (9.14). Both polynomial bases $U_n$ and $P_n$ have leading coefficient $2^n$, so changing from monomials to either basis multiplies the order-$N$ Gram determinant by

$$\left(\prod_{n=0}^{N-1} 2^n\right)^2 = 2^{N(N-1)}.$$

Consequently, the Vandermonde formulas carry precisely the product $4^N 2^{N(N-1)}$. These identities provide a direct factor-by-factor check of every later normalization involving the reflection measures and their determinants.

## 10. Changes of basis and hidden Hankel matrices

Write

$$X_N(x) := (1, x, \dots, x^{N-1})^T. \tag{10.1}$$

There exist explicit lower-triangular matrices $B_E^{(N)}$ and $B_O^{(N)}$ satisfying

$$\Big(U_0(x), \dots, U_{N-1}(x)\Big)^T = B_E^{(N)} X_N(x), \tag{10.2}$$

$$\Big(P_0(x), \dots, P_{N-1}(x)\Big)^T = B_O^{(N)} X_N(x). \tag{10.3}$$

The coefficients of $U_n$ are explicitly

$$U_n(x) = \sum_{j=0}^{\lfloor n/2 \rfloor} (-1)^j \binom{n-j}{j} (2x)^{n-2j}. \tag{10.4}$$

Thus both change-of-basis matrices have diagonal

$$1, 2, 2^2, \dots, 2^{N-1}, \tag{10.5}$$

and hence

$$\det B_E^{(N)} = \det B_O^{(N)} = 2^{N(N-1)/2}. \tag{10.6}$$

Define moments

$$\mu_k^E := \int_{-1}^{1} x^k \, \mathrm{d}\mu_E(x), \qquad \mu_k^O := \int_{-1}^{1} x^k \, \mathrm{d}\mu_O(x), \tag{10.7}$$

and Hankel matrices

$$H_N^E := \left(\mu_{i+j}^E\right)_{i,j=0}^{N-1}, \qquad H_N^O := \left(\mu_{i+j}^O\right)_{i,j=0}^{N-1}. \tag{10.8}$$

**Theorem 10.1** (Congruence to moment Hankel matrices)**.** *Under* RH*,*

$$\mathsf{E}_N = B_E^{(N)} H_N^E (B_E^{(N)})^T, \tag{10.9}$$

$$\mathsf{O}_N = B_O^{(N)} H_N^O (B_O^{(N)})^T. \tag{10.10}$$

*Consequently,*

$$\det \mathsf{E}_N = 2^{N(N-1)} \det H_N^E, \tag{10.11}$$

$$\det \mathsf{O}_N = 2^{N(N-1)} \det H_N^O. \tag{10.12}$$

*Proof.* Let $X_N(x) = (1, x, \dots, x^{N-1})^T$. For the even family put $\Phi_E(x) = (U_0(x), \dots, U_{N-1}(x))^T = B_E^{(N)} X_N(x)$. Then the moment representation gives

$$\mathsf{E}_N = \int \Phi_E(x) \Phi_E(x)^T \, \mathrm{d}\mu_E(x) \tag{10.13}$$

$$= B_E^{(N)} \left( \int X_N(x) X_N(x)^T \, \mathrm{d}\mu_E(x) \right) (B_E^{(N)})^T \tag{10.14}$$

$$= B_E^{(N)} H_N^E (B_E^{(N)})^T. \tag{10.15}$$

Every entry is a finite integral, so pulling the fixed finite matrices through the integral is purely algebraic. The same argument with $\Phi_O = (P_0, \dots, P_{N-1})^T = B_O^{(N)} X_N$ gives the odd identity.

Taking determinants and using that the basis-change matrices are triangular with diagonal $1, 2, \dots, 2^{N-1}$,

$$\det B_E^{(N)} = \det B_O^{(N)} = 2^{N(N-1)/2}. \tag{10.16}$$

Hence

$$\det \mathsf{E}_N = (\det B_E^{(N)})^2 \det H_N^E = 2^{N(N-1)} \det H_N^E, \tag{10.17}$$

and identically for $\mathsf{O}_N$. □

**Corollary 10.2** (Moment relation)**.** *For every $k \geq 0$,*

$$\mu_k^E = 2\big(\mu_k^O + \mu_{k+1}^O\big). \tag{10.18}$$

*Equivalently,*

$$H_N^E = 2\left(H_N^O + H_N^{O,(1)}\right), \qquad H_N^{O,(1)} := \left(\mu_{i+j+1}^O\right)_{i,j=0}^{N-1}. \tag{10.19}$$

*Proof.* The measure identity (9.9) gives, for every $k \geq 0$,

$$\mu_k^E = \int x^k \, \mathrm{d}\mu_E(x) = 2 \int x^k (1+x) \, \mathrm{d}\mu_O(x) = 2(\mu_k^O + \mu_{k+1}^O).$$

Applying this entrywise to the Hankel matrices gives (10.19). □

The first moments may be read directly from the original matrices:

$$\mu_0^E = \lambda_2, \qquad \mu_1^E = \frac{\lambda_3 - \lambda_1}{2}, \qquad \mu_2^E = \frac{\lambda_4}{4}, \tag{10.20}$$

$$\mu_0^O = \lambda_1, \qquad \mu_1^O = \frac{\lambda_2 - 2\lambda_1}{2}, \qquad \mu_2^O = \frac{\lambda_3 - 2\lambda_2 + 3\lambda_1}{4}. \tag{10.21}$$

## 11. Determinants and Vandermonde expansions

Enumerate the positive zero ordinates with multiplicity as $\gamma_1, \gamma_2, \ldots$, and write $\theta_j := \theta_{\gamma_j}$ and $x_j := x_{\gamma_j}$.

**Theorem 11.1** (Cauchy–Binet determinant formulas)**.** *Assume* RH. *Then*

$$\det \mathsf{E}_N = 4^N \sum_{1 \leq j_1 < \cdots < j_N} \Big( \det \big[ \sin(m\theta_{j_k}) \big]_{m,k=1}^{N} \Big)^2, \tag{11.1}$$

*and*

$$\det \mathsf{O}_N = 4^N \sum_{1 \leq j_1 < \cdots < j_N} \Big( \det \big[ \sin((m - \tfrac{1}{2})\theta_{j_k}) \big]_{m,k=1}^{N} \Big)^2. \tag{11.2}$$

*Proof.* We give the even proof; the odd one is identical after replacing the sine frequencies. For $M \geq 1$, let

$$A_{N,M}^E := 2 \big[ \sin(m\theta_j) \big]_{1 \leq m \leq N,\ 1 \leq j \leq M}. \tag{11.3}$$

Then the truncated Gram matrix is

$$G_{N,M}^E = A_{N,M}^E (A_{N,M}^E)^T = 4 \sum_{j=1}^{M} s_j s_j^T, \tag{11.4}$$

where $s_j = (\sin\theta_j, \ldots, \sin N\theta_j)^T$. For each entry $(m,n)$, the series $4\sum_j \sin(m\theta_j)\sin(n\theta_j)$ converges absolutely by Theorem 6.2; hence $G_{N,M}^E \to \mathsf{E}_N$ entrywise. Since the dimension $N$ is fixed, entrywise

convergence is equivalent to convergence in any matrix norm, and the determinant, being a polynomial in the entries, satisfies

$$\det G^E_{N,M} \longrightarrow \det \mathsf{E}_N. \tag{11.5}$$

For each finite $M$, the ordinary Cauchy–Binet formula applied to the $N \times M$ matrix $A^E_{N,M}$ gives, when $M \geq N$,

$$\det G^E_{N,M} = \sum_{J \subset \{1,\dots,M\},\ |J|=N} \det(A^E_{N,M}[:, J])^2 \tag{11.6}$$

$$= 4^N \sum_{1 \leq j_1 < \cdots < j_N \leq M} \Big(\det[\sin(m\theta_{j_k})]^N_{m,k=1}\Big)^2. \tag{11.7}$$

Every summand is nonnegative. Therefore the finite subset sums form an increasing sequence in $M$ and have a limit in $[0, \infty]$. Because their values equal $\det G^E_{N,M}$ and the latter converge to the finite number $\det \mathsf{E}_N$, monotone convergence of these finite sums gives exactly (11.1). The same construction with columns $2(\sin(\theta_j/2), \dots, \sin((N - \frac{1}{2})\theta_j))^T$ gives (11.2). □

**Theorem 11.2** (Vandermonde form)**.** *Under* RH*,*

$$\det \mathsf{E}_N = 4^N 2^{N(N-1)} \sum_{1 \leq j_1 < \cdots < j_N} \left( \prod_{a=1}^{N} \sin^2 \theta_{j_a} \right) \left( \prod_{1 \leq a < b \leq N} (x_{j_b} - x_{j_a})^2 \right), \tag{11.8}$$

$$\det \mathsf{O}_N = 4^N 2^{N(N-1)} \sum_{1 \leq j_1 < \cdots < j_N} \left( \prod_{a=1}^{N} \sin^2(\theta_{j_a}/2) \right) \left( \prod_{1 \leq a < b \leq N} (x_{j_b} - x_{j_a})^2 \right). \tag{11.9}$$

*Proof.* For the even family,

$$\sin(m\theta) = \sin\theta \, U_{m-1}(\cos\theta). \tag{11.10}$$

The leading coefficient of $U_{m-1}$ is $2^{m-1}$, hence

$$\det[U_{m-1}(x_j)]^N_{m,j=1} = 2^{N(N-1)/2} \prod_{j<k} (x_k - x_j). \tag{11.11}$$

Squaring and inserting into (11.1) yields (11.8). The proof for $\mathsf{O}_N$ uses (9.3); $P_{m-1}$ has the same leading coefficient $2^{m-1}$. □

In terms of the zero enumeration with multiplicity,

$$\det \mathsf{E}_N = 4^N 2^{N(N-1)} \sum_{1\le j_1<\cdots<j_N} \prod_{a=1}^{N} \frac{16\gamma_{j_a}^2}{(4\gamma_{j_a}^2+1)^2} \prod_{a<b}(x_{j_b}-x_{j_a})^2, \tag{11.12}$$

$$\det \mathsf{O}_N = 4^N 2^{N(N-1)} \sum_{1\le j_1<\cdots<j_N} \prod_{a=1}^{N} \frac{1}{4\gamma_{j_a}^2+1} \prod_{a<b}(x_{j_b}-x_{j_a})^2. \tag{11.13}$$

If two selected indices correspond to the same multiple zero, the associated Vandermonde factor vanishes; selecting one copy produces the correct multiplicity factor.

**Corollary 11.3** (Arbitrary principal minors)**.** *Let* $I = \{n_1 < \cdots < n_r\} \subset \{1, \dots, N\}$. *Under* RH*,*

$$\det \mathsf{E}_N[I,I] = 4^r \sum_{1\le j_1<\cdots<j_r} \Big(\det\big[\sin(n_a\theta_{j_b})\big]_{a,b=1}^{r}\Big)^2, \tag{11.14}$$

*and*

$$\det \mathsf{O}_N[I,I] = 4^r \sum_{1\le j_1<\cdots<j_r} \Big(\det\big[\sin((n_a-\tfrac12)\theta_{j_b})\big]_{a,b=1}^{r}\Big)^2. \tag{11.15}$$

*Proof.* We prove the even identity. Let $A_E$ be the full zero-column operator and let $A_E[I,:]$ denote the $r \times \infty$ operator obtained by retaining the rows indexed by $I = \{n_1, \dots, n_r\}$. Taking the corresponding principal block of the Gram identity $\mathsf{E}_N = A_E A_E^T$ gives

$$\mathsf{E}_N[I,I] = A_E[I,:]A_E[I,:]^T. \tag{11.16}$$

For a finite truncation to the first $M$ zero indices, Cauchy–Binet gives

$$\det \mathsf{E}_{N,M}[I,I] = 4^r \sum_{1\le j_1<\cdots<j_r\le M} \Big(\det[\sin(n_a\theta_{j_b})]_{a,b=1}^{r}\Big)^2. \tag{11.17}$$

The entries of the truncated principal block converge absolutely to those of $\mathsf{E}_N[I,I]$ by Theorem 6.2, so its determinant converges. The right-hand sums increase monotonically because all terms are nonnegative. Passing to the limit $M \to \infty$ proves (11.14). The odd formula follows identically from the half-integer zero columns. □

## 12. Canonical Cauchy–Binet ensemble and zero influence

The determinant formulas above are sums over all $N$-element zero subsets. It is therefore important to distinguish the full determinant from any one Vandermonde term. In this section the complete Cauchy–Binet sum itself is normalized and used to define the canonical zero-selection law attached to each reflection matrix.

For a positive ordinate $\gamma$, counted with multiplicity, define the column vectors

$$a^E_{N,\gamma} := 2\Big(\sin\theta_\gamma, \sin(2\theta_\gamma), \dots, \sin(N\theta_\gamma)\Big)^T, \tag{12.1}$$

$$a^O_{N,\gamma} := 2\Big(\sin(\tfrac{1}{2}\theta_\gamma), \sin(\tfrac{3}{2}\theta_\gamma), \dots, \sin((N-\tfrac{1}{2})\theta_\gamma)\Big)^T. \tag{12.2}$$

Let $A_E$ and $A_O$ denote the $N \times \infty$ matrices having these columns. Under RH the sine Gram formulas are precisely

$$\mathsf{E}_N = A_E A_E^T, \qquad \mathsf{O}_N = A_O A_O^T. \tag{12.3}$$

The column operators are Hilbert–Schmidt because the traces of the two Gram matrices are finite.

**Theorem 12.1** (Canonical projection ensemble)**.** *Assume* RH *and fix $N$. Let $M_N$ denote either $\mathsf{E}_N$ or $\mathsf{O}_N$, and let $A$ be the corresponding column operator in* (12.3)*. For each $N$-element subset $J$ of the positive zero indices define*

$$\mathbb{P}_N(J) := \frac{\det(A_J)^2}{\det M_N}. \tag{12.4}$$

*Then $\mathbb{P}_N$ is a probability measure. Moreover*

$$K_N := A^T M_N^{-1} A \tag{12.5}$$

*is an orthogonal projection of rank $N$ on the zero-index space, and*

$$\mathbb{P}_N(J) = \det K_N[J,J], \qquad |J| = N. \tag{12.6}$$

*Thus the normalized Cauchy–Binet expansion is a fixed-size projection determinantal process on the positive zero indices, in the standard finite-rank projection-kernel sense; see [20] for general background on determinantal point processes.*

*Proof.* For fixed $N$, regard $A$ as an operator from the zero-index Hilbert space $\ell^2$ to $\mathbb{R}^N$. Its columns are $a_{N,\gamma}$. Since

$$\sum_{\gamma>0} \|a_{N,\gamma}\|_2^2 = \operatorname{tr}\left(\sum_{\gamma>0} a_{N,\gamma} a_{N,\gamma}^T\right) = \operatorname{tr} M_N < \infty, \tag{12.7}$$

$A$ is Hilbert–Schmidt, hence bounded, and the series $AA^T = \sum a_\gamma a_\gamma^T$ converges in trace norm (in finite dimension, equivalently entrywise and in every matrix norm). Under RH, Theorem 6.4 gives $M_N \succ 0$, so $M_N^{-1/2}$ exists.

The infinite Cauchy–Binet identity is obtained by truncation. Let $A^{(M)}$ contain the first $M$ columns. Then $A^{(M)}(A^{(M)})^T \to M_N$ in norm. For $M \geq N$, finite Cauchy–Binet gives

$$\det\Big(A^{(M)}(A^{(M)})^T\Big) = \sum_{J\subset\{1,\dots,M\},\ |J|=N} \det(A_J)^2. \tag{12.8}$$

The left side tends to $\det M_N$ by continuity of the determinant. The right side is monotone increasing in $M$ because every summand is nonnegative; therefore

$$\det M_N = \sum_{|J|=N} \det(A_J)^2. \tag{12.9}$$

Since $\det M_N > 0$, (12.4) is nonnegative and has total mass one.

Now define $B := M_N^{-1/2} A$. As an operator $\ell^2 \to \mathbb{R}^N$ it satisfies

$$BB^* = M_N^{-1/2} AA^T M_N^{-1/2} = I_{\mathbb{R}^N}. \tag{12.10}$$

Thus $B$ is a coisometry. Its adjoint product $K_N := B^*B = A^T M_N^{-1} A$ is bounded and self-adjoint, and

$$K_N^2 = B^*(BB^*)B = B^*B = K_N. \tag{12.11}$$

Hence $K_N$ is the orthogonal projection onto $\operatorname{Ran} B^*$; because $BB^* = I_N$, this range has dimension $N$.

For a finite $N$-element index set $J$, the principal block is $K_N[J,J] = B_J^T B_J$. Since $B_J = M_N^{-1/2} A_J$ is square,

$$\det K_N[J,J] = \det(B_J^T B_J) = \det(B_J)^2 \tag{12.12}$$

$$= \det(M_N^{-1/2})^2 \det(A_J)^2 = \frac{\det(A_J)^2}{\det M_N}, \tag{12.13}$$

which proves (12.6). □

**Definition 12.2** (Zero leverage)**.** For either reflection family define

$$\ell_{N,\gamma} := (a_{N,\gamma})^T M_N^{-1} a_{N,\gamma}. \tag{12.14}$$

If a zero has multiplicity greater than one, each copy is regarded as a separate zero index with the same column vector.

**Corollary 12.3** (Exact one- and two-zero marginals)**.** *Under* RH*,*

$$\mathbb{P}_N(\gamma \in J) = \ell_{N,\gamma}, \tag{12.15}$$

$$\mathbb{P}_N(\gamma, \eta \in J) = \ell_{N,\gamma}\ell_{N,\eta} - K_N(\gamma,\eta)^2, \qquad \gamma \neq \eta, \tag{12.16}$$

$$\sum_{\gamma>0} \ell_{N,\gamma} = N. \tag{12.17}$$

*In particular, the full determinant assigns an intrinsic, normalized influence profile to the zero ordinates.*

*Proof.* We derive the marginals directly. Fix one index $\gamma$ and remove its column from $A$. The Cauchy–Binet normalization for the reduced column set gives

$$\mathbb{P}_N(\gamma \notin J) = \frac{\det(M_N - a_\gamma a_\gamma^T)}{\det M_N}. \tag{12.18}$$

Because $M_N$ is invertible,

$$\det(M_N - a_\gamma a_\gamma^T) = \det M_N \, \det(I - M_N^{-1} a_\gamma a_\gamma^T) \tag{12.19}$$

$$= \det M_N \, (1 - a_\gamma^T M_N^{-1} a_\gamma), \tag{12.20}$$

where the last equality follows from $\det(I - uv^T) = 1 - v^T u$ (choose a basis whose first vector is $u$ when $u \neq 0$, or use the characteristic polynomial of the rank-one operator). Hence

$$\mathbb{P}_N(\gamma \in J) = \ell_{N,\gamma}. \tag{12.21}$$

For two distinct indices $\gamma, \eta$, put $A_S = (a_\gamma, a_\eta)$ and $K_S = A_S^T M_N^{-1} A_S$. The finite-rank determinant identity

$$\det(M_N - A_S A_S^T) = \det M_N \, \det(I_2 - K_S) \tag{12.22}$$

follows by factoring out $M_N$ and using $\det(I - UV) = \det(I - VU)$ for the finite matrices $U = M_N^{-1} A_S$ and $V = A_S^T$; the latter identity follows by taking the determinant of the two Schur complements of $\left(\begin{smallmatrix} I & U \\ V & I \end{smallmatrix}\right)$. Thus

$$\mathbb{P}_N(\gamma, \eta \notin J) = (1 - \ell_\gamma)(1 - \ell_\eta) - K_N(\gamma,\eta)^2. \tag{12.23}$$

Inclusion–exclusion, together with the one-point formula, yields

$$\mathbb{P}_N(\gamma, \eta \in J) = 1 - \mathbb{P}_N(\gamma \notin J) - \mathbb{P}_N(\eta \notin J) + \mathbb{P}_N(\gamma, \eta \notin J) \tag{12.24}$$

$$= \ell_\gamma \ell_\eta - K_N(\gamma, \eta)^2. \tag{12.25}$$

Finally, absolute convergence of the leverage sum follows from positivity and the trace computation

$$\sum_{\gamma>0} \ell_{N,\gamma} = \sum_{\gamma>0} \operatorname{tr}(M_N^{-1} a_\gamma a_\gamma^T) \tag{12.26}$$

$$= \operatorname{tr}\left( M_N^{-1} \sum_{\gamma>0} a_\gamma a_\gamma^T \right) = \operatorname{tr} I_N = N, \tag{12.27}$$

where interchange of sum and trace is valid because the summands are nonnegative and the ambient matrix space is finite dimensional. □

The leverage has an equivalent deterministic interpretation that does not require probabilistic terminology.

**Theorem 12.4** (Deletion law)**.** *Let $M_N^{(-\gamma)}$ be the Gram matrix obtained after removing one copy of the positive zero ordinate $\gamma$ from the zero-side representation. Then*

$$\boxed{\frac{\det M_N^{(-\gamma)}}{\det M_N} = 1 - \ell_{N,\gamma}.} \tag{12.28}$$

*More generally, if $S$ is a finite set of zero indices and $K_N[S,S]$ is the corresponding principal block of* (12.5)*, then*

$$\frac{\det M_N^{(-S)}}{\det M_N} = \det\Big(I - K_N[S,S]\Big). \tag{12.29}$$

*Proof.* Under RH, $M_N \succ 0$ and is therefore invertible. Removing one indexed copy of the zero column $a = a_{N,\gamma}$ changes the Gram sum by exactly

$$M_N^{(-\gamma)} = M_N - aa^T. \tag{12.30}$$

Factoring out $M_N$ gives

$$\det(M_N - aa^T) = \det M_N \, \det(I - M_N^{-1} aa^T). \tag{12.31}$$

The rank-one matrix $M_N^{-1}aa^T$ has at most one nonzero eigenvalue, equal to its trace $a^T M_N^{-1}a = \ell_{N,\gamma}$; hence

$$\det(I - M_N^{-1}aa^T) = 1 - \ell_{N,\gamma}, \tag{12.32}$$

proving (12.28). Notice that $M_N^{(-\gamma)}$ is only positive semidefinite in general, so the ratio is allowed to be zero; no invertibility of the deleted matrix is assumed.

For a finite indexed set $S$ of size $r$, let $A_S$ be the $N \times r$ matrix formed from the corresponding columns. Then

$$M_N^{(-S)} = M_N - A_S A_S^T \tag{12.33}$$

and therefore

$$\frac{\det M_N^{(-S)}}{\det M_N} = \det(I_N - M_N^{-1}A_S A_S^T) \tag{12.34}$$

$$= \det(I_r - A_S^T M_N^{-1} A_S). \tag{12.35}$$

The second equality is Sylvester's finite-dimensional identity $\det(I_N + UV) = \det(I_r + VU)$, which follows by evaluating the determinant of $\left(\begin{smallmatrix} I_N & U \\ -V & I_r \end{smallmatrix}\right)$ using either diagonal block as the Schur pivot. By definition the $r \times r$ matrix $A_S^T M_N^{-1} A_S$ is exactly $K_N[S,S]$, proving (12.29). □

*12.1. Transport of a single zero*

The same framework gives an exact answer to how one moved zero is represented in the Li sequence and how it changes the matrix determinant. This is a sensitivity identity for the zero-side Gram representation; the perturbed zero configuration need not itself arise from an arithmetic $L$-function.

Let one conjugate pair $1/2 \pm i\gamma$ be moved to $1/2 \pm i\gamma'$, leaving all other zeros fixed, and write

$$\theta = 2\arctan\frac{1}{2\gamma}, \qquad \theta' = 2\arctan\frac{1}{2\gamma'}. \tag{12.36}$$

**Proposition 12.5** (Single-zero transport through the Li coefficients)**.** *The contribution of the moved conjugate pair to the change of the Li coefficient is*

$$\boxed{\delta\lambda_n = 2\Big(\cos(n\theta) - \cos(n\theta')\Big).} \tag{12.37}$$

*Consequently the reflection matrices change by*

$$\delta \mathsf{E}_N = a^E_{N,\gamma'}(a^E_{N,\gamma'})^T - a^E_{N,\gamma}(a^E_{N,\gamma})^T, \tag{12.38}$$

$$\delta \mathsf{O}_N = a^O_{N,\gamma'}(a^O_{N,\gamma'})^T - a^O_{N,\gamma}(a^O_{N,\gamma})^T. \tag{12.39}$$

*Thus moving one conjugate zero pair is an update of rank at most two in either reflection family.*

*Proof.* Under RH, the pair $1/2 \pm i\gamma$ contributes

$$2\Big(1 - \cos(n\theta)\Big) \tag{12.40}$$

to $\lambda_n$. Subtraction before and after the move gives (12.37). Using

$$\cos((m+n)\theta) - \cos((m-n)\theta) = -2\sin(m\theta)\sin(n\theta) \tag{12.41}$$

in the definitions of $\mathsf{E}_N$ and $\mathsf{O}_N$ yields (12.38)–(12.39). □

**Theorem 12.6** (Exact determinant response to a moved zero)**.** *Let $M_N$ denote either reflection matrix, let $a = a_{N,\gamma}$ and $a' = a_{N,\gamma'}$ be the corresponding old and new zero columns, and define with respect to the* unperturbed *inverse*

$$\ell = a^T M_N^{-1} a, \qquad \ell' = (a')^T M_N^{-1} a', \qquad \kappa = (a')^T M_N^{-1} a. \tag{12.42}$$

*Then*

$$\boxed{\frac{\det M'_N}{\det M_N} = (1-\ell)(1+\ell') + \kappa^2.} \tag{12.43}$$

*Proof.* Under RH, the unperturbed matrix $M_N$ is positive definite and hence invertible. Proposition 12.5 gives the exact finite-rank update

$$M'_N = M_N + a'(a')^T - aa^T = M_N + UCU^T, \tag{12.44}$$

where $U = (a', a)$ is $N \times 2$ and $C = \mathrm{diag}(1, -1)$. Factoring out $M_N$,

$$\frac{\det M'_N}{\det M_N} = \det(I_N + M_N^{-1} UCU^T). \tag{12.45}$$

Apply Sylvester's identity with $X = M_N^{-1}U$ and $Y = CU^T$:

$$\det(I_N + XY) = \det(I_2 + YX) = \det(I_2 + CU^T M_N^{-1} U). \tag{12.46}$$

No invertibility of $M'_N$ is required for this determinant identity; if the moved configuration makes $M'_N$ singular, both sides simply vanish.

By the definitions of $\ell, \ell', \kappa$ and symmetry of $M_N^{-1}$,

$$U^T M_N^{-1} U = \begin{pmatrix} \ell' & \kappa \\ \kappa & \ell \end{pmatrix}. \tag{12.47}$$

Hence

$$I_2 + C U^T M_N^{-1} U = \begin{pmatrix} 1+\ell' & \kappa \\ -\kappa & 1-\ell \end{pmatrix}, \tag{12.48}$$

and direct evaluation gives

$$\frac{\det M'_N}{\det M_N} = (1-\ell)(1+\ell') + \kappa^2. \tag{12.49}$$

□

**Corollary 12.7** (Infinitesimal zero sensitivity)**.** *For the even family let*

$$\begin{aligned} s_E(\theta) &:= (\sin\theta, \sin 2\theta, \dots, \sin N\theta)^T, \\ \dot{s}_E(\theta) &:= (\cos\theta, 2\cos 2\theta, \dots, N\cos N\theta)^T. \end{aligned} \tag{12.50}$$

*Then a motion of the conjugate pair at ordinate $\gamma$ satisfies*

$$\boxed{\frac{\partial}{\partial\gamma} \log\det \mathsf{E}_N = -\frac{32}{4\gamma^2+1}\, \dot{s}_E(\theta_\gamma)^T \mathsf{E}_N^{-1} s_E(\theta_\gamma).} \tag{12.51}$$

*For the odd family the identical formula holds with*

$$s_O(\theta) := \Big(\sin(\tfrac{1}{2}\theta), \sin(\tfrac{3}{2}\theta), \dots, \sin((N-\tfrac{1}{2})\theta)\Big)^T, \tag{12.52}$$

$$\dot{s}_O(\theta) := \Big(\tfrac{1}{2}\cos(\tfrac{1}{2}\theta), \tfrac{3}{2}\cos(\tfrac{3}{2}\theta), \dots, (N-\tfrac{1}{2})\cos((N-\tfrac{1}{2})\theta)\Big)^T. \tag{12.53}$$

*Proof.* Fix $N$ and one positive ordinate $\gamma_0 > 0$ in the unperturbed zero configuration. The other zero columns are held fixed. For $t > 0$ define

$$\theta(t) := 2\arctan\frac{1}{2t}, \qquad a(t) := 2s(\theta(t)), \tag{12.54}$$

with $s = s_E$ or $s = s_O$ according to the reflection family, and define the one-parameter matrix

$$\widetilde{M}_N(t) := M_N - a(\gamma_0)a(\gamma_0)^T + a(t)a(t)^T. \tag{12.55}$$

This definition avoids any differentiation of the infinite zero sum: only one finite-rank term depends on $t$. Since $\theta$ is $C^\infty$ on $(0,\infty)$ and $s$ is a finite vector of trigonometric functions, $t \mapsto \widetilde{M}_N(t)$ is $C^\infty$ as a map into the real symmetric $N \times N$ matrices.

At $t = \gamma_0$ one has $\widetilde{M}_N(\gamma_0) = M_N \succ 0$ under RH. The smallest eigenvalue is a continuous function of the matrix entries; hence there exists $\varepsilon > 0$, with $(\gamma_0 - \varepsilon, \gamma_0 + \varepsilon) \subset (0,\infty)$, such that $\widetilde{M}_N(t) \succ 0$ throughout this interval. Consequently $\det \widetilde{M}_N(t) > 0$ there and $\log\det \widetilde{M}_N(t)$ is well defined and differentiable.

For any differentiable curve of invertible matrices $A(t)$, Jacobi's formula follows from $\det(A+H) = \det A\, \det(I + A^{-1}H)$ and the first-order expansion $\det(I + \varepsilon B) = 1 + \varepsilon \operatorname{tr} B + o(\varepsilon)$; hence

$$\frac{\mathrm{d}}{\mathrm{d}t} \log\det A(t) = \operatorname{tr}\Big(A(t)^{-1} A'(t)\Big). \tag{12.56}$$

Applying this at $t = \gamma_0$ to (12.55),

$$\widetilde{M}_N'(\gamma_0) = a'(\gamma_0)a(\gamma_0)^T + a(\gamma_0)a'(\gamma_0)^T, \tag{12.57}$$

and therefore, using symmetry of $M_N^{-1}$ and cyclicity of the trace,

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t} \log\det \widetilde{M}_N(t)\bigg|_{t=\gamma_0} &= \operatorname{tr}\Big(M_N^{-1}[a'a^T + aa'^T]\Big) \\ &= 2(a')^T M_N^{-1} a, \end{aligned} \tag{12.58}$$

where $a = a(\gamma_0)$ and $a' = a'(\gamma_0)$. Finally,

$$\theta'(t) = 2\frac{1}{1 + (2t)^{-2}} \left(-\frac{1}{2t^2}\right) = -\frac{4}{4t^2+1}, \tag{12.59}$$

so $a'(t) = 2\theta'(t)\dot{s}(\theta(t))$. Substitution gives

$$\frac{\mathrm{d}}{\mathrm{d}t} \log\det \widetilde{M}_N(t)\bigg|_{t=\gamma_0} = -\frac{32}{4\gamma_0^2+1} \dot{s}(\theta_{\gamma_0})^T M_N^{-1} s(\theta_{\gamma_0}), \tag{12.60}$$

which is (12.51) and its odd analogue. The derivative is thus a local derivative on the open domain $t > 0$ on which the perturbed matrix remains invertible; no statement is made at the boundary $t = 0$. □

**Theorem 12.8** (High-zero leverage decay at fixed matrix order)**.** *Assume* RH *and fix* $N$*. Put*

$$v_E := (1, 2, \dots, N)^T, \qquad c_N^E := v_E^T \mathsf{E}_N^{-1} v_E, \tag{12.61}$$

$$v_O := (\tfrac{1}{2}, \tfrac{3}{2}, \dots, N - \tfrac{1}{2})^T, \qquad c_N^O := v_O^T \mathsf{O}_N^{-1} v_O. \tag{12.62}$$

*Then, as* $\gamma \to \infty$ *through positive zero ordinates,*

$$\ell_{N,\gamma}^E = \frac{4c_N^E}{\gamma^2} + O_N(\gamma^{-4}), \tag{12.63}$$

$$\ell_{N,\gamma}^O = \frac{4c_N^O}{\gamma^2} + O_N(\gamma^{-4}), \tag{12.64}$$

*and*

$$\frac{\partial}{\partial\gamma} \log\det \mathsf{E}_N = -\frac{8c_N^E}{\gamma^3} + O_N(\gamma^{-5}), \tag{12.65}$$

$$\frac{\partial}{\partial\gamma} \log\det \mathsf{O}_N = -\frac{8c_N^O}{\gamma^3} + O_N(\gamma^{-5}). \tag{12.66}$$

*Consequently, if the positive ordinates are enumerated with multiplicity,*

$$\ell_{N,\gamma_j}^{E,O} = O_N\left(\frac{(\log j)^2}{j^2}\right), \qquad \sum_{j>J} \ell_{N,\gamma_j}^{E,O} = O_N\left(\frac{(\log J)^2}{J}\right). \tag{12.67}$$

*Proof.* As $\gamma \to \infty$,

$$\theta_\gamma = 2\arctan\frac{1}{2\gamma} = \frac{1}{\gamma} + O(\gamma^{-3}). \tag{12.68}$$

For fixed $N$ this gives

$$s_E(\theta_\gamma) = \frac{v_E}{\gamma} + O_N(\gamma^{-3}), \qquad \dot{s}_E(\theta_\gamma) = v_E + O_N(\gamma^{-2}), \tag{12.69}$$

$$s_O(\theta_\gamma) = \frac{v_O}{\gamma} + O_N(\gamma^{-3}), \qquad \dot{s}_O(\theta_\gamma) = v_O + O_N(\gamma^{-2}). \tag{12.70}$$

Substitution into (12.14) gives (12.63)–(12.64); substitution into Corollary 12.7 gives (12.65)–(12.66). Finally the Riemann–von Mangoldt formula gives $\gamma_j \asymp j/\log j$, and summation of $(\log j)^2/j^2$ gives (12.67). □

*Remark* 12.9 (What the leverage theorem does and does not say). The decay in Theorem 12.8 is a statement about a *fixed* matrix order $N$. It proves that an individual sufficiently high zero has vanishing influence on that fixed determinant and that the normalized Cauchy–Binet mass has a summable high-zero tail. It does not provide a uniform estimate when $N$ and the zero height grow together, and therefore it does not replace the large-$N$ determinant analysis of Section 13.

## 13. Universal endpoint decay and the Li–Weil determinant law

This section contains two logically different statements. First comes an abstract theorem about positive discrete measures; it does not assume RH. Second comes its specialization to the Li–Cayley zero measures, where RH is assumed in order to obtain the positive moment representation. The abstract theorem does *not* say that the endpoint exponent $\beta$ alone determines the determinant. Its conclusion holds only after the node parameters are restricted to the polynomial scale encoded by hypotheses (ii)–(iii), with the selected weights subleading as in (iv). Once that scale is fixed, logarithmic refinements of the counting law do not change the leading coefficient.

*13.1. An endpoint-exponent theorem for discrete Gram determinants*

Let

$$\mu = \sum_{j\ge 1} w_j\, \delta_{x_j}, \qquad 0 < w_j, \qquad x_j = x(t_j) \uparrow 1, \tag{13.1}$$

be a finite positive measure, with distinct $t_1 < t_2 < \cdots \to \infty$. On the support parameters write $w(t_j) := w_j$. Write

$$D_N[\mu] := \det\left(\int x^{r+s}\, \mathrm{d}\mu(x)\right)_{r,s=0}^{N-1} \tag{13.2}$$

for its monomial Hankel determinant.

**Lemma 13.1** (Discrete Vandermonde entropy)**.** *For $N \ge 2$,*

$$\prod_{1\le a<b\le N} (b-a) = \prod_{k=1}^{N-1} k!, \tag{13.3}$$

*and consequently*

$$\sum_{1\le a<b\le N} \log(b-a) = \frac{1}{2}N^2 \log N - \frac{3}{4}N^2 + O(N\log N). \tag{13.4}$$

*Proof.* For a fixed $b$, the factors with $a < b$ are $(b-1)(b-2)\cdots 1 = (b-1)!$, which proves (13.3). Summing Stirling's formula

$$\log(k!) = k\log k - k + O(\log k)$$

for $1 \le k < N$, and using

$$\sum_{k<N} k\log k = \frac{1}{2}N^2\log N - \frac{1}{4}N^2 + O(N\log N), \qquad \sum_{k<N} k = \frac{1}{2}N^2 + O(N),$$

gives (13.4). □

**Theorem 13.2** (Endpoint-exponent law at polynomial node scale)**.** *Assume that there is a number $\beta > 0$ for which the following conditions hold.*

*(i) There are constants $c, C, t_0 > 0$ such that, for $t \ge t_0$,*

$$c\,t^{-\beta} \le 1 - x(t) \le C\,t^{-\beta}, \qquad c\,t^{-\beta-1} \le x'(t) \le C\,t^{-\beta-1}. \tag{13.5}$$

*(ii) The full parameter sequence is not too dense on the logarithmic scale:*

$$\log t_j \ge (1 - o(1))\log j. \tag{13.6}$$

*(iii) The support is not too sparse: for every sufficiently large $N$ there are $N$ support parameters $s_{N,1} < \cdots < s_{N,N}$ and a number $\eta_N \to 0$ such that*

$$s_{N,N} \le N^{1+\eta_N}, \qquad s_{N,b} - s_{N,a} \ge (b-a)N^{-\eta_N} \quad (1 \le a < b \le N). \tag{13.7}$$

*(iv) The weights on these selected atoms are subleading at the determinant scale:*

$$\sum_{a=1}^{N} \log w(s_{N,a}) \ge -o(N^2\log N). \tag{13.8}$$

*Then*

$$\boxed{\lim_{N\to\infty} \frac{\log D_N[\mu]}{N^2\log N} = -\beta.} \tag{13.9}$$

*More generally, the same limit holds for the Gram determinant in any polynomial basis $p_0, p_1, \ldots$ with $\deg p_n = n$ and leading coefficients $c_n$ satisfying*

$$\sum_{n=0}^{N-1} \log|c_n| = O(N^2). \tag{13.10}$$

**Lemma 13.3** (Upper extremal-polynomial bound)**.** *Under hypotheses* (i)–(ii) *of Theorem 13.2,*

$$\log D_N[\mu] \le -\beta N^2 \log N + o(N^2 \log N).$$

*Proof.* Let $h_n$ be the squared norm of the monic orthogonal polynomial of degree $n$, so $D_N[\mu] = \prod_{n<N} h_n$; see, for example, Chihara [8] for the classical moment-determinant and monic-orthogonal-polynomial framework. The monic competitor $q_n(x) = \prod_{j\le n}(x - x_j)$ vanishes at the first $n$ atoms and satisfies

$$h_n \le \mu(\mathbb{R}) \prod_{j\le n} (1 - x_j)^2.$$

Hypotheses (i)–(ii) give $\sum_{j\le n} \log(1 - x_j) \le -\beta n \log n + o(n \log n)$. Thus $\log h_n \le -2\beta n \log n + o(n \log n)$, and summation over $n < N$ proves the claim. □

**Lemma 13.4** (Selected Vandermonde lower bound)**.** *Under hypotheses* (i), (iii), *and* (iv) *of Theorem 13.2,*

$$\log D_N[\mu] \ge -\beta N^2 \log N - o(N^2 \log N).$$

*Proof.* Retain in Cauchy–Binet the term formed by the selected atoms $y_a = x(s_{N,a})$. Positivity gives

$$D_N[\mu] \ge \left( \prod_{a\le N} w(s_{N,a}) \right) \prod_{a<b} (y_b - y_a)^2.$$

The derivative bound and hypothesis (iii) imply $y_b - y_a \ge c(b - a)N^{-\beta-1-(\beta+2)\eta_N}$. Lemma 13.1 then yields $2\sum_{a<b} \log(y_b - y_a) \ge -\beta N^2 \log N - o(N^2 \log N)$, and hypothesis (iv) shows that the selected weights are subleading. □

**Lemma 13.5** (Polynomial-basis stability)**.** *Let $p_n(x) = c_n x^n +$ lower-degree terms. If $G_N[p]$ is its Gram matrix and $H_N$ is the monomial Gram matrix, then*

$$\det G_N[p] = \left( \prod_{n<N} c_n^2 \right) \det H_N.$$

*Consequently, condition* (13.10) *leaves the coefficient of $N^2 \log N$ unchanged.*

*Proof.* If $C_N$ is the lower triangular coefficient matrix taking the monomial basis to $(p_0, \dots, p_{N-1})$, then $G_N[p] = C_N H_N C_N^T$ and $\det C_N = \prod_{n<N} c_n$. Taking determinants proves both assertions. □

*Proof.* The argument is divided into the three independently checkable modules isolated in Lemmas 13.3–13.5. We now give the complete estimates, including all error terms needed at the $N^2 \log N$ scale.

*Upper bound.* Because $\mu$ is a finite positive measure with infinite support, the monomials $1, x, \dots, x^n$ are linearly independent in $L^2(\mu)$ for every $n$: a nonzero polynomial has only finitely many zeros, whereas the support contains infinitely many distinct points. Hence the monic orthogonal polynomial $\pi_n$ of degree $n$ exists uniquely, and its squared norm

$$h_n := \int |\pi_n(x)|^2 \, \mathrm{d}\mu(x) > 0 \tag{13.11}$$

is the minimum of $\int |q|^2 \, \mathrm{d}\mu$ among all monic polynomials $q$ of degree $n$. Gram–Schmidt, or equivalently triangular change of basis from monomials to the monic orthogonal polynomials, gives the exact product identity

$$D_N[\mu] = \prod_{n=0}^{N-1} h_n. \tag{13.12}$$

For $n \geq 1$ take the monic competitor

$$q_n(x) := \prod_{j=1}^{n} (x - x_j). \tag{13.13}$$

It vanishes at the first $n$ atoms. Since $x_j \uparrow 1$, for $j > n$ and every $1 \leq r \leq n$,

$$0 < x_j - x_r \leq 1 - x_r. \tag{13.14}$$

Therefore

$$|q_n(x_j)| \leq \prod_{r=1}^{n} (1 - x_r), \qquad j > n. \tag{13.15}$$

Writing $M := \mu(\mathbb{R}) < \infty$ and using the extremal property of $\pi_n$,

$$h_n \leq \int |q_n(x)|^2 \, \mathrm{d}\mu(x) \tag{13.16}$$

$$= \sum_{j>n} w_j |q_n(x_j)|^2 \tag{13.17}$$

$$\leq M \prod_{r=1}^{n} (1 - x_r)^2. \tag{13.18}$$

By hypothesis (i), after increasing a fixed initial index if necessary,

$$\log(1 - x_j) \le \log C - \beta \log t_j. \tag{13.19}$$

Hypothesis (ii) means that there exists a sequence $\varepsilon_j \to 0$ such that $\log t_j \ge (1 - \varepsilon_j) \log j$ for all sufficiently large $j$. Hence

$$\log(1 - x_j) \le -\beta \log j + \beta \varepsilon_j \log j + O(1). \tag{13.20}$$

The error sums to $o(n \log n)$. Indeed, for every $\epsilon > 0$ choose $J$ so that $|\varepsilon_j| \le \epsilon$ for $j \ge J$; then

$$\sum_{j=J}^{n} |\varepsilon_j| \log j \le \epsilon \sum_{j=J}^{n} \log j = O(\epsilon n \log n), \tag{13.21}$$

and the finitely many $j < J$ contribute $O_J(1)$. Since

$$\sum_{j=1}^{n} \log j = \log(n!) = n \log n - n + O(\log n), \tag{13.22}$$

we obtain

$$\sum_{j=1}^{n} \log(1 - x_j) \le -\beta n \log n + o(n \log n). \tag{13.23}$$

Thus (13.18) gives

$$\log h_n \le -2\beta n \log n + o(n \log n). \tag{13.24}$$

Summing (13.24) for $0 \le n < N$ is legitimate at the same scale: if $r_n = o(n \log n)$, then for every $\epsilon > 0$ one has $|r_n| \le \epsilon n \log n$ for all sufficiently large $n$, so $\sum_{n<N} r_n = o(N^2 \log N)$. Finally,

$$\sum_{n=1}^{N-1} n \log n = \frac{1}{2} N^2 \log N - \frac{1}{4} N^2 + O(N \log N), \tag{13.25}$$

and therefore

$$\log D_N[\mu] \le -\beta N^2 \log N + o(N^2 \log N). \tag{13.26}$$

*Lower bound.* Fix $N$ large enough that hypothesis (iii) supplies distinct support parameters $s_{N,1} < \cdots < s_{N,N}$, and put $y_a := x(s_{N,a})$. The finite $N \times N$ moment matrix generated by these selected atoms is

$$H_N^{\mathrm{sel}} = V_N W_N V_N^T, \tag{13.27}$$

where $(V_N)_{r,a} = y_a^r$ for $0 \le r \le N-1$ and $W_N = \operatorname{diag}(w(s_{N,1}), \dots, w(s_{N,N}))$. Since the full moment matrix is the sum of this positive semidefinite matrix and the contribution of all remaining atoms, the Cauchy–Binet expansion of the full determinant contains the selected term with positive sign; equivalently, applying Cauchy–Binet first to a finite truncation containing the selected atoms and then passing monotonically to the full measure gives

$$D_N[\mu] \ge \left(\prod_{a=1}^{N} w(s_{N,a})\right) \prod_{1 \le a < b \le N} (y_b - y_a)^2. \tag{13.28}$$

For $a < b$, monotonicity of $x$ and hypothesis (i) give

$$y_b - y_a = \int_{s_{N,a}}^{s_{N,b}} x'(u)\, \mathrm{d}u \tag{13.29}$$

$$\ge c\, s_{N,N}^{-\beta-1}(s_{N,b} - s_{N,a}), \tag{13.30}$$

because $u \le s_{N,N}$ on the interval and $u \mapsto u^{-\beta-1}$ is decreasing. By hypothesis (iii),

$$y_b - y_a \ge c(b-a)N^{-(\beta+1)(1+\eta_N)} N^{-\eta_N} \tag{13.31}$$

$$= c(b-a)N^{-\beta-1-(\beta+2)\eta_N}. \tag{13.32}$$

Taking logarithms and summing over the $\binom{N}{2}$ pairs,

$$2\sum_{a<b} \log(y_b - y_a) \ge 2\sum_{a<b} \log(b-a) \tag{13.33}$$

$$- 2[\beta + 1 + (\beta+2)\eta_N]\binom{N}{2} \log N + O(N^2). \tag{13.34}$$

Lemma 13.1 gives the required pair-product asymptotic. Since $\eta_N \to 0$, (13.34) and (13.4) yield

$$2\sum_{a<b} \log(y_b - y_a) \ge -\beta N^2 \log N - o(N^2 \log N). \tag{13.35}$$

Hypothesis (iv) states precisely that the selected weights do not decrease this bound at the leading scale. Inserting the result into (13.28) gives

$$\log D_N[\mu] \ge -\beta N^2 \log N - o(N^2 \log N). \tag{13.36}$$

Together with (13.26), this proves the limit.

*Polynomial bases.* Let $p_n(x) = c_n x^n +$ terms of degree $< n$, and let $C_N$ be the lower triangular coefficient matrix taking the monomial vector $(1, x, \dots, x^{N-1})^T$ to $(p_0, \dots, p_{N-1})^T$. Its diagonal is $(c_0, \dots, c_{N-1})$, so $\det C_N = \prod_{n<N} c_n$. If $G_N[p]$ denotes the corresponding Gram matrix, then exactly

$$G_N[p] = C_N H_N C_N^T, \tag{13.37}$$

and hence

$$\log\det G_N[p] = \log D_N[\mu] + 2\sum_{n=0}^{N-1} \log|c_n|. \tag{13.38}$$

Hypothesis (13.10) makes the second term $O(N^2) = o(N^2 \log N)$, so the leading coefficient remains $-\beta$. □

*Remark* 13.6 (The node-scale hypotheses are essential). The conclusion is not automatic from the endpoint exponent. Without the node-scale hypotheses, it is false: $\beta$ by itself does not determine the coefficient of $N^2 \log N$. Hypothesis (ii) is used only in the upper bound; hypotheses (iii) and (iv) are used only in the lower bound; hypothesis (i) supplies the endpoint factor in both directions. The following three measures have the *same* endpoint map and the *same* weights,

$$x(t) = 1 - t^{-2}, \qquad w_j = j^{-2},$$

so that $\beta = 2$ in every case, but their normalized determinant asymptotics are different:

| **Node parameters** | **Endpoint hypotheses** | $\dfrac{\log D_N}{N^2 \log N}$ |
|---|---|---|
| $t_j = j$ | (i)–(iv) hold | $\longrightarrow -2$ |
| $t_j = 1 + \log j$ | (ii) fails: parameters are too dense | $\longrightarrow 0$ |
| $t_j = c^j$, $c > 1$ | (iii) fails: parameters are too sparse | $\longrightarrow -\infty$ |

Thus the theorem is a statement about the endpoint exponent *after the parameter scale has been fixed*, not a statement that $\beta$ alone controls arbitrary discrete supports.

For completeness, the last two limits follow directly from the same two estimates used in the proof. If $t_j = 1 + \log j$, the monic competitor gives

$$\log D_N \le -2N^2 \log\log N + O(N^2).$$

For the matching lower scale retain the first $N$ atoms. For $1 \le k < \ell \le N$,

$$\log(\ell/k) \ge \frac{\ell - k}{\ell} \ge \frac{\ell - k}{N},$$

and since $x'(t) = 2t^{-3}$,

$$x(t_\ell) - x(t_k) \ge c\,\frac{\ell - k}{N(1 + \log N)^3}.$$

The integer Vandermonde entropy of Lemma 13.1 cancels the resulting $N^{-1}$ pair scale, while the factor $(1 + \log N)^{-3}$ contributes only $-3N^2 \log\log N + O(N^2)$. The weights contribute $-O(N \log N)$. Hence

$$-3N^2 \log\log N + O(N^2) \le \log D_N \le -2N^2 \log\log N + O(N^2),$$

and the normalized quotient tends to 0.

If $t_j = c^j$, then $1 - x_j = c^{-2j}$, and the monic upper bound gives

$$\log h_n \le -2(\log c)n(n+1) + O(1).$$

Summation yields

$$\log D_N \le -\frac{2}{3}(\log c)N^3 + O(N^2),$$

which forces $\log D_N/(N^2 \log N) \to -\infty$.

Finally, (ii) and (iii) together pin down the full parameter sequence at logarithmic polynomial scale. Indeed (ii) gives $t_N \ge N^{1-o(1)}$, while the existence of $N$ selected support parameters in (iii) gives $t_N \le s_{N,N} \le N^{1+o(1)}$. Hence

$$\boxed{t_N = N^{1+o(1)}.}$$

This is the scale on which the endpoint exponent becomes the leading determinant coefficient. Accordingly, the theorem should be read as

$$\begin{gathered}\text{endpoint exponent } \beta + \text{node clock } t_j = j^{1+o(1)} \\ \Longrightarrow \text{ leading coefficient } -\beta.\end{gathered}$$

It should not be shortened to the assertion that $\beta$ alone governs the determinant.

**Experiment 13.7** (Finite numerical illustration of the three node clocks)**.** To illustrate the preceding distinction, keep the same endpoint map $x(t) = 1-t^{-2}$ and the same weights $w_j = j^{-2}$, and change only the node parameters. A computation using 400-digit arithmetic and Lanczos orthogonalization with full reorthogonalization gives the following values of $\log D_N/(N^2 \log N)$:

| **Node parameters** | $N = 8$ | $N = 16$ | $N = 24$ | **proved limit** |
|---|---|---|---|---|
| $t_j = j$: hypotheses (ii)–(iii) hold | $-1.024$ | $-1.195$ | $-1.273$ | $-2$ |
| $t_j = 1 + \log j$: hypothesis (ii) fails | $-0.862$ | $-0.888$ | $-0.885$ | $0$ |
| $t_j = 1.5^j$: hypothesis (iii) fails | $-1.243$ | $-1.672$ | $-2.118$ | $-\infty$ |

The table is illustrative, not a proof of any limit. In the polynomial-clock case the rigorous $O(N^2)$ remainder gives an $O(1/\log N)$ convergence rate after normalization, so the value $-1.273$ at $N = 24$ is compatible with the limit $-2$. In the logarithmic-clock case the small reversal from $-0.888$ to $-0.885$ is likewise compatible with the proved limit 0; in the exponential-clock case the normalized values continue toward $-\infty$. The rigorous proofs of all three limits are the estimates in Remark 13.6.

*Remark* 13.8 (Role of the four hypotheses). The proof can be read as the following one-line accounting. Hypothesis (ii), combined with the endpoint distance in (i), gives the monic upper estimate

$$\log h_n \le -2\beta n \log n + o(n \log n).$$

For the lower bound, (iii) and the derivative part of (i) give pair gaps of size $(b-a)N^{-(\beta+1)-o(1)}$. The integer Vandermonde contributes $+\frac{1}{2}N^2 \log N$ before squaring, while this scale contributes $-\frac{\beta+1}{2}N^2 \log N$; their difference is $-\frac{\beta}{2}N^2 \log N$, and the square produces $-\beta N^2 \log N$. Hypothesis (iv) only prevents the selected weights from changing that leading order. Thus the identity

$$\frac{1}{2} - \frac{\beta+1}{2} = -\frac{\beta}{2},$$

followed by the square of the Vandermonde, is the entire leading-order balance. Logarithmic refinements such as $t_j \asymp j/\log j$ remain within $j^{1+o(1)}$, so they are invisible at the $N^2 \log N$ scale. Arithmetic information can therefore first appear below that leading scale.

**Theorem 13.9** (Regularly varying endpoint law)**.** *In Theorem 13.2, hypothesis* (13.5) *may be replaced by*

$$1 - x(t) = t^{-\beta} L(t), \qquad x'(t) = \beta t^{-\beta-1} L(t)(1 + o(1)), \tag{13.39}$$

*where $L$ is slowly varying and $\log L(t) = o(\log t)$, provided the remaining node, separation, and weight hypotheses are unchanged. Then*

$$\lim_{N\to\infty} \frac{\log D_N}{N^2 \log N} = -\beta. \tag{13.40}$$

*The same conclusion holds after a polynomial change of Gram basis satisfying* (13.10)*.*

*Proof.* Fix $\varepsilon > 0$. Since $\log L(t) = o(\log t)$, there is $T_\varepsilon$ such that, for $t \geq T_\varepsilon$,

$$t^{-\varepsilon} \leq L(t) \leq t^{\varepsilon}. \tag{13.41}$$

The factor $1 + o(1)$ in the derivative may simultaneously be placed between $1/2$ and $2$.

For the upper bound,

$$1 - x(t) = t^{-\beta} L(t) \leq t^{-(\beta-\varepsilon)}$$

for large $t$. Repeating the monic-polynomial argument in the proof of Theorem 13.2, with $\beta - \varepsilon$ in place of $\beta$, gives

$$\limsup_{N\to\infty} \frac{\log D_N}{N^2 \log N} \leq -(\beta - \varepsilon).$$

For the lower bound, the derivative assumption and (13.41) give

$$x'(t) \geq \frac{\beta}{2}\, t^{-(\beta+1+\varepsilon)}$$

for large $t$. On the selected configuration of hypothesis (13.7), the mean-value estimate used in the lower bound of Theorem 13.2 therefore becomes

$$x(s_{N,b}) - x(s_{N,a}) \geq c_\varepsilon (b - a) N^{-\beta-1-\varepsilon-o(1)}.$$

Lemma 13.1 then yields

$$\liminf_{N\to\infty} \frac{\log D_N}{N^2 \log N} \geq -(\beta + \varepsilon).$$

The selected weights remain subleading by the unchanged weight hypothesis. Since $\varepsilon > 0$ is arbitrary, the limsup and liminf coincide at $-\beta$. The basis statement is unchanged because its contribution is $O(N^2) = o(N^2 \log N)$. □

*Remark* 13.10 (Why regular variation is the natural formulation). The exponent is invariant under the logarithmic reindexings produced by counting laws. For example, if $t_j \asymp j/\log j$, then

$$1 - x(t_j) \asymp j^{-\beta}(\log j)^{\beta},$$

and the extra logarithmic factor is slowly varying. Thus the $T \log T$ density of zeta zeros can alter lower-order terms without changing the endpoint exponent. A formulation demanding $1-x(t_j) \asymp j^{-\beta}$ with hard constants would obscure precisely this invariance.

**Proposition 13.11** (Jacobi translation of the endpoint law)**.** *Let $\mu$ be a probability measure satisfying the endpoint hypotheses, and let its orthonormal polynomials satisfy*

$$xp_n(x) = b_n p_{n+1}(x) + a_n p_n(x) + b_{n-1}p_{n-1}(x), \qquad b_n > 0. \tag{13.42}$$

*For the monomial Hankel determinant $D_N$, one has the exact identity*

$$\boxed{\log D_N = 2\sum_{m=0}^{N-2}(N-1-m)\log b_m.} \tag{13.43}$$

*Consequently the endpoint law is equivalent to*

$$\frac{2}{N^2}\sum_{m<N}(N-m)\log b_m = -\beta \log N + o(\log N), \tag{13.44}$$

*that is, to $b_m = m^{-\beta+o(1)}$ in logarithmic Cesàro mean.*

*Proof.* If $k_n$ is the leading coefficient of the orthonormal polynomial $p_n$, then (13.42) gives $k_n = k_{n-1}/b_{n-1}$. Hence the squared norm of the monic polynomial is

$$h_n = k_n^{-2} = (b_0 \cdots b_{n-1})^2.$$

Since $D_N = \prod_{n=0}^{N-1} h_n$, collecting powers gives (13.43). Finally $\sum_{m<N}(N-m)\log m = \frac{1}{2}N^2 \log N + O(N^2)$, which yields (13.44). □

*Remark* 13.12 (What is universal after the node scale is fixed). Theorem 13.2 and Remark 13.6 show precisely what is universal. The exponent $\beta$ is not sufficient on an arbitrary node sequence. Once hypotheses (ii)–(iii) place the parameters at the logarithmic polynomial scale $t_j = j^{1+o(1)}$, and the

selected weights satisfy (iv), the coefficient is $-\beta$. Consequently $t_j \asymp j$, $t_j \asymp j/\log j$, and other logarithmic refinements compatible with the theorem have the same leading coefficient. The Riemann–von Mangoldt logarithm is therefore invisible at the $N^2 \log N$ scale. For the Li–Cayley map, $\beta = 2$, so the coefficient $-2$ is a geometric consequence of the endpoint map together with the verified node scale, not by itself an arithmetic diagnostic. Finer arithmetic information can first enter below that scale.

*Remark* 13.13 (Relation with earlier discrete Hankel asymptotics). The mere occurrence of an $N^2 \log N$ scale for a discrete Hankel determinant is not new. Monien [27] derived a discrete Vandermonde representation for Hankel determinants of Dirichlet series. For the reciprocal-integer zeta model his Coulomb-gas calculation leads to

$$\log H_n^{(r)}[\zeta] = -n^2 \left( \log(2n) - \frac{3}{2} \right) + o(n^2) \tag{13.45}$$

at the level of the variational analysis presented there. Monien explicitly notes that the passage from the discrete optimizer to the continuous density is assumed and suggests Poissonization as a route to a fully rigorous treatment. Thus (13.45) should not be cited here as a proved general asymptotic theorem from that preprint. Its leading coefficient nevertheless provides an important external consistency check: the nodes $x_m = 1/m$ have endpoint exponent $\beta = 1$, and Theorem 13.2 gives the same rigorous leading term $-n^2 \log n$ under its hypotheses. The present theorem is broader at leading order—arbitrary $\beta$ and non-explicit node sets satisfying coarse separation hypotheses—whereas Monien's special-model variational calculation is finer at order $n^2$.

Haynes and Zudilin [17] prove the one-sided estimate

$$\log |\det(\zeta(a(i+j)+b))_{1 \le i,j \le n}| \le -\frac{a}{2} n^2 \log n + O(n^2), \tag{13.46}$$

which was sufficient for the application they considered. In the notation used here, the support $m^{-a}$ has endpoint exponent $\beta = a$. The endpoint theorem gives both sides and therefore proves the leading asymptotic

$$\log H_n^{(a,b)} = -an^2 \log n + O(n^2) = -\beta n^2 \log n + O(n^2). \tag{13.47}$$

Johansson's discrete orthogonal-polynomial framework [21] is also closely related but does not make Theorem 13.2 an immediate corollary, and the reason is structural rather than a matter of coverage. Every ensemble treated

there is supported on a *uniform* lattice: the configurations range over $\mathbb{N}^N$ or $\{0,\dots,K\}^N$, the Vandermonde is formed in the lattice variable, and the asymptotic analysis proceeds through a constrained equilibrium measure whose saturation constraint $\rho \le 1$ is precisely an expression of that uniform spacing. The detailed results are then obtained for the structured Meixner, Charlier, Krawtchouk, Hahn, and Plancherel weights.

The support relevant here is of a different type. The nodes $x_\gamma$ of (13.67) are nonuniform, do not lie on a lattice, and accumulate at an endpoint. Thus there is no fixed uniform saturation constraint of the form $\rho \le 1$ inherited from unit lattice spacing; any continuum occupancy constraint would instead have to reflect the local nonuniform node density. The hypotheses used above, (13.5), (13.6) and (13.7), are coarse and impose no lattice structure at all. Accordingly, Johansson supplies the rigorous discrete-Coulomb-gas context and is directly relevant to the lower-order constrained-equilibrium questions raised in Remark 13.23, but Theorem 13.2 is logically independent of the results stated there. This is a statement about [21] only; no general priority is claimed against the wider literature on discrete orthogonal polynomials with an accumulation point, on Stieltjes-type moment problems, or on Hankel determinant asymptotics.

These are different matrices from the Li–Weil reflection pair, with Dirichlet-series or lattice nodes rather than Cayley-transformed zero ordinates. Accordingly, no novelty is claimed here for the generic presence of $N^2 \log N$ or for constrained equilibrium methods themselves. The distinction of Theorem 13.2 is the elementary two-sided endpoint-exponent principle under coarse geometric hypotheses and its application to the Li–Weil reflection measures below.

*13.2. The zeta reflection measures and the universal constant $-2$*

**Logical status.** Everything in this subsection assumes RH, because the positive reflection measures and the real ordinates $\gamma$ come from the critical-line Gram representation. No additional unproved spacing hypothesis is used for the first-order $-2$ law: the node-scale and separated- subconfiguration assumptions of Theorem 13.2 are verified below from standard zero counting. Extra hypotheses enter only in the later second-order subsection.

Let $\gamma_1^* < \gamma_2^* < \cdots$ denote the distinct positive ordinates and set

$$\delta_j := 1 - x_{\gamma_j^*} = \frac{2}{4(\gamma_j^*)^2 + 1}. \tag{13.48}$$

Multiplicity is deliberately omitted from this enumeration.

**Lemma 13.14** (Coarse location of distinct ordinates)**.** *There are constants $c > 0$ and $j_0$ such that*

$$\gamma_j^* \geq c\frac{j}{\log j}, \qquad j \geq j_0. \tag{13.49}$$

*This estimate requires no simplicity hypothesis.*

*Proof.* Let $N^*(T)$ count distinct positive ordinates and let $N(T)$ count positive ordinates with multiplicity. Then $N^*(T) \leq N(T)$. The Riemann–von Mangoldt formula gives $N(T) \ll T\log T$, hence $j \leq N(\gamma_j^*) \ll \gamma_j^* \log \gamma_j^*$, which implies (13.49). Multiple zeros can only decrease $N^*(T)$ and therefore move the distinct ordinates to larger heights; the estimate is thus robust with respect to multiplicity. □

**Lemma 13.15** (A deterministically separated zero subsequence)**.** *There are constants $H > 0$, $T_0 > 0$, and distinct positive ordinates $\widetilde{\gamma}_j$ such that, for all $j \geq 1$,*

$$T_0 + 2Hj \leq \widetilde{\gamma}_j \leq T_0 + 2Hj + H. \tag{13.50}$$

*Consequently $\widetilde{\gamma}_j \asymp j$ and*

$$\widetilde{\gamma}_b - \widetilde{\gamma}_a \geq H(b-a), \qquad 1 \leq a < b. \tag{13.51}$$

*Proof.* Write the Riemann–von Mangoldt formula as

$$N(T) = F(T) + R(T), \qquad F(T) := \frac{T}{2\pi}\log\frac{T}{2\pi} - \frac{T}{2\pi}, \qquad R(T) = O(\log T). \tag{13.52}$$

Thus there are absolute constants $A, T_1 > 0$ such that $|R(T)| \leq A\log T$ for $T \geq T_1$. For every fixed $H \geq 1$,

$$F(T+H) - F(T) = \frac{H}{2\pi}\log T + O_H(1), \qquad T \to \infty. \tag{13.53}$$

Choose once and for all $H > 4\pi A$. Then, for sufficiently large $T$,

$$N(T+H) - N(T) \geq \left(\frac{H}{2\pi} - 2A + o(1)\right)\log T > 0.$$

Hence every interval $[T, T+H]$ with $T \geq T_0$ contains a zero for some $T_0 \geq T_1$. Applying this to the alternating boxes $[T_0+2Hj, T_0+2Hj+H]$ gives (13.50); the boxes are disjoint and (13.51) follows immediately. □

**Theorem 13.16** (Li–Weil determinant law)**.** *Assume* RH. *Then, for both reflection families,*

$$\boxed{\lim_{N\to\infty} \frac{\log \Delta_N^E}{N^2 \log N} = \lim_{N\to\infty} \frac{\log \Delta_N^O}{N^2 \log N} = -2.} \tag{13.54}$$

*More quantitatively,*

$$-2N^2 \log N + O(N^2) \le \log \Delta_N^E \le -2N^2 \log N + 2N^2 \log\log N + O(N^2), \tag{13.55}$$

$$-2N^2 \log N + O(N^2) \le \log \Delta_N^O \le -2N^2 \log N + 2N^2 \log\log N + O(N^2). \tag{13.56}$$

*Proof.* For both zero measures the support map is

$$x(t) = 1 - \frac{2}{4t^2+1}, \tag{13.57}$$

so

$$1 - x(t) \asymp t^{-2}, \qquad x'(t) \asymp t^{-3}. \tag{13.58}$$

Thus $\beta = 2$. Lemma 13.14 verifies the full-sequence condition (13.6), while Lemma 13.15 verifies (13.7) with $\eta_N = 0$ up to harmless constants. Along that separated subsequence, the multiplicity of every selected ordinate is at least one and $\widetilde{\gamma}_j \asymp j$. Hence each selected one-point weight in (11.12)–(11.13) is bounded below by a constant multiple of $j^{-2}$. Consequently its logarithmic product is bounded below by $-O(N \log N)$, which is exactly the one-sided information required in (13.8). No upper bound on zero multiplicities is used. The Chebyshev-adapted bases have leading coefficient $2^n$, hence their total basis contribution is $O(N^2)$. Theorem 13.2 therefore gives the limit (13.54).

For the quantitative upper bound, let $h_n$ be the squared norm of the monic orthogonal polynomial for either $\mu_E$ or $\mu_O$. The competitor

$$q_n(x) := \prod_{j=1}^{n} (x - x_{\gamma_j^*}) \tag{13.59}$$

gives

$$h_n \le M \prod_{j=1}^{n} \delta_j^2, \qquad M = \mu([-1,1]). \tag{13.60}$$

Lemma 13.14 yields

$$\delta_j \ll \frac{(\log j)^2}{j^2}, \tag{13.61}$$

and therefore, after restoring the leading coefficient $2^n$ of $U_n$ or $P_n$,

$$\log d_{n+1}^{E,O} \le -4n\log n + 4n\log\log n + O(n). \tag{13.62}$$

Summing gives the upper inequalities in (13.55)–(13.56).

For the quantitative lower bound, retain the separated term $\widetilde{\gamma}_1, \dots, \widetilde{\gamma}_N$ in the nonnegative Cauchy–Binet expansion. Since $x'(t) \asymp t^{-3}$ and (13.51) holds,

$$\widetilde{x}_b - \widetilde{x}_a \gg \frac{b-a}{b^3}. \tag{13.63}$$

Hence

$$2\sum_{a<b}\log(\widetilde{x}_b - \widetilde{x}_a) \ge 2\sum_{a<b}\log(b-a) - 6\sum_{a<b}\log b + O(N^2) \tag{13.64}$$

$$= -2N^2\log N + O(N^2), \tag{13.65}$$

using

$$\sum_{a<b}\log(b-a) = \frac{1}{2}N^2\log N + O(N^2), \qquad \sum_{a<b}\log b = \frac{1}{2}N^2\log N + O(N^2). \tag{13.66}$$

The selected one-point weights can decrease the logarithm of this retained term by at most $O(N\log N)$, while the explicit basis prefactor contributes only $O(N^2)$; this gives the lower inequalities. □

**Corollary 13.17** (Universal Cayley exponent)**.** *For every positive discrete Gram system satisfying the hypotheses of Theorem 13.2 with the Li–Cayley support map*

$$x(t) = \frac{4t^2-1}{4t^2+1}, \tag{13.67}$$

*the leading determinant coefficient is $-2$. In particular, the value $-2$ is unchanged by replacing a $T\log T$ counting law by a $T$ counting law, or by any other logarithmic refinement compatible with the theorem's node hypotheses.*

*Proof.* The map (13.67) has endpoint exponent $\beta = 2$, so the conclusion is Theorem 13.2. □

*Remark* 13.18 (Automorphic Li coefficients). Lagarias defines generalized Li coefficients for principal automorphic $L$-functions and relates them to Weil's quadratic functional [23]. Consequently, whenever a self-dual member of that class is placed under GRH and its canonical reflection compression has the same positive Cayley zero measure with the standard coarse zero-counting and separation properties, Corollary 13.17 gives the same coefficient $-2$. The degree and conductor can change lower-order terms but not the endpoint exponent. This is a conditional universality statement: the present paper does not identify a real even–odd reflection pair for every non-self-dual automorphic representation.

**Corollary 13.19** (Condition-number obstruction)**.** *Assume* RH. *Then*

$$\boxed{\liminf_{N\to\infty} \frac{\log \kappa_2(\mathsf{E}_N)}{N \log N} \geq 2, \qquad \liminf_{N\to\infty} \frac{\log \kappa_2(\mathsf{O}_N)}{N \log N} \geq 2.} \tag{13.68}$$

*More quantitatively,*

$$\log \kappa_2(\mathsf{E}_N),\ \log \kappa_2(\mathsf{O}_N) \geq 2N \log N - 2N \log\log N - O(N). \tag{13.69}$$

*Proof.* For a positive definite $N \times N$ matrix $A_N$,

$$\lambda_{\min}(A_N)^N \leq \det A_N, \tag{13.70}$$

so

$$\log \kappa_2(A_N) \geq \log \lambda_{\max}(A_N) - \frac{1}{N} \log\det A_N. \tag{13.71}$$

For the even family $\lambda_{\max}(\mathsf{E}_N) \geq (\mathsf{E}_N)_{11} = \lambda_2 > 0$, while for the odd family $\lambda_{\max}(\mathsf{O}_N) \geq (\mathsf{O}_N)_{11} = \lambda_1 > 0$. Insert the determinant upper bounds from Theorem 13.16. This gives (13.69), and division by $N \log N$ gives (13.68). □

**Corollary 13.20** (Divergence of the inverse norm under RH)**.** *Assume* RH. *Then*

$$\boxed{\log \|\mathsf{E}_N^{-1}\|_2 \geq 2N \log N - 2N \log\log N - O(N),} \tag{13.72}$$

*and consequently* $\|\mathsf{E}_N^{-1}\|_2 \to \infty$ *faster than* $C^N$ *for every fixed* $C > 1$. *The same statement holds for* $\mathsf{O}_N^{-1}$.

*Proof.* For a positive definite matrix, $\|A_N^{-1}\|_2 = \lambda_{\min}(A_N)^{-1}$ and $\lambda_{\min}(A_N)^N \leq \det A_N$. The determinant upper bound in Theorem 13.16 therefore gives

$$\log \lambda_{\min}(\mathsf{E}_N) \leq -2N \log N + 2N \log\log N + O(N),$$

which is equivalent to (13.72). The odd case is identical. □

*Remark* 13.21 (Congruence warning for first-difference eigenvalues). Let $\widehat{E}_N := \frac{1}{2} D_N \mathsf{E}_N D_N^T$. Then

$$\mathsf{E}_N^{-1} = \frac{1}{2} D_N^T \widehat{E}_N^{-1} D_N. \tag{13.73}$$

Hence a tiny value of $\lambda_{\min}(\widehat{E}_N)$ does force a large inverse after accounting for the congruence, but one must not write $\|\mathsf{E}_N^{-1}\|_2 = 1/\lambda_{\min}(\widehat{E}_N)$. More precisely,

$$\|\mathsf{E}_N^{-1}\|_2 \geq \frac{\sigma_{\min}(D_N)^2}{2\,\lambda_{\min}(\widehat{E}_N)}. \tag{13.74}$$

The direct identity is $\|\mathsf{E}_N^{-1}\|_2 = 1/\lambda_{\min}(\mathsf{E}_N)$.

*Remark* 13.22 (Finite condition-number illustration; no asymptotic claim). Corollary 13.19 uses

$$\lambda_{\min}(A_N)^N \leq \det A_N,$$

which replaces the smallest eigenvalue by the geometric mean of the spectrum. For these matrices the spectrum is extremely spread, so the substitution is lossy. In a direct computation with the measure carried by the first 1200 ordinates and eigenvalues evaluated at 1200 decimal digits, the observed ratios

$$\frac{\log \lambda_{\min}(\mathsf{E}_N)}{N \log N} \tag{13.75}$$

are $-11.9$, $-8.85$, $-7.83$, $-7.31$, $-6.94$, $-6.68$, $-6.48$, $-6.32$ for $N = 2, 4, \ldots, 16$, decreasing slowly. The scale that (13.71) induces for the same ratio (13.75) is only $-2$, since the argument replaces $\log \lambda_{\min}$ by $N^{-1} \log \det$; the sign convention here is that of $\lambda_{\min}$, not of $\kappa_2$. Correspondingly, at $N = 16$ the computation gives $\log_{10} \kappa_2(\mathsf{E}_{16}) \approx 123.9$ against a proved lower bound near 38.5. The corollary is therefore correct but conservative by a factor of about three at these orders. All numbers refer to the measure truncated to the first 1200 ordinates and are reported as computation on that truncation, not as values for the exact infinite matrices, and not as an asymptotic claim.

*13.2.1. Conditional second-order refinements*

This subsection is logically separate from the first-order Li–Weil law. Every statement below that goes beyond the coefficient $-2$ carries hypotheses additional to RH; none of those hypotheses is used in Theorem 13.16.

*Remark* 13.23 (Why the $N^2 \log\log N$ term is unresolved). The upper term $2N^2 \log\log N$ in Theorem 13.16 is not declared to be an artifact. Positivity implies that one favorable Cauchy–Binet subset gives a lower bound, but it does not imply that the logarithm of the full infinite sum equals the logarithm of its largest term; an entropy contribution can survive below the leading scale. Moreover, the Riemann–von Mangoldt clock itself suggests that the displayed logarithmic correction is structurally plausible. If, heuristically, $t_j \asymp j/\log j$ with local spacing $t_{j+1}-t_j \asymp 1/\log j$, then for a general endpoint exponent $\beta$ one has at the pairwise level

$$x(t_b) - x(t_a) \asymp \frac{(b-a)(\log b)^\beta}{b^{\beta+1}}, \tag{13.76}$$

and the corresponding Vandermonde energy has the scale

$$-\beta N^2 \log N + \beta N^2 \log\log N + O(N^2). \tag{13.77}$$

Thus for the Li–Cayley value $\beta = 2$ the coefficient $+2$ in the present upper bound is exactly what the smooth clock predicts. Proving a matching second-order law would require substantially finer information on the dense zero configuration and is deliberately separated from the universal leading theorem. Theorem 13.24 below isolates exactly what that finer information has to be.

The next statement isolates a second-order law under explicit strengthened hypotheses on the node scale, the aggregate gap geometry, and the first-$N$ atom weights. No implication of these hypotheses from RH alone is asserted. Write

$$\Lambda_N := \sum_{1\le a<b\le N} \left[\log(t_b - t_a) - \log\frac{b-a}{\log t_b}\right]. \tag{13.78}$$

**Theorem 13.24** (Conditional second-order law)**.** *Assume the hypotheses of Theorem 13.2 together with the strengthened node scale*

$$\log t_j = \log j - \log\log j + O(1), \tag{13.79}$$

*the aggregate gap regularity hypothesis*

$$\Lambda_N = O(N^2), \tag{13.80}$$

*and the first-$N$ weight condition*

$$\sum_{j=1}^{N} \log w_j \geq -O(N^2). \tag{13.81}$$

*Then*

$$\log D_N[\mu] = -\beta N^2 \log N + \beta N^2 \log\log N + O(N^2). \tag{13.82}$$

*For the Li–Weil reflection measures, assume* RH *and, in addition, that the distinct positive ordinates satisfy*

$$\log \gamma_j^* = \log j - \log\log j + O(1) \tag{13.83}$$

*and that* (13.80) *holds with $t_j = \gamma_j^*$. Then*

$$\log \Delta_N^E,\ \log \Delta_N^O = -2N^2 \log N + 2N^2 \log\log N + O(N^2). \tag{13.84}$$

*No claim is made that* (13.83) *follows from* RH *alone.*

*Proof.* The upper bound is already available. Indeed $1 - x(t_j) \asymp t_j^{-\beta}$ and (13.79) give $\log\bigl(1 - x(t_j)\bigr) = -\beta \log j + \beta \log\log j + O(1)$, so the extremal-polynomial estimate used in Theorem 13.2 yields

$$\log h_n \leq -2\beta n \log n + 2\beta n \log\log n + O(n), \tag{13.85}$$

and summation over $n = 0, \dots, N-1$, together with $\sum_{n<N} n \log n = \frac{1}{2} N^2 \log N + O(N^2)$ and $\sum_{n<N} n \log\log n = \frac{1}{2} N^2 \log\log N + O(N^2)$, gives the upper bound in (13.82).

For the lower bound retain in the nonnegative Cauchy–Binet expansion the single term carried by the first $N$ atoms. By the lower derivative bound in (13.5),

$$\begin{aligned} x_b - x_a &= \int_{t_a}^{t_b} x'(u)\, \mathrm{d}u && (13.86)\\ &\geq c \int_{t_a}^{t_b} u^{-\beta-1}\, \mathrm{d}u && (13.87)\\ &\geq c\,(t_b - t_a) t_b^{-\beta-1}, \qquad 1 \leq a < b \leq N, && (13.88) \end{aligned}$$

because $u \leq t_b$ on $[t_a, t_b]$. Hence

$$2 \sum_{a<b} \log(x_b - x_a) \geq 2 \sum_{a<b} \log(t_b - t_a) - 2(\beta+1) \sum_{a<b} \log t_b + O(N^2). \tag{13.89}$$

By (13.80) and (13.79),

$$\sum_{a<b} \log(t_b - t_a) = \sum_{a<b} \log(b-a) - \sum_{a<b} \log\log t_b + O(N^2) \tag{13.90}$$

$$= \tfrac{1}{2} N^2 \log N - \tfrac{1}{2} N^2 \log\log N + O(N^2). \tag{13.91}$$

while (13.79) gives

$$\sum_{a<b} \log t_b = \tfrac{1}{2} N^2 \log N - \tfrac{1}{2} N^2 \log\log N + O(N^2). \tag{13.92}$$

Combining the preceding estimates gives

$$2 \sum_{a<b} \log(x_b - x_a) \geq -\beta N^2 \log N + \beta N^2 \log\log N - O(N^2). \tag{13.93}$$

Hypothesis (13.81) shows that the first-$N$ one-point weights can lower the logarithm of the retained Cauchy–Binet term by at most $O(N^2)$. Therefore

$$\log D_N[\mu] \geq -\beta N^2 \log N + \beta N^2 \log\log N - O(N^2), \tag{13.94}$$

which matches the upper bound.

For the Li–Weil measures, (13.83) and the explicit weights in (9.15)–(9.16) give

$$\sum_{j \leq N} \log w_E(\gamma_j^*), \quad \sum_{j \leq N} \log w_O(\gamma_j^*) \geq -O(N \log N), \tag{13.95}$$

because every zero multiplicity is at least one. This one-sided lower bound is all that (13.81) requires; no upper bound on multiplicities is assumed. The Chebyshev changes of basis contribute only $O(N^2)$ to the logarithm of the two reflection determinants. Thus the same second-order formula holds for $\Delta_N^E$ and $\Delta_N^O$ under the additional hypotheses stated above. □

*Remark* 13.25 (Where the gap hypothesis actually bites). Hypothesis (13.80) is an upper and a lower constraint, but only the lower one has content, and only for close pairs. On one side, a counting law of the form $N(T+h) - N(T) \ll h \log T + \log T$ forces $t_b - t_a \gg (b-a)/\log t_b$ whenever $b - a \gg \log t_N$, with no further hypothesis; the error term $\log T$ is what prevents the same deduction for smaller $b - a$. Hence (13.80) constrains only the $O(N \log N)$ pairs with $b - a \ll \log t_N$, and it permits those separations to be as small as $\exp(-cN/\log N)$ on logarithmic average. It is logically independent of simplicity, because the sequence $t_j$ contains distinct ordinates and carries no

multiplicity information, and it is far weaker than any conjectured pointwise lower bound on normalized gaps. What it is not is a theorem: no unconditional lower bound of this kind for consecutive ordinates is currently available, which is precisely why (13.82) is stated conditionally.

*Remark* 13.26 (Size of the Cauchy–Binet defect). The gap described in Remark 13.23 can be measured. With the same finite measure on both sides, namely the first 1200 ordinates, and $\log\det$ evaluated by Cholesky at 3000 decimal digits, the difference between $\log \Delta_N^E$ and the logarithm of the single Cauchy–Binet term carried by the first $N$ atoms is as follows.

| $N$ | $\log \Delta_N^E$ | single term | defect | defect$/\lvert\log \Delta_N^E\rvert$ |
|---|---|---|---|---|
| 2 | $-17.336$ | $-20.421$ | 3.085 | 0.178 |
| 4 | $-88.812$ | $-96.619$ | 7.807 | 0.088 |
| 8 | $-414.694$ | $-438.761$ | 24.067 | 0.058 |
| 12 | $-1002.853$ | $-1048.617$ | 45.765 | 0.046 |
| 16 | $-1869.011$ | $-1943.919$ | 74.908 | 0.040 |

Over the computed range the normalized defect decreases, which is consistent with a contribution of lower order relative to $N^2 \log N$; a power-law fit over these orders is close to $N^{3/2}$. No asymptotic exponent is claimed, and no growth rate is asserted beyond the tabulated orders: both statements would require an independent proof, and the positivity of the Cauchy–Binet expansion gives only defect $\geq 0$.

What the table does establish, without extrapolation, is that a single Cauchy–Binet term is already quantitatively inadequate at these orders. Consequently no statement below the $N^2 \log N$ scale should be extracted from one term without control of $\log \sum_J$. This is the quantitative form of the caveat in Remark 13.23.

*Remark* 13.27 (Zero logarithmic capacity). The compact support

$$\{1\} \cup \{x_\gamma : \gamma \in \Gamma_+^*\} \tag{13.96}$$

is countable and therefore has logarithmic capacity zero. This places the two measures outside the usual positive-capacity regular regime of orthogonal-polynomial asymptotics; see Stahl–Totik [36] and Totik's survey [42]. Zero carrier capacity alone does not impose a universal regular $n$th-root law for the leading coefficients; the geometry and the weights of a discrete measure remain decisive. Discrete orthogonal polynomials on infinite node sets with

an accumulation point form a classical subject in their own right; see [45] for a modern asymptotic treatment in a specific model. Under the change of variable $u = (1-x)^{-1}$, the endpoint law $1 - x(t) \asymp t^{-\beta}$ becomes $u \asymp t^{\beta}$, so Theorem 13.2 may also be viewed as a power-counting statement for a Stieltjes-type discrete moment problem. The comparison with earlier discrete Hankel asymptotics in Remark 13.13 is therefore useful: the present claim is not that $N^2 \log N$ decay is new, but that the Li–Weil coefficient is fixed by the Cayley endpoint exponent and can be proved from the coarse hypotheses above.

## 14. $LDL^T$, LU, Cholesky, and Schur complements

Set

$$\Delta_N^E := \det \mathsf{E}_N, \qquad \Delta_N^O := \det \mathsf{O}_N, \qquad \Delta_0^E = \Delta_0^O := 1. \tag{14.1}$$

**Theorem 14.1** ($LDL^T$ decomposition)**.** *Under* RH*, there are unique unit lower-triangular matrices $L_E, L_O$ and positive diagonal matrices $D_E, D_O$ such that*

$$\mathsf{E}_N = L_E D_E L_E^T, \qquad \mathsf{O}_N = L_O D_O L_O^T. \tag{14.2}$$

*Their diagonal pivots are*

$$(D_E)_{nn} = \frac{\Delta_n^E}{\Delta_{n-1}^E}, \tag{14.3}$$

$$(D_O)_{nn} = \frac{\Delta_n^O}{\Delta_{n-1}^O}. \tag{14.4}$$

*Proof.* We prove the statement for a generic real symmetric positive definite matrix $A_N$ whose leading principal blocks are denoted $A_n$, $1 \le n \le N$; the two reflection families then follow from Theorem 6.4. Set $A_0$ equal to the empty matrix and $\det A_0 := 1$.

We argue by induction on $N$. For $N = 1$, write $A_1 = (a_{11})$. Positive definiteness gives $a_{11} > 0$, and the factorization $A_1 = (1)(a_{11})(1)$ is unique. Assume the result for $A_{N-1}$ and write

$$A_N = \begin{pmatrix} A_{N-1} & r \\ r^T & \alpha \end{pmatrix}. \tag{14.5}$$

Since $A_N \succ 0$, its principal block $A_{N-1}$ is positive definite and hence invertible. By the induction hypothesis, $A_{N-1} = L_{N-1} D_{N-1} L_{N-1}^T$ with $L_{N-1}$ unit lower triangular and $D_{N-1}$ positive diagonal. Put

$$y := L_{N-1}^{-1} r, \qquad d_N := \alpha - r^T A_{N-1}^{-1} r, \tag{14.6}$$

and define

$$L_N := \begin{pmatrix} L_{N-1} & 0 \\ y^T D_{N-1}^{-1} & 1 \end{pmatrix}, \qquad D_N := \begin{pmatrix} D_{N-1} & 0 \\ 0 & d_N \end{pmatrix}. \tag{14.7}$$

A direct block multiplication gives $L_N D_N L_N^T = A_N$. Moreover,

$$d_N = \begin{pmatrix} -A_{N-1}^{-1} r \\ 1 \end{pmatrix}^T A_N \begin{pmatrix} -A_{N-1}^{-1} r \\ 1 \end{pmatrix} > 0, \tag{14.8}$$

so $D_N$ is positive diagonal. Uniqueness follows recursively: in any factorization $A_N = L D L^T$ with $L$ unit lower triangular and $D$ diagonal, the leading $(N-1) \times (N-1)$ block fixes $L_{N-1}, D_{N-1}$ by induction; then the last column below the diagonal is fixed by $r = L_{N-1} D_{N-1} \ell$, and the last diagonal entry is fixed by the $(N, N)$ entry.

Finally, $\det L_n = 1$, so

$$\det A_n = \det D_n = \prod_{j=1}^{n} d_j. \tag{14.9}$$

Dividing the identities for $n$ and $n-1$ yields $d_n = \det A_n / \det A_{n-1}$. Taking $A_n = \mathsf{E}_n$ and $A_n = \mathsf{O}_n$ proves (14.3)–(14.4). $\square$

The first pivots are

$$d_{E,1} = \lambda_2, \tag{14.10}$$

$$d_{E,2} = \lambda_4 - \frac{(\lambda_3 - \lambda_1)^2}{\lambda_2}, \tag{14.11}$$

$$d_{O,1} = \lambda_1, \tag{14.12}$$

$$d_{O,2} = \lambda_3 - \frac{(\lambda_2 - \lambda_1)^2}{\lambda_1}. \tag{14.13}$$

The ordinary LU factorization with unit diagonal in the lower factor is

$$\mathsf{E}_N = L_E U_E, \qquad U_E = D_E L_E^T, \tag{14.14}$$

and similarly

$$\mathsf{O}_N = L_O U_O, \qquad U_O = D_O L_O^T. \tag{14.15}$$

Thus the diagonal of $U_E$ and $U_O$ is exactly the pivot sequence in (14.3)–(14.4).

The Cholesky factors are

$$\mathsf{E}_N = R_E R_E^T, \qquad R_E = L_E D_E^{1/2}, \tag{14.16}$$

and

$$\mathsf{O}_N = R_O R_O^T, \qquad R_O = L_O D_O^{1/2}. \tag{14.17}$$

In particular,

$$(R_E)_{nn} = \sqrt{\frac{\Delta_n^E}{\Delta_{n-1}^E}}, \tag{14.18}$$

$$(R_O)_{nn} = \sqrt{\frac{\Delta_n^O}{\Delta_{n-1}^O}}. \tag{14.19}$$

**Proposition 14.2** (Schur-complement recursion)**.** *Write*

$$\mathsf{E}_N = \begin{pmatrix} \mathsf{E}_{N-1} & r_N^E \\ (r_N^E)^T & \lambda_{2N} \end{pmatrix}, \qquad \mathsf{O}_N = \begin{pmatrix} \mathsf{O}_{N-1} & r_N^O \\ (r_N^O)^T & \lambda_{2N-1} \end{pmatrix}. \tag{14.20}$$

*Then under* RH*,*

$$\frac{\Delta_N^E}{\Delta_{N-1}^E} = \lambda_{2N} - (r_N^E)^T \mathsf{E}_{N-1}^{-1} r_N^E > 0, \tag{14.21}$$

$$\frac{\Delta_N^O}{\Delta_{N-1}^O} = \lambda_{2N-1} - (r_N^O)^T \mathsf{O}_{N-1}^{-1} r_N^O > 0. \tag{14.22}$$

*Proof.* We again treat a generic block matrix

$$A_N = \begin{pmatrix} A & r \\ r^T & \alpha \end{pmatrix}, \tag{14.23}$$

where $A = A_{N-1}$ is positive definite. In particular $A$ is invertible. Consider the two block triangular matrices

$$L := \begin{pmatrix} I & 0 \\ -r^T A^{-1} & 1 \end{pmatrix}, \qquad R := \begin{pmatrix} I & -A^{-1} r \\ 0 & 1 \end{pmatrix}. \tag{14.24}$$

Both have determinant one. Direct multiplication gives

$$LA_NR = \begin{pmatrix} A & 0 \\ 0 & \alpha - r^T A^{-1} r \end{pmatrix}. \tag{14.25}$$

Taking determinants therefore yields

$$\det A_N = \det A \left(\alpha - r^T A^{-1} r\right), \tag{14.26}$$

and, since $\det A > 0$,

$$\frac{\det A_N}{\det A} = \alpha - r^T A^{-1} r. \tag{14.27}$$

The scalar on the right is strictly positive because the nonzero vector

$$z := \begin{pmatrix} -A^{-1} r \\ 1 \end{pmatrix} \tag{14.28}$$

satisfies

$$z^T A_N z = \alpha - r^T A^{-1} r > 0 \tag{14.29}$$

by positive definiteness of $A_N$. Substituting $A = \mathsf{E}_{N-1}$, $r = r_N^E$, $\alpha = \lambda_{2N}$ and then $A = \mathsf{O}_{N-1}$, $r = r_N^O$, $\alpha = \lambda_{2N-1}$ proves both formulas and their strict positivity. □

*Remark* 14.3 (Geometric meaning). The quantities in (14.21) and (14.22) are the squared residual norms after orthogonally projecting the new sine mode onto the span of all preceding modes. They quantify the genuinely new Gram energy contributed by frequency $N$ or $N - 1/2$.

## 15. Inverse matrices and reproducing kernels

Whenever the matrices are invertible, the adjugate formula gives

$$(\mathsf{E}_N^{-1})_{ij} = (-1)^{i+j} \frac{\det \mathsf{E}_N^{(j,i)}}{\Delta_N^E}, \tag{15.1}$$

$$(\mathsf{O}_N^{-1})_{ij} = (-1)^{i+j} \frac{\det \mathsf{O}_N^{(j,i)}}{\Delta_N^O}. \tag{15.2}$$

More structure is visible after the moment-basis transformation.

Let

$$u_N(x) := (U_0(x), \dots, U_{N-1}(x))^T, \tag{15.3}$$

$$\mathbf{p}_N(x) := (P_0(x), \dots, P_{N-1}(x))^T. \tag{15.4}$$

Define

$$\mathcal{K}_N^E(x,y) := u_N(x)^T \mathsf{E}_N^{-1} u_N(y), \tag{15.5}$$

$$\mathcal{K}_N^O(x,y) := \mathbf{p}_N(x)^T \mathsf{O}_N^{-1} \mathbf{p}_N(y). \tag{15.6}$$

**Theorem 15.1** (Christoffel–Darboux interpretation)**.** *Under* RH*, $\mathcal{K}_N^E$ and $\mathcal{K}_N^O$ are the reproducing kernels of the polynomial spaces of degree $< N$ in $L^2(\mu_E)$ and $L^2(\mu_O)$, respectively. If $\{q_j^E\}$ and $\{q_j^O\}$ are the corresponding orthonormal polynomials, then*

$$\mathcal{K}_N^E(x,y) = \sum_{j=0}^{N-1} q_j^E(x) q_j^E(y), \tag{15.7}$$

$$\mathcal{K}_N^O(x,y) = \sum_{j=0}^{N-1} q_j^O(x) q_j^O(y). \tag{15.8}$$

*Proof.* We prove the even statement. Let

$$\mathcal{P}_{N-1} := \operatorname{span}\{U_0, \dots, U_{N-1}\} \tag{15.9}$$

inside $L^2(\mu_E)$. Under RH, $\mathsf{E}_N \succ 0$, so these basis functions are linearly independent in $L^2(\mu_E)$ and their Gram matrix is invertible. For fixed $y$, define

$$K_y(x) := u_N(x)^T \mathsf{E}_N^{-1} u_N(y) \in \mathcal{P}_{N-1}. \tag{15.10}$$

Take an arbitrary $f \in \mathcal{P}_{N-1}$ and write $f(x) = c^T u_N(x)$ with $c \in \mathbb{R}^N$. Then, using the moment Gram identity,

$$\langle f, K_y \rangle_{L^2(\mu_E)} = \int c^T u_N(x) u_N(x)^T \mathsf{E}_N^{-1} u_N(y) \, \mathrm{d}\mu_E(x) \tag{15.11}$$

$$= c^T \left( \int u_N(x) u_N(x)^T \, \mathrm{d}\mu_E(x) \right) \mathsf{E}_N^{-1} u_N(y) \tag{15.12}$$

$$= c^T \mathsf{E}_N \mathsf{E}_N^{-1} u_N(y) = f(y). \tag{15.13}$$

Thus $K_y$ has the reproducing property and is the unique reproducing kernel section at $y$. This proves (15.5) as a reproducing kernel; the odd case is identical with $p_N$ and $\mathsf{O}_N$.

Now let $q_0^E, \dots, q_{N-1}^E$ be any orthonormal basis of $\mathcal{P}_{N-1}$. The function

$$\widetilde{K}_N^E(x,y) := \sum_{j=0}^{N-1} q_j^E(x) q_j^E(y) \tag{15.14}$$

has the same reproducing property, because for $f = \sum_j \widehat{f}_j q_j^E$ one has $\langle f, \widetilde{K}_N^E(\cdot, y)\rangle = \sum_j \widehat{f}_j q_j^E(y) = f(y)$. Uniqueness of the Riesz representer in the finite-dimensional Hilbert space therefore gives $\widetilde{K}_N^E = \mathcal{K}_N^E$. This is (15.7); the odd formula follows in the same way. □

In the monomial basis, Theorem 10.1 gives the exact inverse identities

$$\mathsf{E}_N^{-1} = (B_E^{(N)})^{-T} (H_N^E)^{-1} (B_E^{(N)})^{-1}, \tag{15.15}$$

$$\mathsf{O}_N^{-1} = (B_O^{(N)})^{-T} (H_N^O)^{-1} (B_O^{(N)})^{-1}. \tag{15.16}$$

Together with Corollary 8.4, these formulas show that the inverses simultaneously possess moment-kernel and low-displacement-rank descriptions.

*Remark* 15.2 (No universal sign pattern). The off-diagonal entries $\lambda_{m+n} - \lambda_{|m-n|}$ and $\lambda_{m+n-1} - \lambda_{|m-n|}$ do not have an algebraically forced sign. Consequently the matrices are not naturally Stieltjes $M$-matrices, and no universal checkerboard or nonpositive off-diagonal sign pattern for their inverses follows from the present theory.

## 16. Column recursion and the closed two-generator bordering scheme

The rank-two displacement and nesting identities permit the bordering recurrence to be closed algebraically in a finite state. Throughout, $\lambda_0 = 0$ and $\lambda_{-j} = \lambda_j$.

**Definition 16.1** (Bordering column)**.** For $N \geq 1$ let $\mathsf{b}^{(N)}$ be the semi-infinite vector

$$\mathsf{b}_m^{(N)} := \lambda_{m+N} - \lambda_{|m-N|}, \qquad m \geq 1. \tag{16.1}$$

Its first $N-1$ entries form the off-diagonal block of $\mathsf{E}_N$ in (14.20), and $\mathsf{b}_N^{(N)} = \lambda_{2N}$.

**Proposition 16.2** (Column wave equation)**.** *Let $T$ denote the formal nearest-neighbour shift on semi-infinite sequences with Dirichlet condition at 0, that is $(Tv)_m = v_{m+1} + v_{m-1}$ with $v_0 := 0$. No $\ell^2$ assumption is made on the bordering columns. Then for every $N \geq 2$,*

$$\boxed{\mathsf{b}^{(N+1)} + \mathsf{b}^{(N-1)} = T\,\mathsf{b}^{(N)}} \tag{16.2}$$

*as an identity between semi-infinite vectors, valid unconditionally. Equivalently,*

$$\mathsf{b}^{(N)} = U_{N-1}\Big(\tfrac{T}{2}\Big)\,\mathsf{b}^{(1)}, \tag{16.3}$$

*where $U_n$ is the Chebyshev polynomial of the second kind.*

*Proof.* Fix $m \geq 1$. The left-hand side of (16.2) at index $m$ is

$$\Big(\lambda_{m+N+1} - \lambda_{|m-N-1|}\Big) + \Big(\lambda_{m+N-1} - \lambda_{|m-N+1|}\Big), \tag{16.4}$$

and the right-hand side is

$$\Big(\lambda_{m+1+N} - \lambda_{|m+1-N|}\Big) + \Big(\lambda_{m-1+N} - \lambda_{|m-1-N|}\Big), \tag{16.5}$$

where the convention $v_0 = 0$ is automatic because $\mathsf{b}_0^{(N)} = \lambda_N - \lambda_N = 0$. The two positive parts agree termwise. The two negative parts agree as well, since $|m-N-1| = |m-1-N|$ and $|m-N+1| = |m+1-N|$. This proves (16.2). Since $\mathsf{b}^{(0)} = 0$, the recursion $\mathsf{b}^{(N+1)} = T\mathsf{b}^{(N)} - \mathsf{b}^{(N-1)}$ with initial data $\mathsf{b}^{(0)} = 0$, $\mathsf{b}^{(1)}$ is exactly the Chebyshev recursion, giving (16.3). □

*Remark* 16.3 (Interpretation). Under RH the identity (16.2) is the statement $\sin((N+1)\theta) + \sin((N-1)\theta) = 2\cos\theta\,\sin(N\theta)$ transported through the Gram representation (6.11). The content of Proposition 16.2 is that it holds *unconditionally*, as an identity in the Li coefficients: the same discrete wave equation that Section 8 exhibits in the row index also governs the sequence of bordering columns. The pair $\big(\mathsf{b}^{(1)}, \mathsf{b}^{(2)}\big)$ therefore generates the entire semi-infinite array, which is the sense in which the scheme below is a *two-generator* scheme.

**Proposition 16.4** (Closed bordering recursion)**.** *Assume $\mathsf{E}_{N-1}$ is invertible and set $G := \mathsf{E}_{N-1}^{-1}$. Write $e := e_{N-1} \in \mathbb{R}^{N-1}$ for the last standard basis vector, let*

$$\omega_N := \lambda_{2N-1} - \lambda_1, \qquad r^{(N)} := \Big(\mathsf{b}_1^{(N)}, \ldots, \mathsf{b}_{N-2}^{(N)}, 0\Big)^T, \tag{16.6}$$

*so that the off-diagonal block of* $\mathsf{E}_N$ *is*

$$b_N = r^{(N)} + \omega_N\, e. \tag{16.7}$$

*Put* $g := Ge$, $h := Gr^{(N)}$ *and* $q := h + \omega_N g$. *Then the next Schur pivot is*

$$p_N := \frac{\Delta_N^E}{\Delta_{N-1}^E} = \lambda_{2N} - \left(r^{(N)}\right)^T h - 2\omega_N h_{N-1} - \omega_N^2 g_{N-1}. \tag{16.8}$$

*If, in addition,* $p_N \neq 0$*—equivalently,* $\mathsf{E}_N$ *is invertible—then*

$$\mathsf{E}_N^{-1} = \begin{pmatrix} G + p_N^{-1} q q^T & -p_N^{-1} q \\ -p_N^{-1} q^T & p_N^{-1} \end{pmatrix}, \qquad g^{(N)} := \mathsf{E}_N^{-1} e_N = \begin{pmatrix} -q/p_N \\ 1/p_N \end{pmatrix}. \tag{16.9}$$

*In particular* $g_N^{(N)}\, p_N = 1$ *whenever the inverse exists.*

*Proof.* Equation (16.7) is the definition of $r^{(N)}$ together with $\mathsf{b}_{N-1}^{(N)} = \lambda_{2N-1} - \lambda_1 = \omega_N$. Since $G$ is symmetric,

$$b_N^T G b_N = \left(r^{(N)} + \omega_N e\right)^T G \left(r^{(N)} + \omega_N e\right) = \left(r^{(N)}\right)^T h + 2\omega_N\, e^T h + \omega_N^2\, e^T g, \tag{16.10}$$

and $e^T h = h_{N-1}$, $e^T g = g_{N-1}$. Substituting into the Schur complement (14.21) gives (16.8). Formula (16.9) is the standard block inverse of $\left(\begin{smallmatrix} \mathsf{E}_{N-1} & b_N \\ b_N^T & \lambda_{2N} \end{smallmatrix}\right)$ with $q = Gb_N = h + \omega_N g$, which is valid whenever $p_N \neq 0$. □

**Proposition 16.5** (Closed two-vector state update)**.** *Under the hypotheses and notation of Proposition 16.4, assume in addition that* $p_N \neq 0$*, so that* $\mathsf{E}_N^{-1}$ *exists, and let*

$$\rho_N := \left(\mathsf{b}_1^{(N+1)}, \ldots, \mathsf{b}_{N-1}^{(N+1)}\right)^T \in \mathbb{R}^{N-1}. \tag{16.11}$$

*Then the column wave equation gives*

$$\boxed{\rho_N = 2\mathsf{J}_{E,\mathrm{ref}}^{(N-1)} b_N - \mathsf{E}_{N-1} e + \lambda_{2N} e.} \tag{16.12}$$

*If* $q = h + \omega_N g = \mathsf{E}_{N-1}^{-1} b_N$ *and*

$$\eta_N := -\frac{q^T \rho_N}{p_N}, \tag{16.13}$$

*then the second inverse generator*

$$h^{(N)} := \mathsf{E}_N^{-1} r_E^{(N)}, \qquad r_E^{(N)} = (\rho_N^T, 0)^T, \tag{16.14}$$

*is obtained without a new inverse column:*

$$\boxed{h^{(N)} = \begin{pmatrix} 2\mathsf{J}_{E,\mathrm{ref}}^{(N-1)} q + (q_{N-1} - \eta_N) q + p_N g - e \\ \eta_N \end{pmatrix}.} \tag{16.15}$$

*Together with* (16.9)*, equations* (16.8) *and* (16.15) *close the inverse information needed at the next step on the pair* $(g^{(N)}, h^{(N)})$ *and the bordering columns.*

*Proof.* The first identity is the finite restriction of $\mathsf{b}^{(N+1)} + \mathsf{b}^{(N-1)} = T\mathsf{b}^{(N)}$ at the new boundary; the missing $N$-th coupling contributes the last two terms in (16.12). For completeness, the inverse displacement identity gives

$$\mathsf{J}_{E,\mathrm{ref}}^{(N-1)} G - G \mathsf{J}_{E,\mathrm{ref}}^{(N-1)} = \frac{1}{2}(g h^T - h g^T). \tag{16.16}$$

Apply $G$ to (16.12), substitute $q = G b_N = h + \omega_N g$, and use $p_N = \lambda_{2N} - b_N^T q$. The scalar terms combine to give

$$G\rho_N = 2\mathsf{J}_{E,\mathrm{ref}}^{(N-1)} q + q_{N-1} q + p_N g - e. \tag{16.17}$$

Finally multiply $r_E^{(N)} = (\rho_N^T, 0)^T$ by the block inverse (16.9). Its last component is $-q^T \rho_N / p_N = \eta_N$, and its top block is $G\rho_N - \eta_N q$. Equation (16.15) follows. □

**Lemma 16.6** (Interior tridiagonality after elimination)**.** *Let* $A_{n+2} = L_{n+2} D_{n+2} L_{n+2}^T$ *be a leading reflection block, where* $L_{n+2}$ *is unit lower triangular and* $p_1, \ldots, p_{n+1} \neq 0$*. With*

$$B = L_{n+2}^{-1} J_{\mathrm{ref}}^{(n+2)} L_{n+2},$$

*the leading* $(n+1) \times (n+1)$ *block of* $B$ *is tridiagonal and satisfies*

$$B_{ij} p_j = p_i B_{ji}, \qquad 1 \le i, j \le n+1. \tag{16.18}$$

*Proof.* The displacement identity of Theorem 8.3 is supported only in the last row and column. Since $L_{n+2}^{-1} e_{n+2} = e_{n+2}$, conjugation by $L_{n+2}^{-1}$ and $L_{n+2}^{-T}$ gives (16.18). Moreover, $J_{\mathrm{ref}}^{(n+2)} L_{n+2}$ is upper Hessenberg, and left multiplication by $L_{n+2}^{-1}$ preserves that property. Thus $B_{ij} = 0$ for $j > i + 1$; the displayed weighted symmetry and the nonvanishing pivots give the transposed zeros. This proves interior tridiagonality. □

**Lemma 16.7** (Entry computation in the transformed column)**.** *Under the hypotheses of Lemma 16.6, set* $\ell_n = (L_{n+1})_{n+1,n}$, $\tau_n = (L_{n+2})_{n+2,n}$, *and* $\ell_0 = \sigma$, *where* $\sigma = 0$ *for the even family and* $\sigma = 1$ *for the odd family. Then*

$$2B_{n-1,n} = 1, \qquad 2B_{nn} = \ell_n - \ell_{n-1}, \qquad 2B_{n+1,n} = \frac{p_{n+1}}{p_n}. \tag{16.19}$$

*Proof.* Read $J_{\mathrm{ref}}^{(n+2)} L_{n+2} = L_{n+2} B$ at entries $(n-1, n)$ and $(n, n)$. The first gives $2B_{n-1,n} = 1$, while the second gives $2B_{nn} = \ell_n - \ell_{n-1}$; the convention $\ell_0 = \sigma$ absorbs the odd upper-left entry when $n = 1$. Finally $B_{n,n+1} = 1/2$, so (16.18) gives the last identity. □

**Theorem 16.8** (Local $LDL^T$ pivot dynamics)**.** *Let $A$ denote either the even or the odd semi-infinite reflection array, and suppose $p_1, \dots, p_n$ are nonzero. Gaussian $LDL^T$ elimination is then well defined through column $n$; in particular the leading block $A_{n+1}$ has the factorization*

$$A_{n+1} = L_{n+1} \operatorname{diag}(p_1, \dots, p_{n+1})\, L_{n+1}^T, \qquad (L_{n+1})_{jj} = 1, \tag{16.20}$$

*where $p_{n+1}$ is allowed to vanish. The $n$th elimination column in the next leading block is also defined because its construction divides only by $p_1, \dots, p_n$. Define the first two subdiagonals*

$$\ell_n := (L_{n+1})_{n+1,n}, \qquad \tau_n := (L_{n+2})_{n+2,n}, \qquad n \geq 1. \tag{16.21}$$

*Set*

$$\ell_0 = \sigma, \qquad \tau_0 = 0, \qquad \sigma = \begin{cases} 0, & A = \mathsf{E}, \\ 1, & A = \mathsf{O}. \end{cases} \tag{16.22}$$

*Then, for every $n \geq 1$,*

$$\boxed{\frac{p_{n+1}}{p_n} = 1 + (\tau_n - \tau_{n-1}) - \ell_n(\ell_n - \ell_{n-1}).} \tag{16.23}$$

*Equivalently, if*

$$\mathcal{H}_n := \tau_n + n - \frac{1}{2}\ell_n^2 - \frac{1}{2}\sum_{j=1}^{n}(\ell_j - \ell_{j-1})^2, \qquad \mathcal{H}_0 = -\frac{1}{2}\sigma^2, \tag{16.24}$$

*then*

$$\boxed{\mathcal{H}_n - \mathcal{H}_{n-1} = \frac{p_{n+1}}{p_n}.} \tag{16.25}$$

*Thus, once $p_n > 0$ is known, positivity of the next pivot is equivalent to the strict increase $\mathcal{H}_n > \mathcal{H}_{n-1}$.*

*Proof.* We first treat the case $p_{n+1} \neq 0$. All arguments are finite. Take the leading block $A_{n+2}$ and its elimination factorization

$$A_{n+2} = L_{n+2}D_{n+2}L_{n+2}^T, \qquad D_{n+2} = \operatorname{diag}(p_1, \ldots, p_{n+2}).$$

The last pivot $p_{n+2}$ is allowed to vanish; construction of $L_{n+2}$ requires only $p_1, \ldots, p_{n+1} \neq 0$. Let $J_{\mathrm{ref}}^{(n+2)}$ denote $\mathsf{J}_{E,\mathrm{ref}}^{(n+2)}$ or $\mathsf{J}_{O,\mathrm{ref}}^{(n+2)}$, according to the family, and define

$$B := L_{n+2}^{-1}J_{\mathrm{ref}}^{(n+2)}L_{n+2}.$$

Lemma 16.6 shows that the leading $(n+1) \times (n+1)$ block of $B$ is tridiagonal. The underlying intertwining identity is

$$J_{\mathrm{ref}}^{(n+2)}L_{n+2} = L_{n+2}B. \tag{16.26}$$

Lemma 16.7 gives its three relevant entries in (16.19).

Read (16.26) now at entry $(n+1, n)$. Its left-hand side is

$$\frac{1}{2}L_{n,n} + \frac{1}{2}L_{n+2,n} = \frac{1}{2}(1 + \tau_n),$$

where the odd boundary modification has no effect because $n + 1 > 1$. The tridiagonal form of $B$ and (16.19) give for the right-hand side

$$\frac{1}{2}\tau_{n-1} + \frac{1}{2}\ell_n(\ell_n - \ell_{n-1}) + \frac{1}{2}\frac{p_{n+1}}{p_n}.$$

Multiplication by 2 and rearrangement yield

$$\frac{p_{n+1}}{p_n} = 1 + (\tau_n - \tau_{n-1}) - \ell_n(\ell_n - \ell_{n-1}),$$

which is (16.23).

If $p_{n+1} = 0$, perturb only the next diagonal datum of the abstract reflection sequence: $\lambda_{2n+2}$ in the even family and $\lambda_{2n+1}$ in the odd family. The previous pivots $p_1, \dots, p_n$ are unchanged, while $p_{n+1}$ is an affine nonconstant function of this new diagonal entry and is nonzero away from one parameter value. Every coefficient $\ell_j, \tau_j$ appearing in (16.23) is a rational function of the reflection entries with denominators involving only $p_1, \dots, p_n$, so both sides vary continuously as the perturbation is removed. The identity therefore extends to $p_{n+1} = 0$.

Finally,

$$\ell_n(\ell_n - \ell_{n-1}) = \frac{1}{2}(\ell_n^2 - \ell_{n-1}^2) + \frac{1}{2}(\ell_n - \ell_{n-1})^2.$$

Substitution into (16.23) and telescoping gives (16.25). □

**Corollary 16.9** (Defect-energy form)**.** *With $\sigma$ as in Theorem 16.8, define*

$$d_n := 2n + \sigma - \ell_n, \tag{16.27}$$

$$c_n := 2n^2 + (1 + 2\sigma)n - \tau_n, \tag{16.28}$$

$$V_n := (2n + \sigma + 2)d_n - \frac{1}{2}d_n^2 - c_n, \qquad d_0 = c_0 = V_0 = 0. \tag{16.29}$$

*Then*

$$\boxed{\frac{p_{n+1}}{p_n} = V_n - V_{n-1} - \frac{1}{2}(d_n - d_{n-1})^2.} \tag{16.30}$$

*In particular, under the inductive hypothesis $p_1, \dots, p_n > 0$,*

$$p_{n+1} > 0 \quad \Longleftrightarrow \quad V_n - V_{n-1} > \frac{1}{2}(d_n - d_{n-1})^2. \tag{16.31}$$

*Equivalently, the symmetric cell*

$$\mathcal{C}_n = \begin{pmatrix} V_n - V_{n-1} & (d_n - d_{n-1})/\sqrt{2} \\ (d_n - d_{n-1})/\sqrt{2} & 1 \end{pmatrix} \tag{16.32}$$

*satisfies*

$$\det \mathcal{C}_n = \frac{p_{n+1}}{p_n}. \tag{16.33}$$

*Thus the next reflection pivot is controlled by a $2 \times 2$ local cell built only from the first two subdiagonals of the $LDL^T$ elimination factor.*

*Proof.* Substitute the definitions (16.27) and (16.28) into (16.23), and simplify. Equation (16.33) is the Schur-complement identity for the lower-right entry 1. □

**Corollary 16.10** (Cost)**.** *Suppose the inverse-producing steps under consideration have nonzero Schur pivots. Given* $\mathsf{E}_{N-1}^{-1}$ *and* $\mathsf{b}^{(N-1)}, \mathsf{b}^{(N-2)}$*, one step of Propositions 16.2 and 16.4 produces* $\mathsf{b}^{(N)}$*,* $p_N$*,* $g^{(N)}$ *and* $\mathsf{E}_N^{-1}$ *using one banded matrix–vector product and two dense matrix–vector products, that is* $O(N^2)$ *operations and no determinant evaluation. In particular the whole pivot sequence* $p_1, \dots, p_N$ *is obtained in* $O(N^3)$ *operations without ever forming* $\Delta_N^E$*.*

*Proof.* The column-wave update is banded and costs $O(N)$. Forming the required inverse generator vectors uses dense matrix–vector products with an $(N-1)\times(N-1)$ matrix and therefore costs $O(N^2)$; the rank-one block-inverse update also costs $O(N^2)$. Thus one nonsingular bordering step is $O(N^2)$, and summing these costs through order $N$ gives $O(N^3)$. No determinant evaluation is required, because $p_n$ is obtained directly from the Schur formula. □

## 17. Orthogonal polynomials and infinite Jacobi operators

Assume RH. Let $\pi_n^E$ and $\pi_n^O$ denote the monic orthogonal polynomials associated with $\mu_E$ and $\mu_O$. The following elementary argument records the hypotheses behind their three-term recurrences. For either positive measure $\mu = \mu_E$ or $\mu_O$, let $h_n = \|\pi_n\|_{L^2(\mu)}^2 > 0$. Since $x\pi_n$ is monic of degree $n+1$, it has an expansion

$$x\pi_n = \pi_{n+1} + \sum_{j=0}^{n} c_{n,j}\pi_j. \tag{17.1}$$

For $j \le n-2$, symmetry of multiplication by the real variable $x$ gives

$$\langle x\pi_n, \pi_j\rangle = \langle \pi_n, x\pi_j\rangle = 0, \tag{17.2}$$

because $x\pi_j$ has degree at most $n-1$. Thus only the $j = n$ and $j = n-1$ coefficients can remain. Moreover

$$c_{n,n-1} = \frac{\langle x\pi_n, \pi_{n-1}\rangle}{h_{n-1}} = \frac{\langle \pi_n, x\pi_{n-1}\rangle}{h_{n-1}} = \frac{h_n}{h_{n-1}} > 0, \tag{17.3}$$

because $x\pi_{n-1} = \pi_n +$ lower-degree terms. Writing $b_n := c_{n,n}$ and $(a_n)^2 := h_n/h_{n-1}$ gives

$$x\pi_n^E(x) = \pi_{n+1}^E(x) + b_n^E \pi_n^E(x) + (a_n^E)^2 \pi_{n-1}^E(x), \tag{17.4}$$

$$x\pi_n^O(x) = \pi_{n+1}^O(x) + b_n^O \pi_n^O(x) + (a_n^O)^2 \pi_{n-1}^O(x), \tag{17.5}$$

with $a_n^E, a_n^O > 0$ for $n \geq 1$ and the convention $\pi_{-1} = 0$. After normalizing $q_n = \pi_n/\sqrt{h_n}$, multiplication by $x$ has the symmetric tridiagonal matrix with diagonal $b_n$ and positive off-diagonal $a_{n+1}$. Because the measures are supported in $[-1, 1]$, multiplication by $x$ is a bounded self-adjoint operator of norm at most one on each $L^2$ space; once polynomial completeness is established below, the corresponding Jacobi matrices define bounded self-adjoint operators $\mathcal{J}_E$ and $\mathcal{J}_O$ unitarily equivalent to multiplication by $x$.

The first recurrence coefficients are explicit in the Li coefficients:

$$b_0^E = \frac{\lambda_3 - \lambda_1}{2\lambda_2}, \tag{17.6}$$

$$(a_1^E)^2 = \frac{\lambda_2\lambda_4 - (\lambda_3 - \lambda_1)^2}{4\lambda_2^2}, \tag{17.7}$$

$$b_0^O = \frac{\lambda_2 - 2\lambda_1}{2\lambda_1}, \tag{17.8}$$

$$(a_1^O)^2 = \frac{\lambda_1\lambda_3 - (\lambda_2 - \lambda_1)^2}{4\lambda_1^2}. \tag{17.9}$$

**Proposition 17.1** (Jacobi off-diagonals from the principal determinants)**.** *Let*

$$d_n^E := \frac{\Delta_n^E}{\Delta_{n-1}^E}, \qquad d_n^O := \frac{\Delta_n^O}{\Delta_{n-1}^O}. \tag{17.10}$$

*Under* RH*, for every* $n \geq 1$*,*

$$(a_n^E)^2 = \frac{1}{4}\frac{d_{n+1}^E}{d_n^E} = \frac{1}{4}\frac{\Delta_{n+1}^E \Delta_{n-1}^E}{(\Delta_n^E)^2}, \tag{17.11}$$

$$(a_n^O)^2 = \frac{1}{4}\frac{d_{n+1}^O}{d_n^O} = \frac{1}{4}\frac{\Delta_{n+1}^O \Delta_{n-1}^O}{(\Delta_n^O)^2}. \tag{17.12}$$

*Proof.* The basis polynomial $U_{n-1}$, and also $P_{n-1}$, has leading coefficient $2^{n-1}$. Therefore the squared norm $h_{n-1}$ of the corresponding monic orthogonal polynomial is

$$h_{n-1}^E = \frac{d_n^E}{4^{n-1}}, \qquad h_{n-1}^O = \frac{d_n^O}{4^{n-1}}. \tag{17.13}$$

For monic orthogonal polynomials $(a_n)^2 = h_n/h_{n-1}$, which yields (17.11)–(17.12). □

Let $\Gamma_+^*$ denote the set of *distinct* positive ordinates of nontrivial zeros and let $m_\gamma$ denote multiplicity.

The measures $\mu_E$ and $\mu_O$ are convenient unnormalized representing measures. The spectral measure of the first Jacobi basis vector is instead the corresponding probability measure

$$\nu_E := \frac{\mu_E}{\lambda_2}, \qquad \nu_O := \frac{\mu_O}{\lambda_1}, \tag{17.14}$$

because $\mu_E([-1,1]) = \lambda_2$ and $\mu_O([-1,1]) = \lambda_1$. This normalization does not alter the orthogonal polynomials except for their orthonormal scaling, and it does not alter the associated Jacobi operator.

**Theorem 17.2** (Exact spectrum of the infinite Jacobi operators)**.** *Assume* RH. *Then*

$$\operatorname{spec}(\mathcal{J}_E) = \operatorname{spec}(\mathcal{J}_O) = \{1\} \cup \{x_\gamma : \gamma \in \Gamma_+^*\}, \tag{17.15}$$

*where*

$$x_\gamma = \frac{4\gamma^2 - 1}{4\gamma^2 + 1}. \tag{17.16}$$

*Each $x_\gamma$ is an isolated eigenvalue of the multiplication operator with spectral mass determined by $m_\gamma$, while 1 is the unique accumulation point of the spectrum. The inverse transformation is*

$$\gamma = \frac{1}{2}\sqrt{\frac{1+x}{1-x}} \qquad (-1 < x < 1). \tag{17.17}$$

*Moreover,*

$$I - \mathcal{J}_E \in \mathcal{S}_1, \qquad I - \mathcal{J}_O \in \mathcal{S}_1, \tag{17.18}$$

*so both are trace-class perturbations of the identity and have essential spectrum $\{1\}$.*

*Proof.* We treat either measure $\mu = \mu_E$ or $\mu_O$; the support is the same in both cases. Every atom has positive mass, and $x_\gamma \to 1$ as $\gamma \to \infty$. Hence the topological support is exactly

$$K := \operatorname{supp} \mu = \{x_\gamma : \gamma \in \Gamma_+^*\} \cup \{1\}. \tag{17.19}$$

There is no atom at 1; it is present because it is the unique accumulation point of the atoms.

Because $K$ is a compact subset of $\mathbb{R}$, the restrictions of algebraic polynomials form a unital subalgebra of $C(K)$ that separates points. The Stone–Weierstrass theorem therefore makes polynomials uniformly dense in $C(K)$. Continuous functions are dense in $L^2(\mu)$ for every finite Borel measure on a compact metric space, hence polynomials are dense in $L^2(\mu)$. Gram–Schmidt applied to $1, x, x^2, \dots$ therefore produces a complete orthonormal polynomial basis $\{p_n\}_{n\geq 0}$.

Let $\mathcal{F} : \ell^2(\mathbb{N}_0) \to L^2(\mu)$ send the standard vector $e_n$ to $p_n$. Completeness makes $\mathcal{F}$ unitary. Multiplication by $x$ is a bounded self-adjoint operator $M_x$ with $\|M_x\| \leq 1$. Its matrix in the orthonormal polynomial basis is the Jacobi matrix $\mathcal{J}$ coming from the three-term recurrence, so

$$\mathcal{J} = \mathcal{F}^{-1} M_x \mathcal{F}. \tag{17.20}$$

Now identify $L^2(\mu)$ explicitly with a weighted sequence space. If $w_\gamma = \mu(\{x_\gamma\})$, then

$$Tf := \left(\sqrt{w_\gamma}\, f(x_\gamma)\right)_{\gamma \in \Gamma_+^*} \tag{17.21}$$

is unitary from $L^2(\mu)$ onto $\ell^2(\Gamma_+^*)$, and

$$T M_x T^{-1} = \operatorname{diag}(x_\gamma). \tag{17.22}$$

The spectrum of a bounded diagonal operator is the closure of its diagonal entries. Since the only accumulation point is 1,

$$\operatorname{spec}(\mathcal{J}) = \{x_\gamma : \gamma \in \Gamma_+^*\} \cup \{1\}. \tag{17.23}$$

Each $x_\gamma$ is an isolated eigenvalue with one-dimensional eigenspace generated by the normalized atom indicator. The point 1 belongs to the spectrum as a limit point but is not an eigenvalue because $\mu(\{1\}) = 0$. Solving $x = (4\gamma^2 - 1)/(4\gamma^2 + 1)$ for $\gamma > 0$ gives (17.17).

Under the same unitary identification,

$$I - \mathcal{J} \sim \operatorname{diag}(1 - x_\gamma), \qquad 1 - x_\gamma = \frac{2}{4\gamma^2 + 1}. \tag{17.24}$$

To prove trace-class membership without appealing to an unnamed estimate, enumerate distinct ordinates increasingly as $\gamma_j^*$. Lemma 13.14 gives $\gamma_j^* \gg$

$j/\log j$, hence

$$\sum_j (1 - x_{\gamma_j^*}) \ll \sum_{j \geq 2} \frac{(\log j)^2}{j^2} < \infty. \tag{17.25}$$

A positive diagonal operator is trace class exactly when the sum of its diagonal eigenvalues is finite. Therefore

$$\mathrm{Tr}(I - \mathcal{J}) = \sum_{\gamma \in \Gamma_+^*} \frac{2}{4\gamma^2 + 1}, \tag{17.26}$$

and the same expression holds for both $E$ and $O$ because their supports coincide.

For every integer $k \geq 1$, the positive operator $(I - \mathcal{J})^k$ is trace class as well, since $0 \leq 1 - x_\gamma < 1$ and $\sum(1 - x_\gamma) < \infty$. Its eigenvalues are $(1 - x_\gamma)^k$, so for both reflection families

$$\mathrm{Tr}\Big((I - \mathcal{J}_E)^k\Big) = \mathrm{Tr}\Big((I - \mathcal{J}_O)^k\Big) = \sum_{\gamma \in \Gamma_+^*} \left(\frac{2}{4\gamma^2 + 1}\right)^k. \tag{17.27}$$

Finally, trace invariance under change of orthonormal basis gives

$$\sum_{n \geq 0} (1 - b_n^E) = \sum_{n \geq 0} (1 - b_n^O) = \sum_{\gamma \in \Gamma_+^*} \frac{2}{4\gamma^2 + 1}. \tag{17.28}$$

Indeed, for either family the left side is $\sum_n \langle (I - \mathcal{J}) e_n, e_n \rangle = \mathrm{Tr}(I - \mathcal{J})$; monotone convergence is automatic because $I - \mathcal{J} \geq 0$. □

*Remark* 17.3 (Finite truncations are not formulas for the zeros). The $N \times N$ leading Jacobi truncations have eigenvalues equal to the zeros of $\pi_N^E$ or $\pi_N^O$, equivalently Gaussian-quadrature nodes for the corresponding measure. These nodes are generally not points $x_\gamma$. Exact recovery of the support in (17.15) is an infinite inverse-moment statement requiring the full coefficient sequence. Thus (17.17) converts an already recovered spectral support point into an ordinate; it is not a finite closed formula for a zeta zero.

**Proposition 17.4** (Spectral masses and multiplicities)**.** *For $\gamma \in \Gamma_+^*$, the unnormalized representing measures satisfy*

$$\mu_O(\{x_\gamma\}) = \frac{4m_\gamma}{4\gamma^2 + 1}, \tag{17.29}$$

$$\mu_E(\{x_\gamma\}) = \frac{64 m_\gamma \gamma^2}{(4\gamma^2 + 1)^2}. \tag{17.30}$$

*Accordingly, the spectral measures of the first Jacobi basis vector are*

$$\nu_O(\{x_\gamma\}) = \frac{4m_\gamma}{\lambda_1(4\gamma^2+1)}, \tag{17.31}$$

$$\nu_E(\{x_\gamma\}) = \frac{64m_\gamma\gamma^2}{\lambda_2(4\gamma^2+1)^2}. \tag{17.32}$$

*Thus zero multiplicity is encoded by the distinguished cyclic spectral measure, even though the abstract multiplication operator has only one one-dimensional eigenspace for each distinct support point $x_\gamma$.*

*Proof.* The first pair follows from (9.5), (9.6), and (9.15)–(9.16). The second pair follows by the normalization (17.14). □

**Corollary 17.5** (Trace and simplicity defect)**.** *Under* RH*,*

$$\frac{\lambda_1}{2} - \operatorname{Tr}(I - \mathcal{J}_O) = 2\sum_{\gamma\in\Gamma_+^*} \frac{m_\gamma - 1}{4\gamma^2+1} \geq 0. \tag{17.33}$$

*The same trace occurs for $\mathcal{J}_E$. Equality in* (17.33) *holds if and only if every positive ordinate is simple.*

*Proof.* Equation (3.12) with $n = 1$ and (3.10) give

$$\frac{\lambda_1}{2} = 2\sum_{\gamma\in\Gamma_+^*} \frac{m_\gamma}{4\gamma^2+1}. \tag{17.34}$$

Subtract (17.26). Every summand in the resulting series is nonnegative, and it vanishes exactly when $m_\gamma = 1$. □

**Proposition 17.6** (Operator bounds and recovery of the first distinct ordinate)**.** *Assume* RH *and let $\gamma_1$ be the smallest positive zero ordinate. Then*

$$x_{\gamma_1} I \leq \mathcal{J}_E, \mathcal{J}_O \leq I, \tag{17.35}$$

*and*

$$\|I - \mathcal{J}_E\| = \|I - \mathcal{J}_O\| = \frac{2}{4\gamma_1^2+1}. \tag{17.36}$$

*Consequently*

$$\gamma_1 = \frac{1}{2}\sqrt{\frac{2}{\|I-\mathcal{J}_E\|} - 1} = \frac{1}{2}\sqrt{\frac{2}{\|I-\mathcal{J}_O\|} - 1}. \tag{17.37}$$

*This is an identity for the complete infinite operator, not a finite formula for a zeta zero.*

*Proof.* The spectrum in (17.15) is contained in $[x_{\gamma_1}, 1]$, which gives (17.35) by the spectral theorem. The positive compact operator $I - \mathcal{J}$ has largest eigenvalue $1 - x_{\gamma_1} = 2/(4\gamma_1^2 + 1)$, proving (17.36) and its inversion. □

**Theorem 17.7** (Singular values and the Schatten threshold)**.** *Assume* RH*, and enumerate the* distinct *positive ordinates increasingly as*

$$0 < \gamma_1^* < \gamma_2^* < \cdots . \tag{17.38}$$

*Then the singular values of either defect operator $K_E := I - \mathcal{J}_E$ or $K_O := I - \mathcal{J}_O$ are*

$$s_k(K_E) = s_k(K_O) = \frac{2}{4(\gamma_k^*)^2 + 1}. \tag{17.39}$$

*There exist positive constants $c, C$ such that, for all sufficiently large $k$,*

$$c\frac{(\log\log k)^2}{k^2} \le s_k(K_E) = s_k(K_O) \le C\frac{(\log k)^2}{k^2}. \tag{17.40}$$

*Consequently, for every $p > 0$,*

$$I - \mathcal{J}_E,\ I - \mathcal{J}_O \in \mathcal{S}_p \quad \Longleftrightarrow \quad p > \frac{1}{2}. \tag{17.41}$$

*No asymptotic equivalent for* (17.39) *is asserted here, because the operator counts distinct ordinates rather than zeros with multiplicity.*

*Proof.* Equation (17.39) follows from the diagonal multiplication representation and positivity of $K_E, K_O$. Let $N(T)$ count zeros with multiplicity and $N^*(T)$ count distinct positive ordinates. The Riemann–von Mangoldt formula gives $N(T) \asymp T\log T$. Under RH, Littlewood's classical bound

$$S(T) = O\!\left(\frac{\log T}{\log\log T}\right) \tag{17.42}$$

(see [41, Chapter XIV]) implies the corresponding multiplicity bound

$$m(\tfrac{1}{2} + i\gamma) = O\!\left(\frac{\log \gamma}{\log\log \gamma}\right). \tag{17.43}$$

Indeed, write the Riemann–von Mangoldt formula in the form $N(T) = F(T) + S(T) + O(T^{-1})$, where $F$ is continuous. Across an ordinate $\gamma$ of

multiplicity $m_\gamma$, the jump of $N$ is $m_\gamma$; taking one-sided limits at $\gamma$, the continuous terms cancel and

$$m_\gamma = S(\gamma+0) - S(\gamma-0) + O(1) = O\!\left(\frac{\log\gamma}{\log\log\gamma}\right).$$

Hence

$$T\log\log T \ll N^*(T) \ll T\log T. \tag{17.44}$$

Inverting these bounds gives

$$\frac{k}{\log k} \ll \gamma_k^* \ll \frac{k}{\log\log k}, \tag{17.45}$$

which yields (17.40). The upper bound is $p$-summable when $2p > 1$; the lower bound forces divergence when $2p \le 1$. This proves (17.41). □

**Proposition 17.8** (Endpoint exponent and critical Schatten exponent)**.** *Let a positive discrete Gram system satisfy* (13.5) *with exponent $\beta > 0$, and assume the two-sided node scale*

$$\log t_j = (1+o(1))\log j. \tag{17.46}$$

*Let $\mathcal{J}$ denote the Jacobi operator unitarily equivalent to multiplication by $x$ on $L^2(\mu)$, and put $K := I - \mathcal{J}$. Since the support consists of the distinct atoms $x(t_j)$ with unique accumulation point 1, the nonzero eigenvalues, hence the singular values, of the positive compact operator $K$ are the endpoint distances $1 - x(t_j)$. Then the critical Schatten exponent of $K$ is*

$$p_c(K) := \inf\{p > 0 :\ K \in \mathcal{S}_p\} = \frac{1}{\beta}. \tag{17.47}$$

*If in addition*

$$\sum_j \frac{1}{t_j} = \infty, \tag{17.48}$$

*then membership at the critical exponent fails, and*

$$K \in \mathcal{S}_p \quad\Longleftrightarrow\quad p > \frac{1}{\beta}. \tag{17.49}$$

*Proof.* By (13.5),

$$c_1 t_j^{-\beta} \le 1 - x(t_j) \le c_2 t_j^{-\beta} \tag{17.50}$$

for all sufficiently large $j$. From (17.46), for every $\varepsilon > 0$ there is $J_\varepsilon$ such that

$$j^{1-\varepsilon} \le t_j \le j^{1+\varepsilon}, \qquad j \ge J_\varepsilon. \tag{17.51}$$

Therefore, for fixed $p > 0$ and large $j$,

$$c_1^p j^{-\beta p(1+\varepsilon)} \le \Big(1 - x(t_j)\Big)^p \le c_2^p j^{-\beta p(1-\varepsilon)}. \tag{17.52}$$

If $p > 1/\beta$, choose $\varepsilon > 0$ so small that $\beta p(1-\varepsilon) > 1$; the upper comparison in (17.52) gives convergence. If $p < 1/\beta$, choose $\varepsilon > 0$ so small that $\beta p(1+\varepsilon) < 1$; the lower comparison gives divergence. This proves (17.47).

Hypothesis (17.46) alone does not decide the endpoint $p = 1/\beta$. The admissible sequence $t_j = j(\log j)^2$ satisfies (17.46) while $\sum_j t_j^{-1} < \infty$, so in that case $K \in \mathcal{S}_{1/\beta}$. Under the additional hypothesis (17.48) one has

$$\sum_j \Big(1 - x(t_j)\Big)^{1/\beta} \asymp \sum_j \frac{1}{t_j} = \infty, \tag{17.53}$$

so $K \notin \mathcal{S}_{1/\beta}$, and (17.49) follows. □

*Remark* 17.9 (One geometric constant, two invariants). Theorem 13.2 and Proposition 17.8 record the same quantity:

$$-\lim_{N\to\infty} \frac{\log D_N}{N^2 \log N} = \beta = \frac{1}{p_c(I - \mathcal{J})}. \tag{17.54}$$

Once their respective coarse node hypotheses are satisfied, both invariants are fixed by the same endpoint exponent. The determinant additionally uses the separated-subconfiguration and weight hypotheses of Theorem 13.2, whereas the critical Schatten exponent uses the two-sided logarithmic node scale (17.46). Neither invariant records the logarithmic Riemann–von Mangoldt refinement. The determinant is a macroscopic quantity attached to the finite compressions, the critical exponent is an ideal-theoretic invariant of the infinite defect operator, and (17.54) states that, at this coarse level, they encode the same endpoint exponent.

For the Li–Cayley map (13.67) one has $\beta = 2$, so the coefficient $-2$ of Theorem 13.16 and the value $1/2$ of Theorem 17.7 are one geometric constant

seen twice. The divergence hypothesis (17.48) holds in that case: by (17.45) one has $\gamma_k^* \ll k/\log\log k$, so $\sum_k (\gamma_k^*)^{-1} \gg \sum_k (\log\log k)/k = \infty$, and the strict form (17.49) is recovered.

In particular the value $1/2$ here is not the critical line. Replacing the endpoint map by one with exponent $\beta = 3$ moves the critical Schatten exponent to $1/3$ while leaving the zeros untouched, and the value is likewise unrelated to abscissas of convergence of prime series, where a coincidental $1/2$ also occurs. This reinforces Remark 13.12: the leading-order data of the pair is geometric, and arithmetic information can enter only below it.

**Corollary 17.10** (Quadratic Jacobi sum rule)**.** *Under* RH*,*

$$\sum_{n\geq 0}(1-b_n^E)^2 + 2\sum_{n\geq 0}(a_{n+1}^E)^2 = \sum_{\gamma\in\Gamma_+^*}\left(\frac{2}{4\gamma^2+1}\right)^2, \tag{17.55}$$

*and the identical formula holds with $E$ replaced by $O$.*

*Proof.* For $K = I - \mathcal{J}$, the Hilbert–Schmidt norm squared is the sum of the squares of all matrix entries in the Jacobi basis and also the sum of the squares of its eigenvalues. Since the off-diagonal entries occur twice, the stated identity follows. □

**Proposition 17.11** (Local Jacobi inequalities and summability)**.** *Under* RH*, for either Jacobi family,*

$$a_{n+1}^2 \leq (1-b_n)(1-b_{n+1}) \qquad (n\geq 0). \tag{17.56}$$

*Moreover*

$$\sum_{n\geq 0}(1-b_n) < \infty, \qquad \sum_{n\geq 0} a_{n+1} < \infty, \tag{17.57}$$

*and therefore*

$$b_n \longrightarrow 1, \qquad a_n \longrightarrow 0. \tag{17.58}$$

*Proof.* The operator $I - \mathcal{J}$ is positive. Positivity of each consecutive $2\times 2$ principal minor gives (17.56). Equation (17.28) gives the first series in (17.57). Since $1 - b_n \geq 0$,

$$a_{n+1} \leq \sqrt{(1-b_n)(1-b_{n+1})} \leq \frac{(1-b_n)+(1-b_{n+1})}{2}, \tag{17.59}$$

which gives the second series and the limits. □

**Proposition 17.12** (Fredholm determinant of the Jacobi defect)**.** *Under* RH*, the trace-class operators $K_E = I - \mathcal{J}_E$ and $K_O = I - \mathcal{J}_O$ have the same Fredholm determinant*

$$D(z) := \det(I - zK_E) = \det(I - zK_O) = \prod_{\gamma \in \Gamma_+^*} \left(1 - \frac{2z}{4\gamma^2 + 1}\right). \tag{17.60}$$

*The product converges absolutely and locally uniformly, and the zeros of $D$ are*

$$z_\gamma = \frac{4\gamma^2 + 1}{2}, \qquad \gamma \in \Gamma_+^*, \tag{17.61}$$

*one for each distinct positive ordinate. Zeta multiplicity is not encoded as the zero multiplicity of $D$.*

*Proof.* By Theorem 17.2, each defect $K = I - \mathcal{J}$ is positive and trace class, and under the multiplication representation it is diagonal on the normalized atom indicators with eigenvalues

$$\delta_\gamma := 1 - x_\gamma = \frac{2}{4\gamma^2 + 1} > 0, \qquad \gamma \in \Gamma_+^*. \tag{17.62}$$

The only additional spectral point is 0, coming from the accumulation $x_\gamma \to 1$; it contributes no factor to the Fredholm determinant. Since

$$\sum_{\gamma \in \Gamma_+^*} \delta_\gamma < \infty, \tag{17.63}$$

the trace-class Fredholm determinant satisfies

$$\det(I - zK) = \prod_{\gamma \in \Gamma_+^*} (1 - z\delta_\gamma). \tag{17.64}$$

For completeness, local uniform convergence follows directly: on $|z| \le R$, choose a tail for which $R\delta_\gamma \le 1/2$; then $|\log(1 - z\delta_\gamma)| \le 2R\delta_\gamma$, and the Weierstrass test applies. Thus the product defines an entire function and can vanish only when one factor vanishes, namely at

$$z = \delta_\gamma^{-1} = \frac{4\gamma^2 + 1}{2}. \tag{17.65}$$

Each distinct support point supplies one one-dimensional eigenspace of the multiplication operator, so these zeros are indexed by distinct ordinates rather than by zeta multiplicity. The two reflection families have the same support and hence the same eigenvalue list $\{\delta_\gamma\}$, proving equality of their Fredholm determinants. □

*Remark* 17.13 (What the Jacobi operator does and does not retain). The abstract operator $\mathcal{J}$ determines the set of distinct transformed ordinates through its spectrum, but not the zeta multiplicities. The normalized cyclic pair $(\mathcal{J}, e_0)$ determines the probability spectral measure and therefore the relative masses. To recover the integer multiplicities from the odd realization one must also retain its scale $\lambda_1$, equivalently the unnormalized measure $\mu_O$. Thus

$$(\mathcal{J}_O, e_0, \lambda_1) \quad \text{determines} \quad \{(\gamma, m_\gamma) : \gamma \in \Gamma_+^*\}. \tag{17.66}$$

The original semi-infinite matrix $\mathsf{O}$ already retains this scale because $o_{11} = \lambda_1$.

**Proposition 17.14** (Unitary equivalence of the two Jacobi operators)**.** *Under* RH*, $\mathcal{J}_E$ and $\mathcal{J}_O$ are unitarily equivalent.*

*Proof.* From (9.9),

$$\mathrm{d}\mu_E(x) = 2(1+x)\ \mathrm{d}\mu_O(x). \tag{17.67}$$

The actual support is contained in $[x_{\gamma_1}, 1)$ with $x_{\gamma_1} > -1$, so $2(1+x)$ is bounded above and bounded away from zero on the support. Define

$$(Uf)(x) := \sqrt{2(1+x)}\, f(x). \tag{17.68}$$

Then for every $f \in L^2(\mu_E)$,

$$\|Uf\|^2_{L^2(\mu_O)} = \int 2(1+x)|f(x)|^2\ \mathrm{d}\mu_O(x) = \int |f(x)|^2\ \mathrm{d}\mu_E(x), \tag{17.69}$$

so $U$ is an isometry. Because the multiplier $[2(1+x)]^{-1/2}$ is bounded on the support, every $g \in L^2(\mu_O)$ has the preimage $f = g/[2(1+x)]^{1/2} \in L^2(\mu_E)$; hence $U$ is onto and therefore unitary.

Let $M_x$ denote multiplication by $x$. Since both $U$ and $M_x$ are multiplication operators, $UM_x = M_xU$ on all of $L^2(\mu_E)$. Let $\mathcal{F}_E : \ell^2(\mathbb{N}_0) \to L^2(\mu_E)$ and $\mathcal{F}_O : \ell^2(\mathbb{N}_0) \to L^2(\mu_O)$ be the unitary maps sending the standard basis to the corresponding orthonormal polynomial bases. By definition

$$\mathcal{J}_E = \mathcal{F}_E^{-1} M_x \mathcal{F}_E, \qquad \mathcal{J}_O = \mathcal{F}_O^{-1} M_x \mathcal{F}_O. \tag{17.70}$$

Thus $V := \mathcal{F}_O^{-1} U \mathcal{F}_E$ is unitary and

$$V\mathcal{J}_E = \mathcal{F}_O^{-1} U M_x \mathcal{F}_E = \mathcal{F}_O^{-1} M_x U \mathcal{F}_E = \mathcal{J}_O V, \tag{17.71}$$

which proves unitary equivalence. □

**Theorem 17.15** (Darboux factorization of the Jacobi pair)**.** *Assume* RH. *Since* $-1$ *lies strictly below the common spectral support,* $\mathcal{J}_O + I$ *is strictly positive. Let*

$$\mathcal{J}_O + I = LL^* \tag{17.72}$$

*be its positive lower-bidiagonal Cholesky factorization. Then*

$$\mathcal{J}_E + I = L^*L. \tag{17.73}$$

*Writing*

$$L = \begin{pmatrix} r_0 & & & \\ s_0 & r_1 & & \\ & s_1 & r_2 & \\ & & \ddots & \ddots \end{pmatrix}, \qquad r_n > 0, \tag{17.74}$$

*one obtains*

$$1 + b_n^O = r_n^2 + s_{n-1}^2, \qquad a_{n+1}^O = r_n s_n, \tag{17.75}$$

$$1 + b_n^E = r_n^2 + s_n^2, \qquad a_{n+1}^E = s_n r_{n+1}, \tag{17.76}$$

*with* $s_{-1} := 0$.

*Proof.* Let $\{p_n^E\}_{n\geq 0}$ and $\{p_n^O\}_{n\geq 0}$ be the orthonormal polynomial bases in $L^2(\mu_E)$ and $L^2(\mu_O)$, chosen with positive leading coefficients. Consider the bounded operator

$$T : L^2(\mu_E) \longrightarrow L^2(\mu_O), \qquad (Tf)(x) := (x+1)f(x). \tag{17.77}$$

It is bounded because, using $\mathrm{d}\mu_E = 2(1+x)\,\mathrm{d}\mu_O$ and $0 < 1 + x \leq 2$ on the support,

$$\|Tf\|_{L^2(\mu_O)}^2 = \int (1+x)^2 |f(x)|^2 \,\mathrm{d}\mu_O(x) \tag{17.78}$$

$$= \frac{1}{2}\int (1+x)|f(x)|^2 \,\mathrm{d}\mu_E(x) \leq \|f\|_{L^2(\mu_E)}^2. \tag{17.79}$$

For $f \in L^2(\mu_E)$ and $g \in L^2(\mu_O)$,

$$\langle Tf, g\rangle_O = \int (x+1) f\overline{g} \,\mathrm{d}\mu_O = \frac{1}{2}\int f\overline{g} \,\mathrm{d}\mu_E. \tag{17.80}$$

The identity embedding $g \mapsto g$ from $L^2(\mu_O)$ to $L^2(\mu_E)$ is bounded, since

$$\|g\|_{L^2(\mu_E)}^2 = 2\int (1+x)|g|^2 \,\mathrm{d}\mu_O \leq 4\|g\|_{L^2(\mu_O)}^2. \tag{17.81}$$

Hence the adjoint is the everywhere-defined bounded operator

$$T^* g = \frac{1}{2} g. \tag{17.82}$$

Consequently

$$T^* T f = \frac{1}{2}(x+1) f, \qquad T T^* g = \frac{1}{2}(x+1) g. \tag{17.83}$$

Let $R$ be the matrix of $T$ from the $E$ orthonormal basis to the $O$ orthonormal basis,

$$R_{mn} := \langle p_m^O, T p_n^E \rangle_O. \tag{17.84}$$

The polynomial $(x+1)p_n^E$ has degree $n+1$ and, by the same orthogonality calculation used in Proposition 17.17, is orthogonal in $L^2(\mu_O)$ to every polynomial of degree $< n$. Therefore

$$T p_n^E = \alpha_n p_n^O + \beta_n p_{n+1}^O, \tag{17.85}$$

with no other orthonormal-polynomial components. Comparison of positive leading coefficients gives $\beta_n > 0$. The diagonal coefficient is also strictly positive:

$$\alpha_n = \langle p_n^O, (x+1) p_n^E \rangle_O = \frac{1}{2} \langle p_n^O, p_n^E \rangle_E.$$

Writing $p_n^O = (\kappa_n^O / \kappa_n^E) p_n^E +$ lower-degree terms, where $\kappa_n^{E,O} > 0$ are the leading coefficients, and using the $E$-orthogonality of $p_n^E$ to lower-degree polynomials gives

$$\alpha_n = \frac{1}{2} \frac{\kappa_n^O}{\kappa_n^E} > 0.$$

Thus $R$ is lower bidiagonal with positive diagonal and positive subdiagonal. In coefficient space, (17.83) becomes

$$R^* R = \frac{1}{2}(\mathcal{J}_E + I), \qquad R R^* = \frac{1}{2}(\mathcal{J}_O + I). \tag{17.86}$$

Set

$$L := \sqrt{2}\, R. \tag{17.87}$$

Then $L$ is lower bidiagonal and

$$\mathcal{J}_O + I = L L^*, \qquad \mathcal{J}_E + I = L^* L. \tag{17.88}$$

Since the common spectrum is contained in $[x_{\gamma_1}, 1]$ with $x_{\gamma_1} > -1$, both operators $\mathcal{J}_{E,O} + I$ are bounded below by $(1 + x_{\gamma_1})I > 0$. Hence the bidiagonal factor has no zero diagonal entry; with the positive-leading-coefficient convention we take $r_n > 0$. This establishes the asserted Cholesky/Darboux factorization without an appeal to a formal infinite Gaussian elimination.

Finally, write $L_{nn} = r_n$ and $L_{n+1,n} = s_n$, with $s_{-1} = 0$. Direct multiplication gives

$$(LL^*)_{nn} = r_n^2 + s_{n-1}^2, \qquad (LL^*)_{n,n+1} = r_n s_n, \tag{17.89}$$

$$(L^*L)_{nn} = r_n^2 + s_n^2, \qquad (L^*L)_{n,n+1} = s_n r_{n+1}. \tag{17.90}$$

Comparison with the Jacobi entries proves (17.75)–(17.76). □

**Proposition 17.16** (Exact relation between the Weyl functions)**.** *Define, for $z$ off the common spectrum,*

$$m_E(z) := \int \frac{\mathrm{d}\nu_E(x)}{x - z}, \qquad m_O(z) := \int \frac{\mathrm{d}\nu_O(x)}{x - z}. \tag{17.91}$$

*Then*

$$m_E(z) = \frac{2\lambda_1}{\lambda_2}\left[1 + (z+1)m_O(z)\right], \tag{17.92}$$

*and, for $z \neq -1$, equivalently*

$$m_O(z) = \frac{\dfrac{\lambda_2}{2\lambda_1} m_E(z) - 1}{z + 1}. \tag{17.93}$$

*Proof.* From (9.9) and (17.14),

$$\mathrm{d}\nu_E(x) = \frac{2\lambda_1}{\lambda_2}(1 + x)\,\mathrm{d}\nu_O(x). \tag{17.94}$$

Use

$$\frac{1 + x}{x - z} = 1 + \frac{z + 1}{x - z} \tag{17.95}$$

and $\nu_O([-1, 1]) = 1$. □

**Proposition 17.17** (Christoffel relation between orthogonal systems)**.** *Under* RH*,*

$$\pi_n^E(x) = \frac{\pi_{n+1}^O(x) - \dfrac{\pi_{n+1}^O(-1)}{\pi_n^O(-1)}\pi_n^O(x)}{x + 1}. \tag{17.96}$$

*The denominator $\pi_n^O(-1)$ never vanishes.*

*Proof.* The scalar factor 2 in $\mathrm{d}\mu_E = 2(1+x)\,\mathrm{d}\mu_O$ does not affect orthogonality. Let $q$ be any polynomial with $\deg q < n$. Since $\pi_n^E$ is orthogonal to every such $q$ in $L^2(\mu_E)$,

$$0 = \int \pi_n^E(x) q(x)\ \mathrm{d}\mu_E(x) = 2\int (x+1)\pi_n^E(x) q(x)\ \mathrm{d}\mu_O(x). \tag{17.97}$$

Thus the monic polynomial $(x+1)\pi_n^E(x)$ has degree $n+1$ and is orthogonal, with respect to $\mu_O$, to all polynomials of degree at most $n-1$.

Because the monic orthogonal polynomials $\pi_0^O, \dots, \pi_{n+1}^O$ form a basis of the polynomials of degree at most $n+1$, write

$$(x+1)\pi_n^E(x) = \sum_{j=0}^{n+1} c_j \pi_j^O(x). \tag{17.98}$$

Taking the $L^2(\mu_O)$ inner product with $\pi_j^O$ for $j < n$ shows $c_j = 0$. Comparing leading coefficients gives $c_{n+1} = 1$, so

$$(x+1)\pi_n^E(x) = \pi_{n+1}^O(x) + c_n \pi_n^O(x). \tag{17.99}$$

Every zero of $\pi_n^O$ lies in the open convex hull of the support of the positive measure $\mu_O$; the support is contained in $(-1,1)$ and has left endpoint $x_{\gamma_1} > -1$. Hence $\pi_n^O(-1) \neq 0$. Evaluating the preceding identity at $x = -1$ gives $c_n = -\pi_{n+1}^O(-1)/\pi_n^O(-1)$, which is exactly (17.96) after division by $x+1$. The numerator vanishes at $x=-1$, so the quotient is indeed a polynomial of degree $n$. $\square$

**Corollary 17.18** (Determinant bridge)**.** *Under* RH*,*

$$\det \mathsf{E}_N = 2^N (-1)^N \pi_N^O(-1) \det \mathsf{O}_N. \tag{17.100}$$

*Proof.* We first derive the required Christoffel determinant identity from the moment determinants. Let $\mu_k := \int x^k\ \mathrm{d}\mu_O(x)$ and

$$D_N[\mu_O] := \det(\mu_{i+j})_{i,j=0}^{N-1} > 0. \tag{17.101}$$

The monic orthogonal polynomial of degree $N$ has the determinantal representation

$$\pi_N^O(c) = \frac{1}{D_N[\mu_O]} \det \begin{pmatrix} \mu_0 & \mu_1 & \cdots & \mu_N \\ \mu_1 & \mu_2 & \cdots & \mu_{N+1} \\ \vdots & \vdots & & \vdots \\ \mu_{N-1} & \mu_N & \cdots & \mu_{2N-1} \\ 1 & c & \cdots & c^N \end{pmatrix}. \tag{17.102}$$

Indeed, expansion along the last row shows that the right side is monic of degree $N$; integrating it against $c^k \, \mathrm{d}\mu_O(c)$ for $0 \le k < N$ replaces the last row by the already present moment row $(\mu_k, \dots, \mu_{k+N})$, so the determinant vanishes. Thus it is exactly the monic orthogonal polynomial.

Starting from the determinant in (17.102), perform for $j = N-1, N-2, \dots, 0$ the column operation

$$C_{j+1} \leftarrow C_{j+1} - cC_j. \tag{17.103}$$

The determinant is unchanged. The last row becomes $(1, 0, \dots, 0)$, while after deleting its first column the remaining $N \times N$ minor has entries $\mu_{i+j+1} - c\mu_{i+j}$, $0 \le i, j < N$. Expansion along the last row therefore gives

$$D_N[\mu_O] \, \pi_N^O(c) = (-1)^N \det(\mu_{i+j+1} - c\mu_{i+j})_{i,j=0}^{N-1}. \tag{17.104}$$

But the matrix on the right is precisely the moment matrix of the measure $(x-c) \, \mathrm{d}\mu_O(x)$. Hence

$$D_N[(x-c)\mu_O] = (-1)^N \pi_N^O(c) D_N[\mu_O]. \tag{17.105}$$

Taking $c = -1$ proves

$$D_N[(x+1)\mu_O] = (-1)^N \pi_N^O(-1) D_N[\mu_O]. \tag{17.106}$$

Now $\mu_E = 2(x+1)\mu_O$. Multiplying a measure by the scalar 2 multiplies every entry of its $N \times N$ moment matrix by 2, and therefore its determinant by $2^N$. Finally, Theorem 10.1 gives

$$\det \mathsf{E}_N = 2^{N(N-1)} D_N[\mu_E], \qquad \det \mathsf{O}_N = 2^{N(N-1)} D_N[\mu_O]. \tag{17.107}$$

The identical basis factors cancel in the ratio. Combining these identities with (17.106) yields

$$\det \mathsf{E}_N = 2^N (-1)^N \pi_N^O(-1) \det \mathsf{O}_N, \tag{17.108}$$

as claimed. □

**Corollary 17.19** (Bridge between the elimination pivots)**.** *Under* RH*,*

$$d_n^E = -2 \frac{\pi_n^O(-1)}{\pi_{n-1}^O(-1)} d_n^O \qquad (n \ge 1), \tag{17.109}$$

*where $d_n^E, d_n^O$ are defined in* (17.10)*.*

*Proof.* Divide (17.100) for $N = n$ by the same identity for $N = n-1$. □

## 18. The Christoffel margin: exact factorisation of the pivot ratio and a two-sided band

Corollary 17.18 links the two determinant sequences through the single number $\pi_N^O(-1)$. This section identifies that number as a truncated Fredholm determinant of the Jacobi defect, deduces an exact factorisation of the ratio of the two pivot sequences into a universal constant times an arithmetic factor, and turns the resulting monotonicity into a two-sided band which is a necessary condition for RH requiring no arithmetic input at all.

**Definition 18.1** (Christoffel factor)**.** For $N \geq 1$ set

$$F_N := \frac{(-1)^N \pi_N^O(-1)}{2^N}, \qquad F_0 := 1. \tag{18.1}$$

Corollary 17.18 then reads

$$\boxed{\Delta_N^E = 4^N F_N \, \Delta_N^O} \tag{18.2}$$

**Proposition 18.2** ($F_N$ is a truncated Fredholm determinant of the Jacobi defect)**.** *Under* RH*, let $\mathcal{J}_O^{(N)}$ be the leading $N \times N$ truncation of the odd Jacobi operator and let $\mathcal{D}_N := I_N - \mathcal{J}_O^{(N)}$ be its defect (the symbol $\mathcal{K}$ is reserved for the Christoffel–Darboux kernels of Section 15). Then*

$$\boxed{F_N = \det\left(\frac{I_N + \mathcal{J}_O^{(N)}}{2}\right) = \det\Big(I_N - \tfrac{1}{2}\mathcal{D}_N\Big) = \sum_{k=0}^{N} \frac{(-1)^k}{2^k}\, e_k\Big(\mathcal{D}_N\Big),} \tag{18.3}$$

*where $e_k$ denotes the $k$-th elementary symmetric function of the eigenvalues.*

*Proof.* The monic orthogonal polynomial of degree $N$ is the characteristic polynomial of the leading Jacobi truncation, $\pi_N^O(x) = \det\Big(xI_N - \mathcal{J}_O^{(N)}\Big)$; this is the statement recalled in Remark 17.3. Evaluating at $x = -1$,

$$\pi_N^O(-1) = \det\Big(-I_N - \mathcal{J}_O^{(N)}\Big) = (-1)^N \det\Big(I_N + \mathcal{J}_O^{(N)}\Big), \tag{18.4}$$

so $(-1)^N \pi_N^O(-1) = \det\Big(I_N + \mathcal{J}_O^{(N)}\Big)$ and (18.1) gives the first equality in (18.3). The second is $I + \mathsf{J} = 2I - (I - \mathsf{J})$ divided by 2, and the third is the expansion of $\det(I - zK)$ in elementary symmetric functions at $z = 1/2$. $\square$

**Theorem 18.3** (Margin factorisation)**.** *Let* $p_N^E := \Delta_N^E/\Delta_{N-1}^E$ *and* $p_N^O := \Delta_N^O/\Delta_{N-1}^O$ *be the Cholesky pivots* (14.3)–(14.4)*, and define the* margin

$$\kappa_N := \frac{p_N^E}{p_N^O}. \tag{18.5}$$

*Then, under* RH*,*

$$\boxed{\kappa_N = 4\,\frac{F_N}{F_{N-1}}} \qquad (N \geq 1). \tag{18.6}$$

*Within any positive even–odd pair with the same Christoffel relation* $\mathrm{d}\mu_E = 2(1+x)\,\mathrm{d}\mu_O$ *and the same polynomial normalizations used here, the constant* 4 *is independent of the particular measure. The measure-dependent information in this factorization sits in* $F_N/F_{N-1}$*.*

*Proof.* Divide (18.2) at $N$ by the same identity at $N-1$. □

**Theorem 18.4** (Monotonicity, band, and Fredholm limit)**.** *Let* $\mathcal{D} = I - \mathcal{J}_O$ *be the Jacobi defect of Proposition 17.11, a positive trace-class operator satisfying* $\operatorname{Tr}\mathcal{D} \leq \lambda_1/2$*, with equality if and only if every positive zero ordinate is simple. Under* RH*:*

*(i)* $F_N$ *is nonincreasing in* $N$ *and* $0 < F_N \leq 1$*;*

*(ii)* $\det\Big(I - \frac{1}{2}\mathcal{D}\Big) \leq F_N \leq 1$ *for every* $N$*, and* $F_N \to \det\Big(I - \frac{1}{2}\mathcal{D}\Big)$ *as* $N \to \infty$*, where*

$$\det\Big(I - \tfrac{1}{2}\mathcal{D}\Big) = \prod_{\gamma \in \Gamma_+^*} \frac{4\gamma^2}{4\gamma^2 + 1}; \tag{18.7}$$

*(iii) consequently*

$$\boxed{4\det\Big(I - \tfrac{1}{2}\mathcal{D}\Big) \;\leq\; \kappa_N \;\leq\; 4 \qquad (N \geq 1),} \tag{18.8}$$

*and the weaker but entirely explicit form*

$$4 - \lambda_1 \;\leq\; \kappa_N \;\leq\; 4. \tag{18.9}$$

*Proof.* Write $A := \frac{1}{2}(I + \mathcal{J}_O) = I - \frac{1}{2}\mathcal{D}$ and $A_N$ for its leading $N \times N$ truncation, so that $F_N = \det A_N$ by Proposition 18.2. By Theorem 17.2 the

spectrum of $\mathcal{J}_O$ lies in $[x_{\gamma_1}, 1]$, hence $0 \prec A \preceq I$ and every diagonal entry of $A$ in the Jacobi basis satisfies $0 < A_{nn} \leq 1$.

(i) Positivity of $F_N$ is positivity of the principal minors of the positive definite operator $A$. For monotonicity, bordering gives

$$\det A_N = \det A_{N-1}\Big(A_{NN} - v^T A_{N-1}^{-1} v\Big) \leq \det A_{N-1}\, A_{NN} \leq \det A_{N-1}, \tag{18.10}$$

where $v$ is the last off-diagonal block and $A_{N-1}^{-1} \succ 0$. Also $F_1 = A_{11} \leq 1$.

(ii) Let $\mu_1 \geq \mu_2 \geq \cdots$ be the eigenvalues of $\mathcal{D}$ and $\mu_1^{(N)} \geq \cdots \geq \mu_N^{(N)}$ those of the compression $\mathcal{D}_N = P_N \mathcal{D} P_N$. Cauchy interlacing for compressions gives $\mu_j^{(N)} \leq \mu_j$ for $1 \leq j \leq N$, and $0 \leq \mu_j < 1$ for all $j$ because $\|\mathcal{D}\| = 1 - x_{\gamma_1} < 1$ by (17.36). Hence

$$F_N = \prod_{j=1}^{N}\Big(1 - \tfrac{1}{2}\mu_j^{(N)}\Big) \geq \prod_{j=1}^{N}\Big(1 - \tfrac{1}{2}\mu_j\Big) \geq \prod_{j\geq 1}\Big(1 - \tfrac{1}{2}\mu_j\Big) = \det\Big(I - \tfrac{1}{2}\mathcal{D}\Big), \tag{18.11}$$

each factor being in $(0, 1]$. Since $\mathcal{D}$ is trace class, $\mathcal{D}_N \to \mathcal{D}$ in trace norm and the Fredholm determinant is trace-norm continuous, so $F_N \to \det(I - \mathcal{D}/2)$. Formula (18.7) is $D(1/2)$ in the notation of Proposition 17.12, since $1 - \frac{1}{4\gamma^2+1} = \frac{4\gamma^2}{4\gamma^2+1}$.

(iii) Combining (i), (ii) with (18.6) gives $\kappa_N = 4F_N/F_{N-1} \leq 4$ and $\kappa_N \geq 4\det(I - \mathcal{D}/2)$, because $F_N \geq \det(I - \mathcal{D}/2)$ and $F_{N-1} \leq 1$. Finally, for a positive operator with eigenvalues in $[0, 1)$ one has $\prod(1 - \mu_j/2) \geq 1 - \frac{1}{2}\sum \mu_j$, and $\operatorname{Tr}\mathcal{D} \leq \lambda_1/2$ by Corollary 17.5, whence $\det(I - \mathcal{D}/2) \geq 1 - \lambda_1/4$ and $4\det(I - \mathcal{D}/2) \geq 4 - \lambda_1$. □

**Corollary 18.5** (An arithmetic-free necessary condition)**.** *If* RH *holds, then for every* $N \geq 1$*,*

$$\Delta_N^E\, \Delta_{N-1}^O \leq 4\, \Delta_{N-1}^E\, \Delta_N^O.$$

*Equivalently* $F_N \leq F_{N-1}$*. No zero, prime, or Li coefficient beyond those defining the two determinants enters this inequality.*

*Proof.* This is the upper half of (18.8), whose only input is $\mathcal{J}_O \preceq I$, that is, that the odd reflection measure is supported in $(-\infty, 1]$. Under RH the support is $\{x_\gamma\} \cup \{1\} \subset (0, 1]$. □

## 19. Hankel determinants of zeta values

**Logical status.** This section is independent of RH and of the Li–Weil zero measures. It is an external application of the abstract endpoint theorem to a positive Dirichlet-series moment measure.

Fix an integer $a \geq 1$ and an integer $b \geq 0$, and set $H_n^{(a,b)} := \det_{1\leq i,j\leq n}\Big(\zeta(a(i+j)+b)\Big)$. Expanding each entry,

$$\zeta(a(i+j)+b) = \sum_{m\geq 1} m^{-b}\Big(m^{-a}\Big)^{i+j}, \tag{19.1}$$

so $H_n^{(a,b)}$ is the monomial Hankel determinant of the positive discrete measure

$$\nu = \sum_{m\geq 1} m^{-(b+2a)}\,\delta_{u_m}, \qquad u_m = m^{-a}, \tag{19.2}$$

in indices $r, s = 0, \dots, n-1$. Two immediate consequences of positivity: $H_n^{(a,b)} > 0$ for every $n$, so no nonvanishing argument is required; and the Cauchy–Binet lower bound of Theorem 13.2 is available.

The atoms of $\nu$ accumulate at 0 rather than at 1. Apply the reflection $x \mapsto 1-x$: the change of basis from $\{1, x, \dots, x^{N-1}\}$ to $\{1, (1-x), \dots, (1-x)^{N-1}\}$ is triangular with diagonal entries $\pm 1$, so the Gram determinants coincide. The reflected measure has atoms $x_m = 1 - m^{-a} \uparrow 1$ with support map

$$x(t) = 1 - t^{-a}, \qquad 1 - x(t) = t^{-a}, \qquad x'(t) = a\,t^{-a-1}. \tag{19.3}$$

**Theorem 19.1** (Full leading asymptotics of $H_n^{(a,b)}$)**.** *For all positive integers $a$ and all integers $b \geq 0$,*

$$\boxed{\log \det_{1\leq i,j\leq n}\Big(\zeta(a(i+j)+b)\Big) = -a\,n^2\log n + O(n^2).} \tag{19.4}$$

*In particular $\log H_n^{(a,b)}/(n^2\log n) \to -a$. The statement is unconditional.*

*Proof.* Hypothesis (i) of Theorem 13.2 holds with $\beta = a$ and constants $c = C = 1$ for the first pair and $c = C = a$ for the second. Hypothesis (ii) holds with $t_j = j$ and no error at all. Hypothesis (iii) holds with $s_{N,k} = k$ and $\eta_N = 0$: then $s_{N,N} = N$ and $s_{N,\ell} - s_{N,k} = \ell - k$. Hypothesis (iv) holds because $\sum_{m\leq N} \log m^{-(b+2a)} = -(b+2a)\,N\log N + O(N) = o(N^2\log N)$.

For the sharper error term, run the two bounds of Theorem 13.2 with these exact data. In the upper bound, $\log(1-x_j) = -a\log j$ exactly, so

$\sum_{j\le n}\log(1-x_j) = -a\,n\log n + O(n)$ and $\log h_n \le -2a\,n\log n + O(n)$; summing gives $\log H_n^{(a,b)} \le -a\,n^2\log n + O(n^2)$. For the lower bound select the first $n$ reflected atoms $y_k = 1 - k^{-a}$. The mean-value theorem gives, for $1 \le k < \ell \le n$,

$$y_\ell - y_k = k^{-a} - \ell^{-a} = a\xi_{k\ell}^{-a-1}(\ell - k) \ge a\,n^{-a-1}(\ell - k),$$

with $k < \xi_{k\ell} < \ell \le n$. Hence the selected Cauchy–Binet term is at least

$$\left(\prod_{k=1}^{n} k^{-(b+2a)}\right) a^{n(n-1)} n^{-(a+1)n(n-1)} \left(\prod_{1\le k<\ell\le n}(\ell - k)\right)^2.$$

Taking logarithms and applying Lemma 13.1 gives

$$\log H_n^{(a,b)} \ge -an^2\log n + O(n^2);$$

the weight product contributes only $O(n\log n)$. □

**Corollary 19.2** (Quantitative denominator growth)**.** *Let $a \ge 1$ and $b \ge 1$ be integers, and suppose all the numbers $\zeta(an+b)$, $n \ge 1$, are rational. Let $q_n$ be the common denominator of $\zeta(a+b), \dots, \zeta(an+b)$. Then*

$$\liminf_{n\to\infty} \frac{\log q_n}{n\log n} \;\ge\; \frac{a}{2}. \tag{19.5}$$

*Consequently, either $\zeta(an+b)$ is irrational for infinitely many $n$, or $q_n$ grows at least like $n^{(a/2+o(1))n}$.*

*Remark* 19.3. The determinant theorem allows $b = 0$, because its smallest zeta argument is $2a + b > 1$. The denominator corollary starts at $\zeta(a+b)$; the restriction $b \ge 1$ excludes the pole that would occur at $a = 1$, $b = 0$.

*Proof.* By definition of the common denominators, $q_k \mid q_n$ for $k \le n$, and

$$q_{n+1}q_{n+2}\cdots q_{2n}\,H_n^{(a,b)} = \det_{1\le i,j\le n}\Big(q_{i+n}\zeta(a(i+j)+b)\Big) \in \mathbb{Z}.$$

By positivity $H_n^{(a,b)} > 0$ for every $n$, so the integer is nonzero and $q_{n+1}\cdots q_{2n}\,H_n^{(a,b)} \ge 1$ for all $n$. Hence $\log(q_{n+1}\cdots q_{2n}) \ge -\log H_n^{(a,b)} = a\,n^2\log n + O(n^2)$ by Theorem 19.1. Since $q_k \le q_{2n}$ for $k \le 2n$, the left side is at most $n\log q_{2n}$, so $\log q_{2n} \ge a\,n\log n + O(n)$. Writing $N = 2n$ gives $\log q_N \ge \frac{a}{2}N\log N + O(N)$. □

## 20. Conclusions

The conclusions retain the separation between results independent of the Riemann Hypothesis (RH) and results obtained after assuming it.

**Unconditional reflection structure.** The Li sequence is encoded by two Toeplitz–Hankel reflection arrays with exact reconstruction formulas, a discrete wave equation, rank-two displacement, closed bordering recursions, and local $LDL^T$ pivot dynamics. The reflected Li–Weil pairing is likewise an unconditional identity. These statements do not use positivity of the Weil form or the location of the nontrivial zeros. The even family alone gives a matrix criterion equivalent to RH through the transformed Bombieri–Lagarias multiset; equivalence to RH is not the same as assuming RH in the proof of the algebraic identities.

**Positive and spectral structure under** RH**.** Once RH is assumed, the reflection arrays become positive sine Gram systems on the Cayley images of the zeros. They admit positive Chebyshev moment measures, Cauchy–Binet–Vandermonde expansions, a fixed-order projection determinantal process with exact zero-influence laws, and infinite Jacobi operators whose spectral support recovers the distinct ordinates. The normalized cyclic spectral measure together with its scale retains multiplicity. The Jacobi defect has critical Schatten exponent $1/2$, and the even and odd systems are linked by exact Christoffel/Darboux and Fredholm determinant structures. Finite Jacobi truncations remain quadrature approximations and are not formulas for individual zeta zeros.

**Endpoint law and its precise scope.** The abstract endpoint theorem is logically independent of RH. It does not assert that $\beta$ alone controls every discrete determinant: the counterexamples of Remark 13.6 show that the same $\beta = 2$ can give normalized limits $0$, $-2$, or $-\infty$ when the node clock is changed. Once the parameters are fixed at the polynomial scale $t_j = j^{1+o(1)}$ in the precise two-sided sense of hypotheses (ii)–(iii), with subleading selected weights, the leading coefficient is exactly $-\beta$. For the Li–Cayley map this yields $-2$ under RH. Logarithmic Riemann–von Mangoldt refinements can affect lower orders but not that leading coefficient. The displayed second-order law belongs to the separate conditional refinement of Subsubsection 13.2.1; it is not a consequence of RH alone.

**Independent zeta-value Hankel application.** The zeta-value Hankel family is a separate unconditional application of the same endpoint mecha-

nism. Its support has $\beta = a$ on the exact node scale $t_j = j$, and therefore

$$\log H_n^{(a,b)} = -an^2 \log n + O(n^2).$$

Thus the paper contributes to a collection of matrix themes already represented in *Linear Algebra and its Applications*: Hankel–Jacobi interaction [37], structured low-rank Hankel analysis [15], compound-matrix determinantal identities [1], and determinant-sensitive fixed-rank perturbation theory [2]. The present contribution joins these themes through a single Li–Weil reflection construction, while retaining the precise logical separation between its unconditional, RH-conditional, and additional-hypothesis layers.

The common theme is a separation of roles: the algebraic reflection structure is unconditional; positive zero-side geometry is conditional on RH; and the endpoint coefficient is universal only after the node scale has been fixed. This separation is the organizing principle behind the main contributions listed in Subsection 1.1.

## Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

No external data set was used. All numerical data generated for the finite illustrations are reported in the manuscript.

## Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used Codex for LaTeX compilation and double-checking of numerical tests. The author reviewed and edited the output as needed and takes full responsibility for the content of the published article.

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